

## 1.0 History of Statistics

The Word statistics have been derived from Latin word “**Status**” or the Italian word “**Statista**”, meaning of these words is “**Political State**” or a Government. Shakespeare used a word Statist in his drama Hamlet (1602). In the past, the statistics was used by rulers. The application of statistics was very limited but rulers and kings needed information about lands, agriculture, commerce, population of their states to assess their military potential, their wealth, taxation and other aspects of government.

Gottfried Achenwall used the word statistik at a German University in 1749 which means that political science of different countries. In 1771 W. Hooper (Englishman) used the word statistics in his translation of Elements of Universal Erudition written by Baron B.F Bieford, in his book statistics has been defined as the science that teaches us what is the political arrangement of all the modern states of the known world. There is a big gap between the old statistics and the modern statistics, but old statistics also used as a part of the present statistics. During the 18th century the English writer have used the word statistics in their works, so statistics has developed gradually during last few centuries. A lot of work has been done in the end of the nineteenth century.

At the beginning of the 20th century, William S Gosset was developed the methods for decision making based on small set of data. During the 20th century several statistician are active in developing new methods, theories and application of statistics. Now these days the availability of electronics computers is certainly a major factor in the modern development of statistics.

### 1.1 Definition of Statistics.

**Prof. A.L. Bowley gives the three definitions of Statistics which as follows:**

1. Statistics may be called the science of counting. This definition is too narrow because it covers only one aspect of the science, namely collection of data.
2. Statistics may be called the science of averages. This definition also is not satisfactory because averages are not only one of the devices used in statistical analysis.
3. Statistics is the science of the measurement of social organism, regarded as a whole in all its manifestations. This definition also is not satisfactory because it confines the scope of statistics only to sociology.

Boddington defines statistics as “the science of estimates and probabilities.” This definition also is unacceptable because estimates and probabilities are only a part of statistical method.

Croxton and Cowden say, “Statistics may be defined as a science of collection, presentation, analysis and interpretation of numerical data.” This definition clearly points out four stages in statistical investigation, namely: (1) collection of data (2) presentation of data (3) analysis of data and (4) interpretation of data.

However, to the above stages one more stage may be added and that is the organization of data. Thus, **statistics may be defined as the science of collection, organization, presentation, analysis and interpretation of numerical data.**

### 1.2 Definition of Statistics by Singular and Plural Sense:

In singular perspective the meaning of statistics is that statistics suggests those principles, formula and functions through which the calculated subjects are expressed.

In plural perspective statistics means the expression of the calculated affairs of our day to day life. For example – the amount of recent per capita(average) expenditure of the officials workers of an industry, the annual birth rate or death rate of Bangladesh etc.

### 1.3 Types of Statistics:

- (i) Descriptive Statistics
- (ii) Inferential or Inductive Statistics

#### Descriptive Statistics & Inferential or Inductive Statistics:

Descriptive & Inductive Statistics refer to the application of statistical methods & theories. Descriptive Statistics deals with collection, presentation, analysis & interpretation of statistical data without involving generalizations. On the other hand, generalization, predictions, estimations & decisions made from collected & analyzed data using statistical methods & theories come within the fold of Inductive Statistics. When we compute average with the sole purpose of describing a given phenomenon, we are using the mean as a tool of descriptive statistics, whereas if we make use of the average for making prediction or general conclusion, we are using it in inductive sense.

For example: Average salary of the employees of a factory may be described by calculating the mean salary of all the employees of the factory. This involves only description of mean salary. But if this mean averages of a particular factory is used as an estimate of average salary of the employees of all the factories in an industry instead of computing the average for all the employees in the industry then we use the mean in inductive sense.

### 1.4 Functions of Statistics

1. It present facts in a definite form.
2. It simplifies mass of figures.
3. It facilitate comparison.
4. It helps in formulating and testing hypothesis.
5. It helps in prediction.
6. It helps in the formulation of suitable policies.

### 1.5 Scope and Importance of Statistics:

**1. *Statistics and planning:*** Statistics is indispensable into planning in the modern age which is termed as “the age of planning”. Almost all over the world the govt. are re-storing to planning for economic development.

**2. *Statistics and economics:*** Statistical data and techniques of statistical analysis have to immensely useful involving economical problem. Such as wages, price, time series analysis, demand analysis.

**3. *Statistics and business:*** Statistics is an irresponsible tool of production control. Business executive are relying more and more on statistical techniques for studying the much and desire of the valued customers.

**4. *Statistics and industry:*** In industry statistics is widely used inequality control. In production

engineering to find out whether the product is confirming to the specifications or not. Statistical tools, such as inspection plan, control chart etc.

**5. *Statistics and mathematics:*** Statistics are intimately related recent advancements in statistical technique are the outcome of wide applications of mathematics.

**6. *Statistics and modern science:*** In medical science the statistical tools for collection, presentation and analysis of observed facts relating to causes and incidence of diseases and the result of application various drugs and medicine are of great importance.

**7. *Statistics, psychology and education:*** In education and physiology statistics has found wide application such as, determining or to determine the reliability and validity to a test, factor analysis etc.

**8. *Statistics and war:*** In war the theory of decision function can be a great assistance to the military and personal to plan “maximum destruction with minimum effort.”

**6. *Accounting:*** Statistical data are also employed in accounting particularly in auditing function, the technique of sampling and destination is frequently used.

**7. *Control:*** The management control process combines statistical and accounting method in making the overall budget for the coming year including sales, materials, labor and other costs and net profits and capital requirement.

## **1.6 Limitations of Statistics**

The important limitations of statistics are:

1. Statistics does not deal with individual items
2. Statistics deals only with quantitative characteristics.
3. Statistical results are true only on an average.
4. Statistics can be misused.
5. Statistical results are not accurately correct.
6. Statistical laws are not exact.

**1.7 Population:** In Statistics, Aggregate of all possible outcomes of the universe which we investigated is called population.

**There are two types of statistical population.**

**(1) Finite Population (2) Infinite Population**

(1) **Finite Population:** A **population** is said to be **finite** if it consists of a **finite** or fixed number of elements. For **example:** Heights of all students enrolled in Dhaka university.

(2) **Infinite population:** A **population** is said to be **infinite** if there is no limit to the number of elements it can contain.

**For example:** The **population** of stars in space, the number of red blood cells in a person's body, and so on.

**1.8 Sample:** A representative part of population is called sample.

**For example:** A small piece of a tumor that is taken to test in a lab.

**Suitable example of Population and Sample:**

We are interested to make 1 kg rice. We put 1 kg rice with water in a pan. Later on we take 5 or 6 pieces of rice in the spoon to see either it is boiled or not. This 5 or 6 pieces of rice is our sample and 1 kg rice is our population.

**1.9 Variable:** A quantitative characteristics (but not always quantitative) which varies from one individual item to another is called variable.

There are two types of variables:

**(1) Qualitative Variable**

**(2) Quantitative Variable**

(1) **Qualitative Variable:** If the Variable is nonnumeric then it is called qualitative Variable.

**Example:** Color, religion, sex etc.

**Quantitative Variable:** If the Variable is numeric then it is called quantitative Variable.

**Example:** Height, weight, age etc.

There are two types of quantitative Variable:

**(1) Discrete Variable (2) Continuous Variable**

**(1) Discrete Variable:** Discrete variable can assume only certain values and there are usually “gaps” between the values.

**Example:** No. of workers in a factory, no. of bedrooms in a house etc.

**(2) Continuous Variable:** Continuous variable can assume any value within a specific range.

**Example:** Height, weight, age etc.

# Data Collection

**Population:** In Statistics, Aggregate of all possible outcomes of the universe which we investigated is called population.

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**(2) Quantitative Variable**

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**(2) Continuous Variable:** Continuous variable can assume any value within a specific range.

**Example:** Height, weight, age etc.

**Data:** Raw materials of Statistics are called data.

**Example:** We are interested to find the average income of parents of DIIT students. For this purpose, we take sample of 300 parents out of 2000 parents (Say). So, raw material is 300. That is total data = 300. After that we find average by using this data.

**In general, there are two types of data.**

**(1) Qualitative data**

**(2) Quantitative data**

**(1) Qualitative data:** If the data is nonnumeric then it is called qualitative data.

**Example:** Color, religion, sex etc.

**(2) Quantitative data:** If the data is numeric then it is called quantitative data.

**Example: Height, weight, age etc.**

**There are two types of quantitative data:**

**(1) Discrete data (2) Continuous data**

**(1) Discrete data: Discrete data can assume only certain values and there are usually “gaps” between the values.**

**Example: No. of workers in a factory, no. of bedrooms in a house etc.**

**(2) Continuous data: Continuous data can assume any value within a specific range.**

**Example: Height, weight, age etc.**

In Statistics, the variables or numbers are defined and categorized using different **scales of measurements**. Each **level of measurement** scale has specific properties that determine the various use of statistical analysis. Here, we will learn four types of scales such as nominal, ordinal, interval and ratio scale.

**What is the Scale?**

A scale is a device or an object used to measure or quantifies any event or another object.

## **Levels of Measurements**

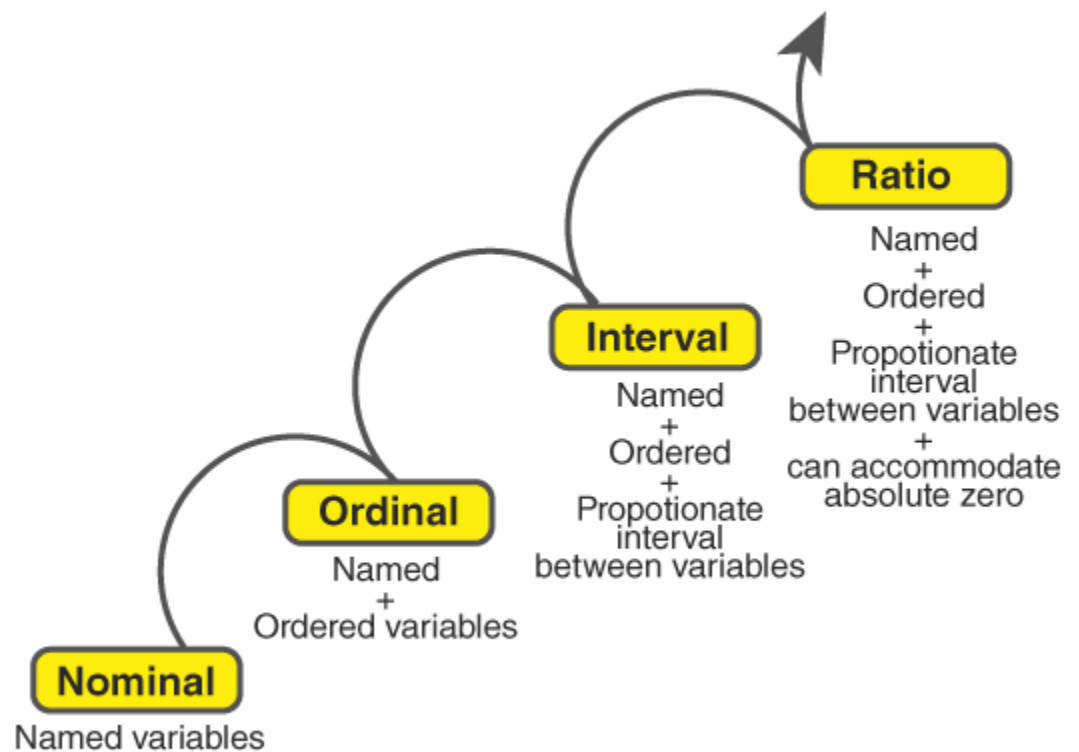
There are four different scales of measurement. The data can be defined as being one of the four scales. The four types of scales are:

- Nominal Scale
- Ordinal Scale
- Interval Scale
- Ratio Scale



# LEVELS OF MEASUREMENT

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## Levels of Measurement

The level of measurement refers to the relationship among the values that are assigned to the attributes for a variable.

There are typically four levels of measurement that are defined:

- Nominal
- Ordinal
- Interval
- Ratio

In **nominal** measurement the numerical values just “name” the attribute uniquely. No ordering of the cases is implied. For example, jersey numbers in basketball are measures at the nominal level. A player with number 30 is not more of anything than a player with number 15, and is certainly not twice whatever number 15 is.

In **ordinal** measurement the attributes can be rank-ordered. Here, distances between attributes do not have any meaning. For example, on a survey you might code Educational Attainment as 0=less than high school; 1=some high school. 2=high school degree; 3=some college; 4=college degree; 5=post college. In this measure, higher numbers mean *more* education. But is distance from 0 to 1 same as 3 to 4? Of course not. The interval between values is not interpretable in an ordinal measure.

In **interval** measurement the distance between attributes *does* have meaning. For example, when we measure temperature (in Fahrenheit), the distance from 30-40 is same as distance from 70-80. The interval between values is interpretable. Because of this, it makes sense to compute an average of an interval variable, where it doesn't make sense to do so for ordinal scales. But note that in interval measurement ratios don't make any sense - 80 degrees is not twice as hot as 40 degrees (although the attribute value is twice as large).

Finally, in **ratio** measurement there is always an absolute zero that is meaningful. This means that you can construct a meaningful fraction (or ratio) with a ratio variable. Weight is a ratio variable. In applied social research most “count” variables are ratio, for example, the number of clients in past six months. Why? Because you can have zero clients and because it is meaningful to say that “...we had twice as many clients in the past six months as we did in the previous six months.”

It's important to recognize that there is a hierarchy implied in the level of measurement idea. At lower levels of measurement, assumptions tend to be less restrictive and data analyses tend to be less sensitive. At each level up the hierarchy, the current level includes all of the qualities of the one

below it and adds something new. In general, it is desirable to have a higher level of measurement (e.g., interval or ratio) rather than a lower one (nominal or ordinal).

## Nominal Scale

A nominal scale is the 1<sup>st</sup> level of measurement scale in which the numbers serve as “tags” or “labels” to classify or identify the objects. A nominal scale usually deals with the non-numeric variables or the numbers that do not have any value.

### Characteristics of Nominal Scale

- A nominal scale variable is classified into two or more categories. In this measurement mechanism, the answer should fall into either of the classes.
- It is qualitative. The numbers are used here to identify the objects.
- The numbers don't define the object characteristics. The only permissible aspect of numbers in the nominal scale is “counting.”

### Example:

An example of a nominal scale measurement is given below:

What is your gender?

M- Male

F- Female

Here, the variables are used as tags, and the answer to this question should be either M or F.

## Ordinal Scale

The ordinal scale is the 2<sup>nd</sup> level of measurement that reports the ordering and ranking of data without establishing the degree of variation between them. Ordinal represents the “order.” Ordinal data is known as qualitative data or categorical data. It can be grouped, named and also ranked.

### Characteristics of the Ordinal Scale

- The ordinal scale shows the relative ranking of the variables
- It identifies and describes the magnitude of a variable
- Along with the information provided by the nominal scale, ordinal scales give the rankings of those variables
- The interval properties are not known
- The surveyors can quickly analyses the degree of agreement concerning the identified order of variables

### Example:

- Ranking of school students – 1st, 2nd, 3rd, etc.
- Ratings in restaurants
- Evaluating the frequency of occurrences
  - Very often
  - Often
  - Not often

- Not at all
- Assessing the degree of agreement
  - Totally agree
  - Agree
  - Neutral
  - Disagree
  - Totally disagree

## Interval Scale

The interval scale is the 3<sup>rd</sup> level of measurement scale. It is defined as a quantitative measurement scale in which the difference between the two variables is meaningful. In other words, the variables are measured in an exact manner, not as in a relative way in which the presence of zero is arbitrary.

### Characteristics of Interval Scale:

- The interval scale is quantitative as it can quantify the difference between the values
- It allows calculating the mean and median of the variables
- To understand the difference between the variables, you can subtract the values between the variables
- The interval scale is the preferred scale in Statistics as it helps to assign any numerical values to arbitrary assessment such as feelings, calendar types, etc.

### Example:

- Likert Scale
- Net Promoter Score (NPS)
- Bipolar Matrix Table

## Ratio Scale

The ratio scale is the 4<sup>th</sup> level of measurement scale, which is quantitative. It is a type of variable measurement scale. It allows researchers to compare the differences or intervals. The ratio scale has a unique feature. It possesses the character of the origin or zero points.

### Characteristics of Ratio Scale:

- Ratio scale has a feature of absolute zero
- It doesn't have negative numbers, because of its zero-point feature
- It affords unique opportunities for statistical analysis. The variables can be orderly added, subtracted, multiplied, divided. Mean, median, and mode can be calculated using the ratio scale.
- Ratio scale has unique and useful properties. One such feature is that it allows unit conversions like kilogram – calories, gram – calories, etc.

### Example:

An example of a ratio scale is:

What is your weight in Kg?

- Less than 55 kg
- 55 – 75 kg
- 76 – 85 kg
- 86 – 95 kg
- More than 95 kg

## Four Different Levels of Measurement

In descending order of precision, the four different levels of measurement are:

**Nominal**—Latin for name only (Republican, Democrat, Green, Libertarian)

**Ordinal**—Think ordered levels or ranks (small—8oz, medium—12oz, large—32oz)

**Interval**—Equal intervals among levels (1 dollar to 2 dollars is the same interval as 88 dollars to 89 dollars)

**Ratio**—Let the “o” in ratio remind you of a zero in the scale (Day 0, day 1, day 2, day 3, ...)

The first level of measurement is **nominal level of measurement**. In this level of measurement, the numbers in the variable are used only to classify the data. In this level of measurement, words, letters, and alpha-numeric symbols can be used. Suppose there are data about people belonging to three different gender categories. In this case, the person belonging to the female gender could be classified as F, the person belonging to the male gender could be classified as M, and transgendered classified as T. This type of assigning classification is nominal level of measurement.

The second level of measurement is the **ordinal level of measurement**. This level of measurement depicts some ordered relationship among the variable’s observations. Suppose a student scores the highest grade of 100 in the class. In this case, he would be assigned the first rank. Then, another classmate scores the second highest grade of an 92; she would be assigned the second rank. A third student scores a 81 and he would be assigned the third rank, and so on. The ordinal level of measurement indicates an ordering of the measurements.

The third level of measurement is the **interval level of measurement**. The interval level of measurement not only classifies and orders the measurements, but it also specifies that the distances between each interval on the scale are equivalent along the scale from low interval to high interval. For example, an interval level of measurement could be the measurement of anxiety in a student between the score of 10 and 11, this interval is the same as that of a student who scores between 40 and 41. A popular example of this level of measurement is temperature in centigrade, where, for example, the distance between 94°C and 96°C is the same as the distance between 100°C and 102°C.

The fourth level of measurement is the **ratio level of measurement**. In this level of measurement, the observations, in addition to having equal intervals, can have a value of zero as well. The zero in the scale makes this type of measurement unlike the other types of measurement, although the properties are similar to that of the interval level of measurement. In the ratio level of measurement, the divisions between the points on the scale have an equivalent distance between them.

**According to source, there are two types of data.**

**(1) Primary data (2) Secondary data**

**Primary Data:** The data are collected through first hand investigation without any internal records or any published sources are called primary data.

**Example:** Survey data, interview data, questionnaire data etc.

**Secondary Data:** The data are collected from any medium which has already been collected by others are called secondary data.

**Example:** Journal, thesis, published statistics etc.

## **Methods of Collecting Primary Data**

Primary data can be collected by the following sources:

- (i) Observation method
- (ii) Interview method,
- (iii) Questionnaire method, and
- (iv) Schedule method.

## **Methods of Collecting Secondary Data**

Secondary data can be collected by the following sources:

- (1) Journals,
- (2) Reports,
- (3) Institutional publications,
- (4) Public library,
- (5) Information bureau,

There are several ways of collecting primary data.

**These may be:**

- (i) Observation method
- (ii) Interview method,
- (iii) Questionnaire method, and
- (iv) Schedule method.

### **i. Observation Method:**

In observation method, the information is sought by way of investigator's own direct observation without asking from the respondent.

The main advantage of this method is that it is free from subjective biasness, as it is free from respondent's willingness. It is, however, an expensive and time consuming method. Moreover, the information provided by this method is very limited and some of the more busy people like executives may not be accessible to direct observation.

## **ii. Interview Method:**

**Primary data may be collected either through personal interviews or through telephonic interviews:**

- (a) In the personal interviews the interviewer asks questions generally in a face to face contact.

Through interview method more and reliable information may be obtained. Personal information can be obtained easily under this method. It is, however, a very expensive and time consuming method, especially when large and widely spread geographical sample is taken. Certain types of respondents, such as officials, executives or people of high income groups, may not be easily accessible.

In this method, the respondent may give wrong and imaginary information. For effective interview there should be a good rapport with respondents which is often very difficult to develop. For a good



result the interviewer's approach should be friendly, courteous, conversational and unbiased for which a proper training is required.

(b) In telephonic interviews contact is made with the respondents through telephone.

**The main merits of telephonic interviews are:**

(i) It is more flexible and faster than other methods.

(ii) It is cheaper and less time consuming.

(iii) Recall is easy and replies can be recorded without causing embarrassment to respondents.

(iv) At times, access can be made to respondents who otherwise cannot be contacted for one reason or the other.

(v) No staff is required and wider representation of sample is possible.

This method also has several weaknesses. For example, the surveys are restricted to respondents who have telephonic facilities and little time is given to respondents for considered answers. It is not suitable for intensive surveys where comprehensive answers are required for various questions.

**iii. Questionnaire Method:**

In this method a questionnaire is mailed to the person concerned with a request to answer the questions and return the questionnaire. This

method is most extensively applied in various researches of human and economic geography.

**The main merits of this method are given below:**

(i) There is low cost even when the universe is large and is widespread geographically.

(ii) It is free from bias of interviewer as answers are respondent's own words.

(iii) Respondents, who are not easily approachable, can also be reached conveniently. Moreover, respondents are given enough time to give well thought answers.

**The main demerits of this method are:**

(i) Low rate of return of the duly filled questionnaires.

(ii) It can be used only when the respondents are educated and cooperative.

(iii) The control of the questionnaire may be lost once it is sent.

(iv) It is difficult to know whether willing respondents are truly representative.

(v) This method is likely to be the slowest of all.

**iv. Schedule Method:**

This method of data collection is very much like the collection of data through questionnaires, with little difference that lies in the fact that schedules (proforma containing a set of questions) are being filled in by the enumerators who are specially appointed for this purpose. Enumerators explain the aims and objects of the investigation and also remove the difficulties which any respondent may feel in understanding the implications of a particular question.

This method is very useful in extensive enquiries and can lead to fairly reliable results. It is, however, very expensive and is usually adopted in investigations conducted by governmental agencies or by some organizations. Population census all over the world is conducted through this method.

For the selection of an appropriate method, the objective, nature and scope of study, the availability of time and funds, and the level of precision required are to be taken into consideration.

### **Collection of Secondary data**

**A researcher can obtain secondary data from various sources. Secondary data may either be published data or unpublished data.**

**Published data are available in**

**(i) Publications of government**

**(ii) Technical and trade journal**

**(iii) Reports of various business banks etc.**

**(iv) Public records**

**(v) Statistical or historical documents**

**Unpublished data may be found in letters, diaries, unpublished biographies or work.**

**Before using secondary data, it must be checked for the following characteristics:**

- (1) Reliability of data- Who collected the data? From what source? Which methods? Time? Possibility of bias? Accuracy?**
- (2) Suitability of data-The object, scope and nature of the original enquiry must be studied and then carefully scrutinize the data for suitability.**
- (3) Adequacy-Adequacy of the data is to be judged in the light of the requirements of the survey and the geographical area covered by the available data.**

**Distinctions between primary data and secondary data**

| <b>Primary Data</b>                          | <b>Secondary data</b>                   |
|----------------------------------------------|-----------------------------------------|
| (1) It is first hand conducted data.         | (1) It is second level collected data.  |
| (2) It is direct data.                       | (2) It is indirect data.                |
| (3) It may be unprocessed.                   | (3) It is always processed.             |
| (4) It is collect from the origin of source. | (4) It is collected through media.      |
| (5) It is expensive.                         | (5) It is not expensive.                |
| (6) It takes huge time.                      | (6) It does not take huge time.         |
| (7) It creates secondary data                | (7) It creates decisions.               |
| (8) It gives very recent information.        | (8) It gives late information.          |
| (9) It suitable for researchers.             | (9) It suitable for analyzers.          |
| (10) It is suitable for original study.      | (10) It is suitable for relative study. |

**Questionnaire:** A formal list of a question is called questionnaire.

## **Questionnaire on branding**

### **Brand Perception**

**Name**\_\_\_\_\_

**Address**\_\_\_\_\_

**Phone Number**\_\_\_\_\_

**Email ID**\_\_\_\_\_

**1. Describe what does “brand” mean to you?**

\_\_\_\_\_

**2. Do you buy only branded product?**

**(a) Yes always**

**(b)Never**

**(c ) Only when quality is important.**

**(d)Rarely**

**3. Are you a loyal customer for the product you buy?**

**(a) Yes always. I stick to the same product.**

**(b)Never. I keep experimenting with new product.**

**(c )Only for quality products. Where it is a commodity. I choose the least- priced products.**

**4.Are you a price sensitive consumer?**

**(a) Yes**

**(b)No**

**5. Will you stick to the same products? If their price is increased. (And you are loyal customer to the product)?**

**(a) Up to a certain limit in price increase.**

**(b)Yes. Irrespective of the price increase.**

**(c)No. I may look for different product.**

**6. Which of the following, according to you, help build a good brand image.**

**(a) Quality**

**(b)Communication strategies.**

**(c )Competitive pricing**

**(d)Good value added service**

**(e) Free trials and discounts**

**(f)Others**

**If others please specify\_\_\_\_\_**

**7. Are a popular celebrities good brand ambassadors and is investing in them a good strategy?**

**(a) Sometimes. May be**

**(b) No. No**

### **Qualities & characteristics of a good questionnaire**

- Need to know what needs to be measured – Having a clear picture and understanding of what data needs to be collected contributes to the quality of data collection
- Should know how to word/frame question and words should be neutral and should not be leading – Whatever your opinion may be, own opinion should never be reflected in the questions. This is done both intentionally and unintentionally but should be taken care of.
- Emphasis on right word/phrase should be kept in mind – The language should be clear so that the required data can be received. This also makes the question and the requirement of survey easy to understand and thus help in getting a better response and answer.
- Define and qualify terms – This is most essential when a technical survey or a field specific survey is being done. If you think that the audience being surveyed might not know about some terms than they must be defined to get any proper response. This will increase the quality and decrease the bounce rate or number of questions which were left unanswered.
- Avoid double negatives or more than 1 negative word in question – Use of negative word has a psychological effect and can influence the answer.
- Sufficient or adequate alternatives should be provided – Available options should have the most expected answers
- Multiple questions in question should be avoided – One question should have one answer. If more than one question needs to be asked than it should be made a separate question to improve clarity of questions.
- Word requiring emphasis should be emphasized – It helps in making a point and question clear

- options like good/bad/fair/average should be quantified through photographs or other mean – These are very vague terms and interpretations of these varies from person to person
- Unwanted assumptions should be avoided – A survey is about getting factual data and assumptions should be avoided.

**How could you make a questionnaire/good questionnaire? (If marks is 2 or 3 then you can shortly answered as like as below :)**

### **Designing a Questionnaire:**

- Keep it short and simple.
- Keep it clean and visually appealing.
- Keep **questions** brief and easy to understand.
- Use open-ended **questions** appropriately.
- Questions should be logically arranged.
- Questions of a sensitive nature should be avoided.
- Avoid using leading **questions**.
- Avoid using compound **questions**.



# Presentation of Data

## Methods of Collecting Data

**Investigation Method:** In this method data are collected through direct and indirect investigations. It is the basic methods of collecting data.

**Questionnaire Method:** In this method data are collected by fulfilling questionnaire. It is the modern method of collecting data.

**Interview Method:** In this method data are collected through tele-media or direct interviewing the persons. It is suitable for internal affairs.

**Published Media Method:** In this method data are collected from known published media. This method gives only secondary data. It is the maximum used method of collecting data.

**Unpublished Media Method:** In this method data are collected from individual, organizational and unpublished media. This method gives both primary and secondary data. It is also widely used method of collecting data.

## Methods of Collecting Primary Data

Primary data can be collected by the following sources:

- (1) Direct investigation,
- (2) Direct interviews,
- (3) Telephone interviews,
- (4) Filling questionnaire,
- (5) Personal observations,
- (6) Counting by enumerators,
- (7) Mailing,
- (8) Surveys,
- (9) Census, etc.

## Methods of Collecting Secondary Data

Secondary data can be collected by the following sources:

- (1) Journals,
- (2) Reports,
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- (4) Public library,
- (5) Information bureau,
- (6) Govt. agencies,

- (7) International agencies (UNO, FAO, WHO etc),
- (8) National and international news media,
- (9) Statistical books, etc.

## Properties of a Good Questionnaire

A good questionnaire has the following properties:

- (1) It should be easier to understand.
- (2) It should be clear to express the facts.
- (3) It should be concise form.
- (4) The number of questions should be as few as possible.
- (5) Questions should be logically arranged.
- (6) Questions should be short and simple.
- (7) Questions of a sensitive nature should be avoided.
- (8) Instructions should have to the information.
- (9) It should be just in time.
- (10) Questions should be capable of objective answer i.e., Yes or No.

## Primary Data and Secondary Data

**Primary Data:** The data are collected through first hand investigation without any internal records or any published sources are called primary data.

**Example:** Survey data, interview data, questionnaire data etc.

**Secondary Data:** The data are collected from any medium which has already been collected by others are called secondary data.

**Example:** Journal, thesis, published statistics etc.

## Uses of Primary Data

The common uses of primary data are mentioned below:

- (1) They are used as the basis of measurements.
- (2) They are used for original study.
- (3) They are direct investigations.
- (4) They are used for very recent information.
- (5) They are used for reducing uncertainty.

## Uses of Secondary Data

The common uses of secondary data are mentioned below:

- (1) They are used for decisions making.
- (2) They are suitable for multipurpose uses.
- (3) They are used for reference purposes.
- (4) They are used for documentary project.
- (5) They are used for minimizing costs.

### Distinctions between primary data and secondary data

| Primary Data                                 | Secondary data                          |
|----------------------------------------------|-----------------------------------------|
| (1) It is first hand conducted data.         | (1) It is second level collected data.  |
| (2) It is direct data.                       | (2) It is indirect data.                |
| (3) It may be unprocessed.                   | (3) It is always processed.             |
| (4) It is collect from the origin of source. | (4) It is collected through media.      |
| (5) It is expensive.                         | (5) It is not expensive.                |
| (6) It takes huge time.                      | (6) It does not take huge time.         |
| (7) It creates secondary data                | (7) It creates decisions.               |
| (8) It gives very recent information.        | (8) It gives late information.          |
| (9) It suitable for researchers.             | (9) It suitable for analyzers.          |
| (10) It is suitable for original study.      | (10) It is suitable for relative study. |

### Frequency Distribution

**Frequency Distribution:** A grouping of data into categories showing the number of observations in each mutually exclusive category.

**Example;** Numbers in different groups, runs per over, weaves per meter etc.

### Types of Frequency Distribution

Types of frequency distribution depend on three bases, i.e., (a) Discrete or Discontinuous frequency distribution, (b) Continuous frequency distribution, and (c) Relative frequency distribution.

**(a) Discrete or Discontinuous Frequency Distribution:** A frequency distribution containing discontinuous variables is called discrete or discontinuous frequency distribution. A discrete variable is that which can vary only by finite 'jumps', and cannot manifest every conceivable fractional value (i.e., 2.01, 50.88 etc).

**(b) Continuous frequency distribution:** A frequency distribution containing continuous variables is called continuous frequency distribution. A continuous variable is capable of manifesting every conceivable fractional value within the range of possibilities (i.e., 10 – 20, 20 – 30, 30 – 40 etc).

**(c) Relative frequency distribution:** A frequency distribution containing the considerations of relative measures is called relative frequency distribution.

### Steps of Constructing Frequency Distribution

The following steps can be taken to construct a frequency distribution table:

**(Step – 1) Finding Highest and Lowest Values:** Firstly, we have to find the highest and the lowest value of the data set.

**(Step – 2) Finding Number of Classes:** Secondly, we have to find the number of classes by using Sturges rule  $k = 1 + 3.322 \log N$ .

**(Step – 3) Finding Class Interval:** Thirdly, we have to find the class interval by using Sturges rule,

$$\text{Class Interval} = \frac{\text{Highest Value} - \text{Lowest Value}}{1 + 3.322 \log N}.$$

**(Step – 4) Finding Class Boundaries:** Fourthly, we have to find the upper and lower class boundaries by using exclusive method (10 – 20, 20 – 30, 30 – 40 etc.) or inclusive method (10 – 19, 20 – 29, 30 – 39 etc.).

**(Step – 5) Making Tally Marks:** Fifthly, we have to make tally marks (////) by taking all data to the respective classes.

**(Step – 6) Finding Frequencies:** Sixthly, we have to find the frequencies for each class by counting the number of tallies of each class.

**(Step – 7) Tabulating Frequency Distribution:** Finally, we have to tabulate the frequency distribution by arranging classes in the first column, tally marks in the second column, frequencies in the third column and so on.

## Stem and Leaf Display

**Origin:** Stem and leaf display is invented by a famous American statistician John. Wilder Tukey in 1977.

**Definition:** The method of presenting data set using the leftmost digits as a stem and the ones on the right as leaves is called stem and leaf display. A basic stem-and-leaf display contains two columns separated by a vertical line. The left column contains the stems and the right column contains the leaves.

**Example:** For two digital data set, first digit is stem and second digit is leaf.

**Illustration:** Let a data set be 27, 29, 30, 22, 48, 31, 12, 50, 35, 49, 38. Taking the data set, we can make a stem and leaf display as follow:

| Stem | Leaf       |
|------|------------|
| 1    | 2          |
| 2    | 2, 7, 9    |
| 3    | 0, 1, 5, 8 |
| 4    | 8, 9       |
| 5    | 0          |

To gist up, stem and leaf display is a technique of presenting data set so that the concentrations in the data become visible.

## Five Number Summary

**Origin:** Five number summary is also invented by a famous American statistician John. Wilder Tukey in 1977.

**The Five Elements:** Five number summary has five elements. i.e.,

- (1) The median of the data (Med.);
- (2) The lower quartile ( $Q_1$ );
- (3) The upper quartile ( $Q_3$ );
- (4) The smallest observation ( $X_{\text{smallest}}$ );
- (5) The largest observation ( $X_{\text{largest}}$ ).

**Model:** A model of five number summary is as under:



Fig: A model of box plot

To end up, a five number summary helps us to determine the central measure of a data set in a summarized form.

## Classification

**Definition:** Classification is the process to group or arrange data separately in aspect of their common characteristics. After collection, and editing of data an important step towards processing the data is classification.

**Example:** It is well known that the letters collected in a post office are sorted into different lots or a geographical basis, i.e., in accordance with their destinations as Dhaka, Chittagong, Rajshahi, Khulan etc.

## Tabulation

**Definition:** Tabulation is the process to arrange the data according to their characteristics in separate columns vertically and rows horizontally. The purpose of a table is to simplify the presentation, and to facilitate comparisons.

**Example:** The students of Life College tabulated as follows:

## Types of Classification

Broadly, the data can be classified on the following four bases:

- (1) **Geographical Classification** → i.e., area-wise, e.g., cities, districts, roads etc. Geographical classifications are usually listed in alphabetical order for easy reference.
- (2) **Chronological Classification** → i.e., on the basis of time. When data are observed over a period of time the type of classification is known as chronological classification. Time series are usually listed in chronological order normally starting with the earliest period.

**(3) Qualitative Classification** → i.e., according to some attributes. In qualitative classification, data are classified on the basis of some attributes or qualities such as sex, color of eye, color of hair, literacy, nationality, religion, etc.

**(4) Quantitative Classification** → i.e., in terms of magnitudes. Quantitative classification refers to the classification of data according to some characteristics that can be measured, such as height, weight, income, sales etc.

## Illustration of a bivariate frequency distribution

**Bivariate Distribution:** When the data incurred two types of variables then the frequency of that distribution is called bivariate distribution.

Let us take sale and profit (in crores of Tk.) of 5 companies as follows:

| Sales | Profits |
|-------|---------|
| 12    | 2       |
| 23    | 4       |
| 33    | 6       |
| 38    | 7       |
| 55    | 9       |

Now we can put these data in a bivariate frequency distribution as follows:

**Table showing sales and profit (in crores of Tk)**

| Profits \ Sales | 0 – 2 | 2 – 4 | 4 – 6 | 6 – 8 | 8 – 10 | Total |
|-----------------|-------|-------|-------|-------|--------|-------|
| 10 – 20         | I     | 0     | 0     | 0     | 0      | 1     |
| 20 – 30         | 0     | 0     | I     | 0     | 0      | 1     |
| 30 – 40         | 0     | 0     | 0     | II    | 0      | 2     |
| 40 – 50         | 0     | 0     | 0     | 0     | 0      | 0     |
| 50 – 60         | 0     | 0     | 0     | 0     | I      | 1     |
| Total           | 1     | 0     | 1     | 2     | 1      | 5     |

## Steps/Parts/Rules of Tabulation

The following steps/parts/rules are needed for constructing tabulation:

**(1) Table Number:** For more than one, tables should be numbered.

**(2) Date:** Table may be dated at time of completion.

**(3) Title:** Each table must have a suitable and short title describing the contents of the table.

**(3) Master Caption:** Master caption refers to the vertical data of all columns through homogeneous variables.

**(4) Column Heading:** Column heading is the basic title of similar ranges of data set.

**(5) Subcolumn Heading:** It divides the column heading in different groups.

**(6) Column Number:** Each column may be numbered for explanations.

**(7) Unit Head:** It refers to the measurement units for each column or row.

**(8) Cell Insides:** It is optional. It refers each cell may include an inside for alternative frequencies.

**(9) Stub:** Stubs are the designations of row headings.

**(10) Substub:** Stubs may be divided into different subrows called substubs.

**(11) Body:** Body is the entire statistical data set.

**(12) Head Notes:** It is the explanatory statement of the data set placed below the title of the table and may enclosed by brackets.

**(13) Foot Notes:** Difficult items in the table are explained for better understanding which placed below the table.

**(14) Source Notes:** Data sources are listed below the footnotes.

**(15) References:** References are listed below the source notes.

## Steps/Parts/Rules of Classification

The following steps/parts/rules are needed for constructing classification:

- (1) Classes should preferably be between 0 to 10, 5 to 15 (numerically). A to D, E to G, AA, BB (alphabetically), Dhaka, Jessore (geographically) etc.
- (2) As far as possible we should avoid odd values of class intervals, i.e., 3,5,7,11,13, etc.
- (3) The starting point should be the lower limit of first class as zero or 5 or multiple of 5.
- (4) It should ensure continuity and to get correct class-interval which includes fractional numbers, i.e., 9.5, 10.99, etc.
- (5) Class should same size as possible.
- (6) It should reduce cost, time, labor, space, etc.
- (7) It should brief to active works, and to run the processes.

## Histogram Versus Frequency Polygon

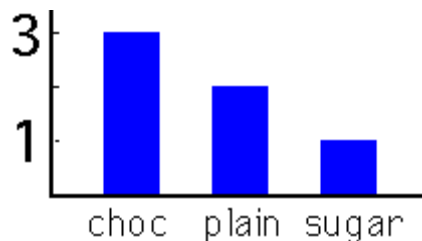
Both histogram and frequency polygon are two methods of presenting data. Despite they have the following distinctions:

| Histogram                                    | Frequency Polygon                             |
|----------------------------------------------|-----------------------------------------------|
| (1) It is a rectangular data presenting way. | (1) It is a multiangular data presenting way. |
| (2) It needs class intervals.                | (2) It needs midpoints.                       |
| (3) It shows rectangular area.               | (3) It shows angles.                          |
| (4) It gives importance on bars.             | (4) It gives importance on lines.             |
| (5) It is widely used.                       | (5) Its uses are limited.                     |
| (6) It is suitable for continuous variables. | (6) It is suitable for discrete variables.    |
| (7) It is not suitable for comparing among   | (7) It is suitable for comparing among        |

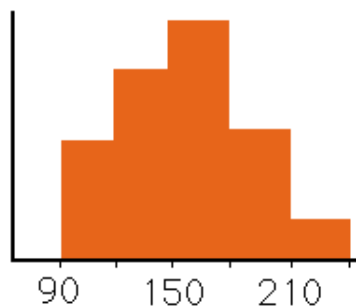
|                                                      |                                              |
|------------------------------------------------------|----------------------------------------------|
| more than one distribution.                          | more than one distribution.                  |
| (8) It is easy to understand.                        | (8) It is difficult to understand.           |
| (9) It needs both equal and unequal class intervals. | (9) It needs only equal midpoints.           |
| (10) It is very attractive to general people.        | (10) It is not attractive to general people. |

## Bar Diagram Versus Histogram

There are two differences, one is in the type of data that is presented and the other in the way they are drawn. In bar graphs are usually used to display "categorical data", that is data that fits into categories. For example suppose that I offered to buy donuts for six people and three said they wanted chocolate covered, 2 said plain and one said with icing sugar. I would present this in a bar graph as:



Histograms on the other hand are usually used to present "continuous data", that is data that represents measured quantity where, at least in theory, the numbers can take on any value in a certain range. A good example is weight. If you measure the weights of a group of adults you might get numbers between say 90 pounds and 240 pounds. We usually report our weights as pounds or to the nearest half pound but we might do so to the nearest tenth of a pound or however accurate the scale is. The data would then be collected into categories to present a histogram. For example:



The above might be a histogram for heights (with the appropriate scale on the vertical axis). Here the data has been collected into categories of width 30 pounds.

The difference in the way that bar graphs and histograms are drawn is that the bars in bar graphs are usually separated where in histograms the bars are adjacent to each other. This is not always true however. Sometimes you see bar graphs with no spaces between the bars but histograms are never drawn with spaces between the bars.



# Symbols

| Symbols           | Meaning                         |
|-------------------|---------------------------------|
| ////              | right tally                     |
| \\\\              | left tally                      |
|                   | straight tally                  |
| <del>    </del>   | right tally bar                 |
| <del>\\\\\\</del> | left tally bar                  |
| <del>    </del>   | straight tally bar              |
| O                 | point of origin                 |
| X                 | x-axis/horizontal axis/run axis |
| Y                 | y-axis/vertical axis/rise axis  |
| f                 | frequency                       |
| N                 | total frequencies               |
| n                 | number of samples               |
| □                 | micro square                    |
| □                 | macro square                    |
| W                 | Breaking line/false base        |
| R                 | range                           |
| c/i/k             | class interval                  |
| H.V.              | highest value                   |
| L.V.              | lowest value                    |
| T.V.              | target value                    |
| logN              | log of numbers                  |
| #                 | number                          |
| Q                 | quartiles                       |

## Business Statistics

|      |               |
|------|---------------|
| D    | deciles       |
| P    | percentiles   |
| F    | fractiles     |
| Med. | mediana       |
| '    | prime/foot    |
| "    | Inch          |
| •    | dot position  |
| ≈    | approximately |
| °    | degree        |
|      | tally mark    |
| →    | right arrow   |
| ↑    | up arrow      |

# Formulas

(1) **Stem = Leftmost Digit(s)**

(2) **Leaf = Rightmost Digit**

(3) **For pie chart, 1% frequency =  $3.6^\circ$**

(4) **Number of Classes (K) =  $1 + 3.322 \log N$  [Sturge's rule]**

(5) **Class Interval (C) =  $\frac{\text{Highest Value} - \text{Lowest Value}}{1 + 3.322 \log N}$**

\* 'Stem and Leaf Display' is invented by John Tukey in 1977.

\* 'Box Plot Whisker' is originated from 'Five Number Summary'.

\* 'Five Number Summary' is the simplest presentation of a data set.

\* '1% frequency =  $3.6^\circ$ ' is used for geometric simplification.

(6) A box plot (five number summary) has five elements. i.e.,

(1) **The median of the data (Med.);**

(2) **The lower quartile ( $Q_1$ );**

(3) **The upper quartile ( $Q_3$ );**

(4) **The smallest observation ( $X_{\text{smallest}}$ );**

(5) **The largest observation ( $X_{\text{largest}}$ ).**

(7) A box plot model is exhibited below:



(8) Graph sheet style:

**Area =  $8'' \times 10''$**

**Scaling space = 5 micros based macros.**

(9) Graph sheet scale:

**Horizontal axis = 5 micros based units**

**Vertical axis = 5 micros based units**

(10) Graph sheet legend:

**Dots legend as required**

**Lines legend as required**

**Curve legend as required**

## **Presentation of Data**

### **Some Important definitions:**

**Class limit:** The class limits are the lowest and the highest values that can be included in the class.

Example: Let the class be 20-40. The lowest value of this class is 20 and the highest 40. The lowest value and the highest value are called lower limit and upper limit respectively. Here, Lower limit=20 and upper limit=40.

**Class Interval:** The span of a class i.e. the difference between the upper limit and the lower limit is known as class interval.

Example: Let the class be 20-40. Here class interval=20.

**Frequency:** The number of observations corresponding to the particular class is known as frequency.

Example: Let the marks of the class =20-40 corresponding to the number of students is 7. So, Frequency of this class=7.

**Frequency distribution:** In statistics, a frequency distribution is a table that displays the frequency of various outcomes in a sample.

Example: Let us take the following frequency distribution table:

| Marks<br>(out of 100) | Number of Students<br>f |
|-----------------------|-------------------------|
| 0-20                  | 5                       |
| 20-40                 | 7                       |
| 40-60                 | 10                      |
| 60-80                 | 2                       |
| 80-100                | 1                       |
| Total                 | N=25                    |

**Exclusive Method/Continuous frequency distribution:** When the class interval is so fixed that the upper limit of one class is the lower limit of the next class then it is known as exclusive method.

Example: Let us take the following frequency distribution table:

| Marks<br>(out of 100) | Number of Students |
|-----------------------|--------------------|
| 0-20                  | 5                  |
| 20-40                 | 7                  |
| 40-60                 | 10                 |
| 60-80                 | 2                  |
| 80-100                | 1                  |
| Total                 | N=25               |

Here class interval=20 .Let us take the classes =0-20 & 20-40.Upper limit of 1<sup>st</sup> class=20 and lower limit of the next class=20 and so on. This is the example of exclusive method.

**Inclusive method/ discrete frequency distribution:** If upper limit of one class is included in that class itself then it is known as inclusive method.

Example of inclusive method is as follows:

| Marks<br>(out of 100) | Number of Students |
|-----------------------|--------------------|
| 0-19                  | 5                  |
| 20-39                 | 7                  |
| 40-59                 | 10                 |
| 60-79                 | 2                  |
| 80-99                 | 1                  |
| Total                 | N=25               |

**Mid- Point/Mid value:** If we add lower limit and upper limit of each class & then divided by 2 , we get mid-point.

Example: Frequency distribution table

| Marks<br>(out of 100) | Number of Students | Mid Point |
|-----------------------|--------------------|-----------|
| 0-20                  | 5                  | 10        |
| 20-40                 | 7                  | 30        |
| 40-60                 | 10                 | 50        |
| 60-80                 | 2                  | 70        |
| 80-100                | 1                  | 90        |
| Total                 | N=25               |           |

**Frequency density:** It is the ratio of frequency of a certain class and the class interval of that specific class.

$$\text{Frequency density} = \frac{\text{Frequency of a certain class}}{\text{Class interval of that specific class}}$$

**Relative Frequency:** It is the ratio of frequency of a certain class and total frequency.

$$\text{Relative Frequency} = \frac{\text{Frequency of a certain class}}{\text{Total Frequency}}$$

Example of relative frequency and frequency density:

| Marks<br>(out of 100) | Number of Students | Relative Frequency   | Frequency density     |
|-----------------------|--------------------|----------------------|-----------------------|
| 0-20                  | 5                  | $\frac{5}{25} = 0.2$ | $\frac{5}{20} = 0.25$ |
| 20-40                 | 7                  | 0.28                 | 0.35                  |
| 40-60                 | 10                 | 0.4                  | 0.5                   |
| 60-80                 | 2                  | 0.08                 | 0.1                   |
| 80-100                | 1                  | 0.04                 | 0.05                  |
| Total                 | N=25               |                      |                       |

**Less than and More than Cumulative Frequency:** If we find out less than cumulative frequency we add frequency from top to bottom .But if we find out more than cumulative frequency, we add frequency from bottom to top.

Example:

**Frequency distribution table**

| Marks<br>(out of 100) | Number of Students<br>f | Cumulative Frequency(c.f) |           |
|-----------------------|-------------------------|---------------------------|-----------|
|                       |                         | Less than                 | More than |
| 0-20                  | 5                       | 5                         | 25        |
| 20-40                 | 7                       | 12                        | 20        |
| 40-60                 | 10                      | 22                        | 13        |
| 60-80                 | 2                       | 24                        | 3         |
| 80-100                | 1                       | 25                        | 1         |
| Total                 | N=25                    |                           |           |

# Frequency Distribution

**Frequency Distribution:** A grouping of data into categories showing the number of observations in each mutually exclusive category.

**Example:** Numbers in different groups, runs per over, weaves per meter etc.

## Types of Frequency Distribution

Types of frequency distribution depend on three bases, i.e., (a) Discrete or Discontinuous frequency distribution, (b) Continuous frequency distribution, and (c) Relative frequency distribution.

**(a) Discrete or Discontinuous Frequency Distribution:** A frequency distribution containing discontinuous variables is called discrete or discontinuous frequency distribution. A discrete variable is that which can vary only by finite 'jumps', and cannot manifest every conceivable fractional value (i.e., 2.01, 50.88 etc).

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**(Step – 7) Tabulating Frequency Distribution:** Finally, we have to tabulate the frequency distribution by arranging classes in the first column, tally marks in the second column, frequencies in the third column and so on.

**We present data in two ways:**

(1) Statistical table (2) Statistical Chart

**(1) Statistical table:**

A table is a systematic arrangement of statistical data in columns and rows. Rows are horizontal arrangement, whereas columns are vertical ones.

A statistical table is presentation of numbers in a logical arrangement, with some brief explanation to show what they are.

However, before, tabulating data it is often necessary to first classify them.

## **Classification**

**Definition:** Classification is the process to group or arrange data separately in aspect of their common characteristics. After collection, and editing of data an important step towards processing the data is classification.

**Example:** It is well known that the letters collected in a post office are sorted into different lots or a geographical basis, i.e., in accordance with their destinations as Dhaka, Chittagong, Rajshahi, Khulan etc.

**Types of classification:** Broadly, the data can be classified on the following four bases:

- (i) Geographical, i.e. area wise e.g. cities, districts etc.
- (ii) Chronological, i.e. on the basis of time.
- (iii) Qualitative, i.e. according to some attributes.
- (iv) Quantitative i.e. in terms of magnitudes.

## Steps/Parts/Rules of Tabulation

The following steps/parts/rules are needed for constructing tabulation:

**(1) Table Number:** For more than one, tables should be numbered.

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**Model:** A model of five number summary is as under:

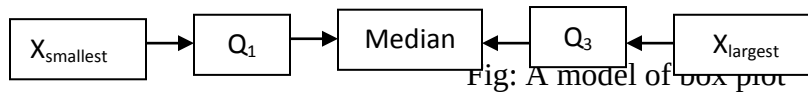


Fig: A model of box plot

To end up, a five number summary helps us to determine the central measure of a data set in a summarized form.

## Problems & Solutions

### Problem 1

BBA (Professional) 2002

A set of data consists of 230 observations that range from Tk 235 to Tk 567. What class interval would you recommend?

**Solution:**

$$\begin{aligned}
 \text{The recommended class interval (c)} &= \frac{\text{Highest Value} - \text{Lowest Value}}{1 + 3.322 \log N} \\
 &= \frac{567 - 235}{1 + 3.322 \log 230} \\
 &= \frac{332}{1 + (3.322 \times 2.3617)} \\
 &= 37.53 \\
 &\approx 40 \quad [\text{Ans.}]
 \end{aligned}$$

### Problem 2

BBA (Professional) 2003

The ages of the sample of the students attending the School of Business this semester are:

19 17 15 20 23 41 33 21 18 20 18 33 32 29 24  
 19 18 20 17 22 55 19 22 25 28 30 44 19 20 39

Construct a Relative Frequency Distribution with intervals 15-19, 20-24, 25-29, 30-34, and 35 and older.

**Solution:**

**Relative Frequency Distribution Table**

| Class        | Tally  | f      | r.f. = f/N   |
|--------------|--------|--------|--------------|
| 15 – 19      | ////// | 10     | 10/30 = 0.33 |
| 20 – 24      | ////   | 9      | 9/30 = 0.30  |
| 25 – 29      | ///    | 3      | 3/30 = 0.10  |
| 30 – 34      | ////   | 4      | 4/30 = 0.13  |
| 35 and older | ////   | 4      | 4/30 = 0.14  |
|              |        | N = 30 | Σr.f. = 1    |

**Problem 3**

BBA (Professional) 2004

The profits (in lakh taka) of 30 companies for the year 2003-2004 are given below:

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 20 | 22 | 35 | 42 | 37 | 42 | 48 | 53 | 49 | 65 |
| 39 | 48 | 67 | 18 | 16 | 23 | 37 | 35 | 49 | 63 |
| 65 | 55 | 45 | 58 | 57 | 69 | 25 | 29 | 58 | 65 |

Classify the above data taking a suitable class-interval.

Solution: Let us determine the suitable class interval with the help of the following formula:

$$\text{Class interval, } i = \frac{\text{Range}}{1 + 3.322 \log N}$$

Here, Range = Highest value - Lowest value  
= 69 - 16

Range = 53

Total no. of observations, N = 30

$$i = \frac{53}{1 + 3.322 \log 30}$$

$$\Rightarrow i = \frac{53}{1 + 3.322(1.4771)}$$

$$\therefore i = 8.97 \approx 9$$

In general, class interval is multiple of 5. So we should avoid the values like 3, 4, 6, 7, 8, 9 etc

Here we take class interval = 10 (say) and the first class will be 15-25.

Frequency distribution of profits of 30 companies

| Profits<br>(In lakh<br>Taka) | Tally | No. of companies<br>(f) |
|------------------------------|-------|-------------------------|
| 15-25                        | ###   | 5                       |
| 25-35                        | //    | 2                       |
| 35-45                        | ####  | 7                       |
| 45-55                        | ####  | 6                       |
| 55-65                        | ###   | 5                       |
| 65-75                        | ###   | 5                       |
| Total                        |       | N=30                    |

**Problem 11**

BBA (Honours) 2009

The following data represent the marks in business mathematics (x) and business statistics (y) of 13 students. Prepare a bi-variate frequency distribution table from the following data:

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| x: | 32 | 15 | 21 | 27 | 58 | 37 | 47 | 58 | 46 | 50 | 40 | 54 | 62 |
| y: | 75 | 48 | 54 | 70 | 92 | 53 | 74 | 85 | 65 | 62 | 73 | 77 | 40 |

**Solution:**

Here Class interval data x series (c) =  $\frac{HV - LV}{1 + 3.322 \log N} = \frac{62 - 15}{1 + 3.322 \log 13} \approx 10$

and Class interval data y series (c) =  $\frac{HV - LV}{1 + 3.322 \log N} = \frac{92 - 40}{1 + 3.322 \log 13} \approx 10$

**Bi-variate Frequency Distribution Table**

| Bus. Math (x) \ Bus. Stat (y) | 10-20 | 20-30 | 30-40 | 40-50   | 50-60 | 60-70 | Total |
|-------------------------------|-------|-------|-------|---------|-------|-------|-------|
| 40-50                         | /(1)  |       |       |         |       | /(1)  | 2     |
| 50-60                         |       | /(1)  | /(1)  |         |       |       | 2     |
| 60-70                         |       |       |       | /(1)    | /(1)  |       | 2     |
| 70-80                         |       | /(1)  | /(1)  | /// (3) |       |       | 5     |
| 80-90                         |       |       |       |         | /(1)  |       | 1     |
| 90-100                        |       |       |       |         | /(1)  |       | 1     |
| Total                         | 1     | 2     | 2     | 4       | 3     | 1     | N=13  |

Solution: Let us determine the suitable class interval with the help of the following formula:

$$\text{Class interval, } i = \frac{\text{Range}}{1 + 3.322 \log N}$$

In general, class interval is multiple of 5. So we should avoid the values like 3, 4, 6, 7, 8, 9 etc.

Here we take class interval = 10 (say) and the first class will be 15-25.

$$\text{Class interval, } i = \frac{\text{Range}}{1 + 3.322 \log N}$$

$$\begin{aligned} \text{Here, Range} &= \text{Highest value} - \text{Lowest value} \\ &= 69 - 16 \end{aligned}$$

$$\text{Range} = 53$$

$$\text{Total no. of observations, } N = 30$$

$$i = \frac{53}{1 + 3.322 \log 30}$$

$$\Rightarrow i = \frac{53}{1 + 3.322(1.4771)}$$

$$\therefore i = 8.97 \approx 9$$

In general, class interval is multiple of 5.

So we should avoid the values like 3,4,6,7,8,9 etc.

Here we take class interval =10(say) and the first class will be 15-25.





## Stem and Leaf Display:

**Origin:** Stem and leaf display is invented by a famous American Statistician John Wilder Tucky in 1977.

**Definition:** The method of presenting data set using the leftmost digits as a stem and the ones on the right as leaves is called stem and leaf display.

**Example:** For two digital data set, first digit is stem and second digit is leaf.

**Illustration:** Let a data set be 27, 29, 30, 22, 48, 31, 12, 50, 35, 49, and 38. Taking the data set, we can make a stem and leaf display as follow:

| Stem | Leaf       |
|------|------------|
| 1    | 2          |
| 2    | 2, 7, 9    |
| 3    | 0, 1, 5, 8 |
| 4    | 8, 9       |
| 5    | 0          |

Key: 1|2 means 12, 5|0 means 50.

To gist up, stem and leaf display is a technique of presenting data set so that the concentrations in the data become visible.

**Problem 8 by Abdul Aziz (Page-1-67):** The obtained marks of 30 candidates in the B.C.S examination are given below:

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 34 | 33 | 36 | 80 | 31 | 77 | 46 | 81 | 76 | 46 | 86 | 40 | 42 | 60 | 42 |
| 32 | 64 | 46 | 76 | 40 | 56 | 54 | 57 | 66 | 57 | 56 | 70 | 50 | 44 | 63 |

- Construct a frequency distribution table by using suitable class interval.
- Calculate the absolute frequency, relative frequency, cumulative frequency, relative cumulative frequency and frequency density
- Construct a stem and leaf diagram for this data.

**Solution:** To find suitable class interval, we can use Sturges rule which as follows:

$$\text{Class Interval, } c = \frac{\text{Range}}{1 + 3.322 \log N}$$

Here, Range = *Highest Value* - *Lowest Value*

$$\text{Range} = 86 - 31 = 55$$

Total number of observations,  $N = 30$

$$\begin{aligned} \Rightarrow \text{Class Interval, } c &= \frac{55}{1 + 3.322 \log 30} \\ &= \frac{55}{1 + 3.322(1.4771)} \\ &= \frac{55}{1 + 4.91} \\ &= \frac{55}{5.91} \end{aligned}$$

Formula which as follows:  $\therefore \text{Class Interval, } C = 9.31 \approx 9$

In general, class interval is multiple of 5. So, we should avoid the values like 4, 6, 7, 8, 9 etc.

Here we take class interval = 10 and the first class will be 30-40.

**Table for constructing frequency distribution**

| Class  | Tally | Frequency/Absolute Frequency | Relative Frequency (%)            | Cumulative Frequency | Relative Cumulative Frequency (%)  | Frequency density    |
|--------|-------|------------------------------|-----------------------------------|----------------------|------------------------------------|----------------------|
| 30-40  | ###   | 5                            | $\frac{5}{30} \times 100 = 16.67$ | 5                    | $\frac{5}{30} \times 100 = 16.67$  | $\frac{5}{10} = 0.5$ |
| 40-50  | ####  | 8                            | 26.67                             | 13                   | $\frac{13}{30} \times 100 = 43.33$ | 0.8                  |
| 50-60  | ###/  | 6                            | 20                                | 19                   | 63.33                              | 0.6                  |
| 60-70  | ////  | 4                            | 13.33                             | 23                   | 76.67                              | 0.4                  |
| 70-80  | ////  | 4                            | 13.33                             | 27                   | 90                                 | 0.4                  |
| 80-90  | ///   | 3                            | 10                                | 30                   | 100                                | 0.3                  |
| Total\ |       | N=30                         |                                   |                      |                                    |                      |

### Construction of stem and leaf diagram

| Stem | Leaf            | Frequency |
|------|-----------------|-----------|
| 3    | 4,6,1,2,3       | 5         |
| 4    | 6,2,4,6,0,2,6,0 | 8         |
| 5    | 4,6,0,6,7,7     | 6         |
| 6    | 6,0,3,4         | 4         |
| 7    | 6,7,6,0         | 4         |
| 8    | 6,0,1           | 3         |

### Construction of stem and sorted leaf diagram

| Stem | Leaf                   | Frequency |
|------|------------------------|-----------|
| 3    | 1, 2, 3, 4, 6          | 5         |
| 4    | 0, 0, 2, 2, 4, 6, 6, 6 | 8         |
| 5    | 0, 4, 6, 6, 7, 7       | 6         |
| 6    | 0, 3, 4, 6             | 4         |
| 7    | 0, 6, 6, 7             | 4         |
| 8    | 0, 1, 6                | 3         |

**Key:**3|1 means 31,8|6 means 86.

**Statistical chart:** A chart can take the shape of either a diagram or a graph.

**Diagram:** For representing data diagrams are more commonly used than graphs.

### Types of diagram

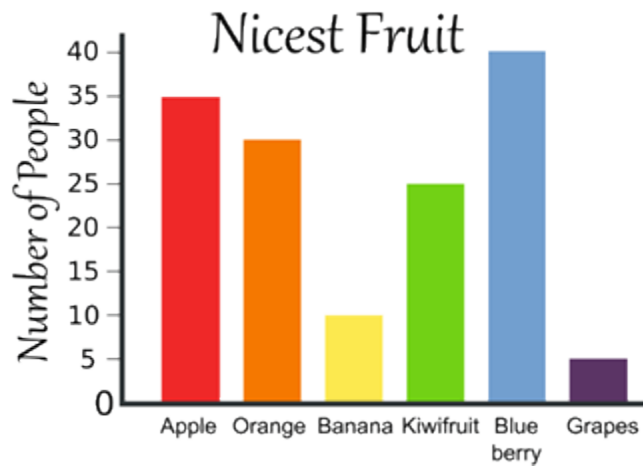
- I. One dimensional diagram, e.g. bar diagram
- II. Two dimensional diagram e.g. Rectangles, squares and circle.
- III. Pictogram and Cartogram

#### I. One dimensional or Bar Diagram

A bar is a thick line whose width is shown only for attention.

A **bar chart** is a type of graph in which each column (plotted either vertically or horizontally) represents a categorical variable or a discrete ungrouped numeric variable.

A **Bar Graph** (also called Bar Chart) is a graphical display of data using bars of different heights.



## Types of Graphs

### 1. Line graphs:

In the X axis we take time and in the Y axis the value of the variable.



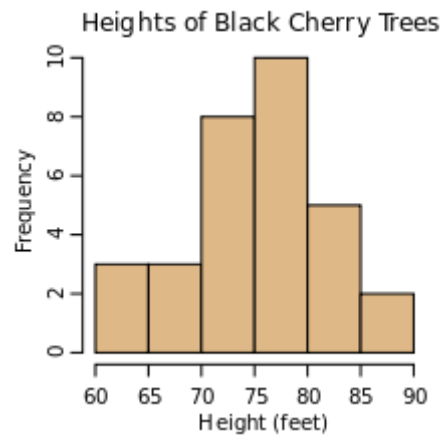
## **2. Graphs of frequency distribution**

1. Histogram
2. Frequency Polygon
3. Frequency Curve
4. Less than and more than Cumulative frequency curve or ogives.

### **1. Histogram:**

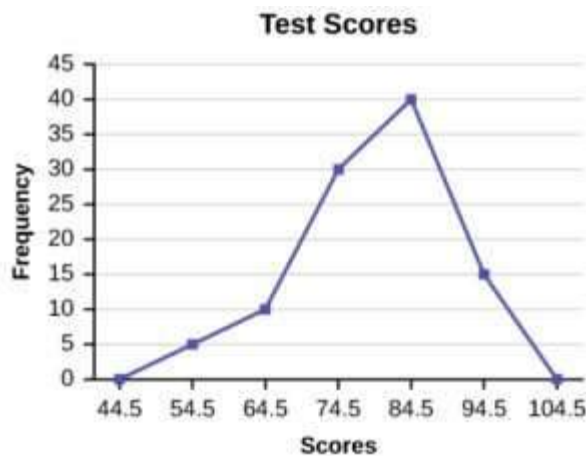
A histogram is a bar graph that represents a frequency distribution. The width

represents the interval and the height represents the corresponding frequency. There are no spaces between the bars.



## 2. Frequency Polygon:

A **frequency polygon** is a graph constructed by using lines to join the midpoints of each interval. The heights of the points represent the **frequencies**. A **frequency polygon** can be created from the histogram or by calculating the midpoints from the **frequency** distribution table.



Mid Points →

**3. Frequency Curve:** The **frequency-curve** for a distribution can be obtained by drawing a smooth and free hand **curve** through the mid-points in the X axis and in the Y axis we take frequency. We can also draw frequency curve by using histogram.

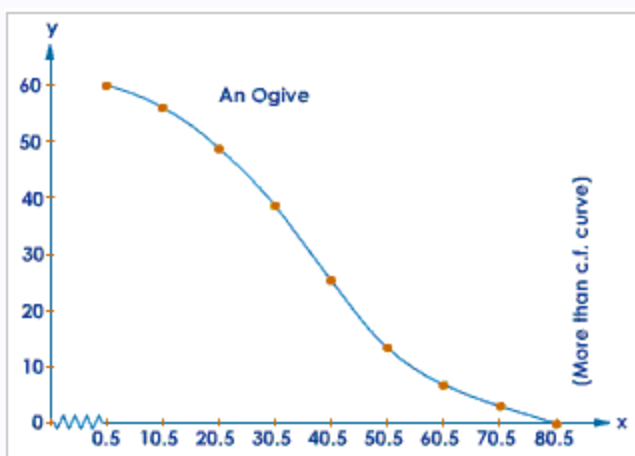
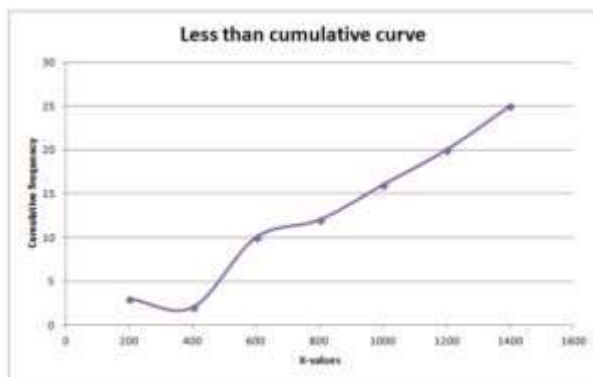
## Difference between frequency polygon and frequency curve

The main difference is that when we draw frequency polygon, we can add the points by using scale but when we draw frequency curve, we can add the points by using hand.

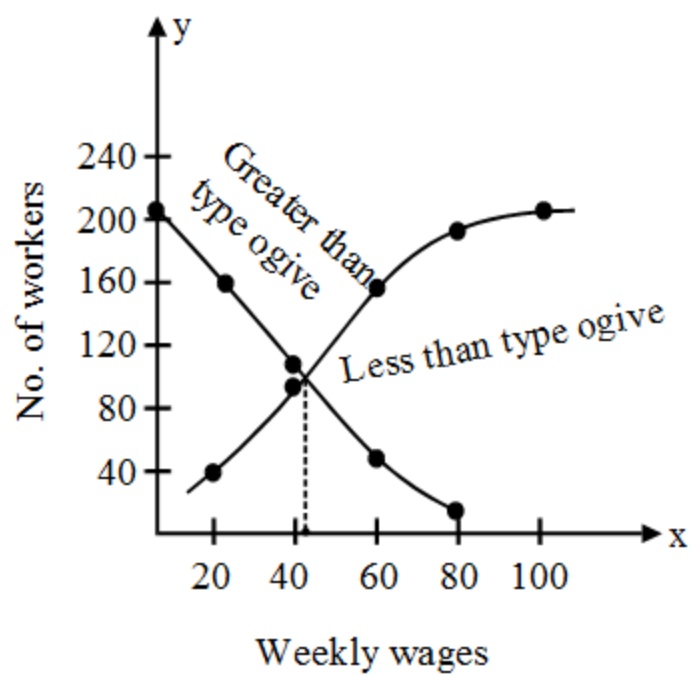
### 4. Less than and more than cumulative frequency curve:

When we draw less than cumulative frequency curve, in the X axis we take upper limit of each class and in the Y axis we take less than cumulative frequency. On the other hand, When we draw more than cumulative frequency curve, in the X axis we take lower limit of each class and in the Y axis we take more than cumulative frequency.

If we draw less than and more than cumulative frequency curve, we get one intersect point. From this intersect point; if we draw perpendicular to the X-axis, we get median.

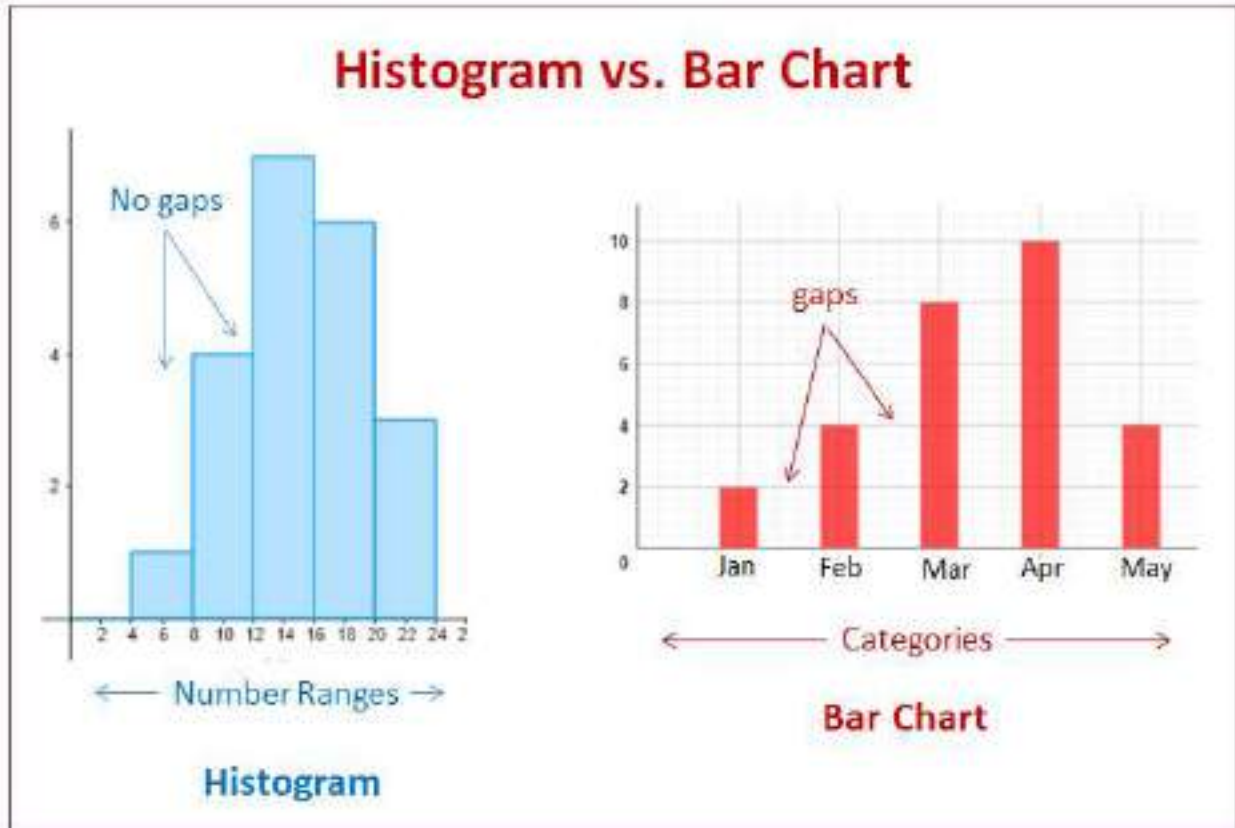






## Distinguish between Bar diagram and Histogram

| <b>BASIS FOR COMPARISON</b> | <b>HISTOGRAM</b>                                                                                                           | <b>BAR GRAPH</b>                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Meaning                     | Histogram refers to a graphical representation, that displays data by way of bars to show the frequency of numerical data. | Bar graph is a pictorial representation of data that uses bars to compare different categories of data. |
| Indicates                   | Distribution of non-discrete variables                                                                                     | Comparison of discrete variables                                                                        |
| Presents                    | Quantitative data                                                                                                          | Categorical data                                                                                        |
| Spaces                      | Bars touch each other, hence there are no spaces between bars                                                              | Bars do not touch each other, hence there are spaces between bars.                                      |
| Elements                    | Elements are grouped together, so that they are considered as ranges.                                                      | Elements are taken as individual entities.                                                              |
| Can bars be reordered?      | No                                                                                                                         | Yes                                                                                                     |
| Width of bars be same       | Need not to                                                                                                                | Same                                                                                                    |



## Distinguish between diagram and a graph

- (i) Diagrams are very attractive to eyes and hence, they are invariably used in advertisement and publicity. But graphs are not attractive to eyes and hence, generally, they are not used in advertisement and publicity.
- (ii) Diagrams do not add anything to the meaning of the data and hence, they cannot be used by statisticians or research scholars in the further analysis of data. But graphs make the data more meaningful and hence, they are very much used by statisticians and research scholars in the analysis and interpretation of data.
- (iii) Diagrams create effective and long lasting impression in the minds of the onlookers but graphs do not create such impression.
- (iv) Diagrams help us in making comparison between the data but graphs help us in studying the cause and effect relationship between two variables.
- (v) Diagrams are better drawn on plain papers but graphs are accurately drawn on graph papers.
- (vi) Diagrams are rarely used to present frequency distributions. But graphs are mostly used to present such distributions.
- (vii) Diagrams never present any mathematical relationship between the variables. But graphs always present mathematical relationship between the variables.

# Measures of Central Tendency

## Introduction:

According to Prof Bowley “Measures of central tendency (averages) are statistical constants which enable us to comprehend in a single effort the significance of the whole.” measure of central tendency is a measure that tells us where the middle of a bunch of data lies. One of the most widely used set of summary figures is known as measures of location which are often referred to as averages, central tendency or central location. An average is a single value which is considered as the most representative or typical value for a given set of data. Such a value is neither the smallest nor the larger value, but is a number whose value is somewhere in the middle of the group. For this reason an average is frequently referred to as a measure of central tendency or central value.

## Objectives of Averaging /Measure of central tendency:

The main objectives of Measure of Central Tendency are

- 1) To get one single value that describes the characteristic of the entire data.
- 2) To facilitate comparisons between data.

## Different measures of Central Tendency

The following are the important measures of central tendency:

- (1) Arithmetic Mean
- (2) Geometric mean
- (3) Harmonic Mean
- (4) Median
- (5) Mode

### (1) Arithmetic mean or mean:

Arithmetic mean or simply the mean of a variable is defined as the sum of the observations divided by the number of observations. If the variable X assumes n values  $X_1, X_2, \dots, X_N$  then the mean,  $\bar{X}$ , is given by

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_N}{N}$$

$$\bar{X} = \frac{\sum X}{N}$$

This formula is for the ungrouped or raw data.

Computing Arithmetic Mean for grouped data:

(a) Direct Method:

When the arithmetic mean is computed from grouped data, the mid-point of each class is taken as the representative value of that class. The various mid- values are multiplied by their respective frequencies, the products are added, and the sum of the products is then divided by the total number of

observations to obtain the arithmetic mean. Symbolically, let  $x_1, x_2, x_3, \dots, x_N$  be the mid values and their corresponding frequencies are  $f_1, f_2, f_3, \dots, f_N$  respectively. Then arithmetic mean,

$$\bar{X} = \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 + \dots + f_N x_N}{f_1 + f_2 + f_3 + \dots + f_N} = \frac{\sum fx}{N}$$

- (b) Short cut method: The method of computation of arithmetic mean presented by direct method becomes laborious and time consuming and thus we look for a convenient method to take advantage of the both. This is the so called short cut method. Symbolically,

$$\bar{X} = A + \frac{\sum fd}{N} i$$

Here,  $A$  = Assumed mean or Arbitrary value

$f$  = Frequency

$d$  = New Variable

$N$  = Total no. of observations

$i$  = Class interval

### Merits of Mean:

- 1) Arithmetic mean rigidly defined by Algebraic Formula.
- 2) It is easy to calculate and easy to understand.
- 3) It is based on all observations of the given data.
- 4) It is capable of further mathematical treatment, hence it is widely used in statistical analysis.
- 5) Arithmetic mean can be computed even if the detailed distribution is not known but some of the observation and number of the observation are known.
- 6) It is least affected by the fluctuation of sampling.
- 7) For every kind of data mean can be calculated.

### Demerits of Arithmetic mean :

- 1) It can neither be determined by inspection or by graphical location.
- 2) Arithmetic mean cannot be computed for qualitative data like data on

intelligence honesty and smoking habit etc.

3) It is too much affected by extreme observations and hence it is not adequately represent data consisting of some extreme point.

4) Arithmetic mean cannot be computed when class intervals have open ends.

5) If any one of the data is missing then mean cannot be calculated.

## Uses:

1) Mean is used in fields such as business, engineering and computer science.

2) It is used in report card or in our population.

3) In addition to mathematics and statistics, the arithmetic mean is used frequently in fields such as economics, sociology, and history, though it is used in almost every academic field to some extent. For example, per capita GDP gives an approximation of the arithmetic average income of a nation's population.

4) While the arithmetic mean is often used to report central tendencies, it is not a robust statistic, meaning that it is greatly influenced by outliers.

5) It's used to compute the variance and SD (Standard Deviation). It's good for inferential statistics.

## Weighted Arithmetic mean:

For calculating simple mean, we suppose that all the values or the sizes of items in the distribution have equal importance. But, in practical life this may not be so. In case some items are more Important than others, a simple average computed is not representative of the distribution. Proper weightage has to be given to the various items. For example, to have an idea of the change in cost of living of a certain group of persons, the simple average of the prices of the commodities consumed by them will not do because all the commodities are not equally important, e.g. rice, wheat and pulses are more important than tea, confectionery etc., It is the weighted arithmetic average which helps in finding out the average value of the series after giving proper weight to each group.

**Definition:**

The average whose component items are being multiplied by certain values known as “weights” and the aggregate of the multiplied results are being divided by the total sum of their “weight”.

If  $X_1, X_2, X_3, \dots, X_n$  be the values of a variable  $X$  with respective weights of  $w_1, w_2, w_3, \dots, w_n$  assigned to them, then

**Direct Method:** Weighted Arithmetic Mean,

$$\overline{X}_w = \frac{w_1 X_1 + w_2 X_2 + w_3 X_3 + \dots + w_n X_n}{w_1 + w_2 + w_3 + \dots + w_n} = \frac{\sum wX}{\sum w}$$

**Short cut Method:** Weighted Arithmetic Mean,

$$\overline{X}_w = A_{w_w} + \frac{\sum wd}{\sum w}$$

**Uses of the weighted mean:**

Weighted arithmetic mean is used in:

- Construction of index numbers.
- Comparison of results of two or more universities where number of students differ.
- Computation of standardized death and birth rates.

**MEDIAN:**

Median is the middle value of the distribution i.e. median of a distribution is the value of the variable which divides it into two equal parts when the data are arranged in an ascending or descending order of magnitude

Thus, in an ungrouped frequency distribution if the  $n$  values are arranged in ascending or descending order of magnitude, the median is the middle value if  $n$  is odd. When  $n$  is even, the median is the mean of the two middle values.

**For Odd number:**

Suppose we have the following series:

15, 19, 21, 7, 10, 33, 25, 18 and 5

We have to first arrange it in either ascending or descending order. These figures are arranged in an ascending order as follows:

5,7,10,15,18,19,21,25,33

We know that, for odd number,

$$\text{Median} = \frac{n+1}{2} \text{th item}$$

Here, total no. of observations,  $n = 9$

$$\therefore \text{Median} = \frac{9+1}{2} \text{th} = 5\text{th item} = 18$$

**For even number:**

Suppose we have the following series:

15, 19, 21, 7, 10, 33, 25, 18, 5, 23

These figures are arranged in an ascending order as follows:

5, 7, 10, 15, 18, 19, 21, 23, 25, 33

Here, total number of observations,  $n = 10$

We know that, for even number



*Median = Mean* value of  $\frac{n}{2}th$  and  $(\frac{n}{2} + 1)th$  item

$$\frac{n}{2} = \frac{10}{2} = 5th \text{ item} = 18 \text{ \& } (\frac{n}{2} + 1)th = (\frac{10}{2} + 1)th = 6th \text{ item} = 19$$

$$\therefore \text{Mean of 18 and 19} = \frac{18+19}{2} = 18.5$$

$$\therefore \text{Median} = 18.5$$

### **Grouped Frequency Distribution**

First obtain the cumulative frequencies for the data. Then mark the class corresponding to which a cumulative frequency is greater than  $(N)/2$  for the first time. (N is the total number of observations.) Then that class is median class. Then median is evaluated by interpolation formula.

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} i$$

Where, L = lower limit of the median class,

N= Number of observations

f= frequency of the median class

p.c.f=Preceding cumulative frequency of the median class

## **Merits of Median :**

- 1) It is easy to calculate and easy to understand.
- 2) It is well defined an ideal average should be.
- 3) It can also be computed in case of frequency distribution with open ended classes.
- 4) It is not affected by extreme values and also interdependent of range or dispersion of the data.
- 5) It can be determined graphically.
- 6) It is proper average for qualitative data where items are not measured but are scored.
- 7) It is only suitable average when the data are qualitative & it is possible to rank various items according to qualitative characteristics.
- 8) It can be calculated easily by watching the data.
- 9) In some cases median gives better result than mean.

## **Demerits :**

- 1) For computing median data needs to be arranged in ascending or descending order.
- 2) It is not based on all the observations of the data.
- 3) It cannot be given further algebraic treatment.
- 4) It is affected by fluctuation of sampling.
- 5) It is not accurate when the data is not large.
- 6) In some cases median is determined approximately as the mid-point of two observations whereas for mean this does not happen.

## Uses:

- 1) When extreme values are not given then it is used to measure the location.(for skewed distribution).
- 2) When measurement scale is ordinal that time median can be used.
- 3)For skewed data, median is used.
- 4)In some cases median gives us the accurate value than mean. For example, if we are considering the salary of people, if one's salary is more than its mean then so in such cases median is used.

## Mode:

The most frequent score in the distribution A, where a single score is most frequent has one mode and is called unimodal. A distribution that consists of only one of each score has no modes. When there are ties for the most frequent score, the distribution is bimodal if two scores tie or multimodal if more than two scores tie.

## Computation of the mode:

### (a)Ungrouped or Raw Data:

For ungrouped data or a series of individual observations, mode is often found by mere inspection.

#### Example:

2, 7, 10, 15, 10, 17, 8, 10, 2

Mode =  $M_0$ =10

### (b)Grouped data:

For grouped frequency distributions, the modal class is the class with the largest frequency. After identifying modal class mode is evaluated by using interpolated formula. This formula is applicable when classes are of equal width.

$$Mode = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} i$$

Where, L= Lower limit of the modal class

$\Delta_1$ = The difference between frequency of the modal class and pre modal class

$\Delta_2$ = The difference between frequency of the modal class and post modal class

I = Class interval of the modal class.

## **Merits of Mode:**

- 1) It is readily comprehensible and easy to compute. In some case it can be computed merely by inspection.
- 2) It is not affected by extreme values. It can be obtained even if the extreme values are not known.
- 3) Mode can be determined in distributions with open classes.
- 4) Mode can be located on the graph also.
- 5) If all the data are not given then also mode can be calculated.
- 6) It is easy to understand.
- 7) It is the most observed data point.
- 8) Sometimes we get more than one mode and sometimes there is no mode, that gives us the characteristics of the data.

## **Demerits:**

- 1) It is ill defined. It is not always possible to find clearly defined mode. In some cases, we may come across distributions with two modes. Such distributions are called Bimodal. If a distribution has more than two modes, it is said to be Multimodal.
- 2) It is not based upon all the observation.
- 3) Mode can be calculated by various formulae as such the value may differ from one to other. Therefore, it is not rigidly defined.
- 4) It is affected to a greater extent fluctuations of sampling.

## **Uses:**

- 1) Mode is used by manufacturers of ready- made garments, shoes and accessories in common use etc. The readymade garment manufacturers make those sizes more which are used by most of the persons than the other sizes. Similarly, the makers of shoes will make that size maximum which the majority people use and others in less quantity. Mode is used when the data is categorized.

## **Geometric Mean(G.M) :**

Geometric Mean is the nth root of their product.

If  $x_1, x_2, \dots, x_n$  are observations then geometric mean,

$$G.M = (x_1 * x_2 * \dots * x_n)^{\frac{1}{n}}$$

$$\Rightarrow \log G.M = \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n)$$

$$\Rightarrow \log G.M = \frac{1}{n} \sum \log x$$

$$\Rightarrow G.M = \text{Anti log} \left[ \frac{1}{n} \sum \log x \right]$$

For grouped data:

$$\text{Geometric mean, } G.M = \text{Anti log} \left[ \frac{1}{N} \sum f \log x \right]$$

## **Merits of Geometric mean :**

1. It is rigidly defined
2. It is based on all items
3. It is very suitable for averaging ratios, rates and percentages
4. It is capable of further mathematical treatment.
5. Unlike AM, it is not affected much by the presence of extreme values

## **Demerits of Geometric mean:**

1. If any of the observations is zero, the geometric mean becomes zero
2. If any of the observation is negative, the geometric mean becomes imaginary
3. A geometric mean is not easily understandable by a non-mathematical person

## **Uses of geometric mean:**

The **geometric mean** is a type of average , usually **used for** growth rates, like population growth or interest rates. While the arithmetic **mean** adds items, the **geometric mean** multiplies items. Also, we can only get the **geometric mean** for positive numbers.

## Harmonic mean (H.M):

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic mean of the reciprocal of the given values.

For ungrouped data: If  $x_1, x_2, \dots, x_n$  are  $n$  observations, then Harmonic Mean,

$$H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum \frac{1}{x}}$$

For grouped data:

$$\text{Harmonic Mean, } H.M = \frac{N}{f_1 \frac{1}{x_1} + f \frac{1}{x_2} + \dots + f_{k2} \frac{1}{x_k}} = \frac{N}{\sum f \frac{1}{x}} = \frac{N}{\sum \frac{f}{x}}$$

## Merits of H.M:

1. It is rigidly defined.
2. It is defined on all observations.
3. It is amenable to further algebraic treatment.
4. It is the most suitable average when it is desired to give greater weight to smaller observations and less weight to the larger ones.

## Demerits of H.M:

1. It is not easily understood.
  2. It is difficult to compute.
  3. It is only a summary figure and may not be the actual item in the series
- It gives greater importance to small items and is therefore, useful only when small items have to be given greater weightage.

## Uses of Harmonic Mean:

**Harmonic means** are often **used** in averaging things like rates (e.g., the average travel speed given a duration of several trips). The weighted **harmonic mean** is **used** in finance to average multiples like the price-earnings ratio because it gives equal weight to each data point.

**# What do you mean by central tendency? Mention the objectives of measuring central tendency.**

According to Lawrence J.Kaplan” one of the most widely used set of summary figures is known as measure of location which are often referred to as average, central tendency or central location. The purpose for computing an average value for a set of observations is to obtain a single value which is representative of all the items.”

**Objectives of measuring central tendency:**

There are two main objectives of the study of central tendency:

- (i) To get single value that describes the characteristics of the entire data.
- (ii) To facilitate comparison.

**Mathematical Properties of Arithmetic Mean**

1.The algebraic sum of the deviations of all the observations from arithmetic mean is zero. i.e.  $\sum (X - \bar{X}) = 0$

Example: Let us take the followings:

|       | SL. Number | X              | $X - \bar{X} = X - 30$   |
|-------|------------|----------------|--------------------------|
|       | 1          | 10             | -20                      |
|       | 2          | 20             | -10                      |
|       | 3          | 30             | 0                        |
|       | 4          | 40             | 10                       |
|       | 5          | 50             | 20                       |
| Total | N=5        | $\sum X = 150$ | $\sum (X - \bar{X}) = 0$ |

$$\text{Mean, } \bar{X} = \frac{\sum X}{N} = \frac{150}{5} = 30$$

2. The sum of the squared deviations of all the observations from arithmetic mean is minimum. The following example would clarify the point:

|       | SL.<br>Number | X             | $X - \bar{X} = X - 4$    | $(X - \bar{X})^2$           |
|-------|---------------|---------------|--------------------------|-----------------------------|
|       | 1             | 2             | -2                       | 4                           |
|       | 2             | 3             | -1                       | 1                           |
|       | 3             | 4             | 0                        | 0                           |
|       | 4             | 5             | 1                        | 1                           |
|       | 5             | 6             | 2                        | 4                           |
| Total | N=5           | $\sum X = 20$ | $\sum (X - \bar{X}) = 0$ | $\sum (X - \bar{X})^2 = 10$ |

3. If we have two groups such as:

1st Group

Mean =  $\bar{X}_1$

Total no. of observations =  $N_1$

2nd Group

Mean =  $\bar{X}_2$

Total no. of observations =  $N_2$

Then Combined Mean,  $\bar{X}_{12} = \frac{N_1 \bar{X}_1 + N_2 \bar{X}_2}{N_1 + N_2}$

4. Arithmetic mean depends on change of origin and scale of measurement.



## **#What are the properties /requisites for an Ideal/ Good measure of central tendency? Which measure do you think best and why?**

According to Professor Yule, the following are the characteristics to be satisfied by an ideal measure of central tendency:

- (i) It should be rigidly defined.
- (ii) It should be easy to understand and easy to calculate.
- (iii) It should be based on all the observations.
- (iv) It should be suitable for further mathematical treatment.
- (v) It should be affected as little as possible by fluctuations of sampling.  
In addition to the above criteria, we may add the following (which is not due to Prof.Yule)
- (vi) It should not be affected much by extreme values.  
I think arithmetic mean is the best measure of central tendency. Because, Arithmetic mean follows most of the properties of Prof. Yule .But other measures such as median, mode; geometric mean and harmonic mean do not follow all the properties of Prof.Yule.

## **# When Geometric mean is preferred to arithmetic mean?**

Geometric mean is preferred to arithmetic mean in the following cases:

- (i)The geometric mean is used to calculate averages of ratios and percentage.
- (ii)It is also used for computing average rate of increase or decrease.
- (iii)It is useful for the construction of index number.

## **# When median is preferred to arithmetic mean?**

Median is preferred to arithmetic mean in the following cases:

- (i)Median is very useful when our observations are not quantitative.
- (ii) It is very useful for open end class interval.
- (iii)It is useful for comparing two or more sets of qualitative data.

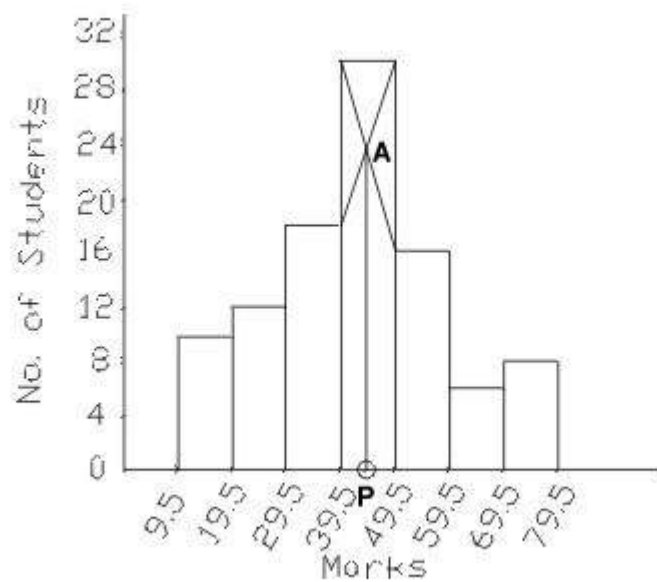
## Finding Mode Graphically

| Marks<br>inclusive series | Conversions into<br>exclusive series | No. of students<br>(frequency) |
|---------------------------|--------------------------------------|--------------------------------|
| (x)                       |                                      | (f)                            |
| 10-19                     | 9.5-19.5                             | 10                             |
| 20-29                     | 19.5-29.5                            | 12                             |
| 30-39                     | 29.5-39.5                            | 18                             |
| 40-49                     | 39.5-49.5                            | 30                             |
| 50-59                     | 49.5-59.5                            | 16                             |
| 60-69                     | 59.5-69.5                            | 6                              |
| 70-79                     | 69.5-79.5                            | 8                              |

The following steps must be followed to find the mode graphically.

1. Represent the given data in the form of a Histogram. The height of the rectangles in the histogram is marked by the frequencies of the class interval as shown in the graph .Identify the highest rectangle. This corresponds to the modal class of the series.
2. Join the top corners of the modal rectangle with the immediately next corners of the adjacent rectangles. The two lines must be cutting each other. This might be difficult to visualize so look at the graph given below.
3. Let the point where the joining lines cut each other be 'A'. Draw a perpendicular line from point A onto the x-axis. The point 'P' where the perpendicular will meet the x-axis will give the mode.

### The Histogram:



In this case the value of point P turns out to be 44.12

### Finding Median Graphically

| Marks   | Conversion into exclusive series | No. of students | Cumulative Frequency |
|---------|----------------------------------|-----------------|----------------------|
| (x)     |                                  | (f)             | (C.M)                |
| 410-419 | 409.5-419.5                      | 14              | 14                   |
| 420-429 | 419.5-429.5                      | 20              | 34                   |
| 430-439 | 429.5-439.5                      | 42              | 76                   |
| 440-449 | 439.5-449.5                      | 54              | 130                  |
| 450-459 | 449.5-459.5                      | 45              | 175                  |
| 460-469 | 459.5-469.5                      | 18              | 193                  |
| 470-479 | 469.5-479.5                      | 7               | 200                  |

The median value of a series may be determined through the graphic presentation of data in the form of Ogives. This can be done in 2 ways.

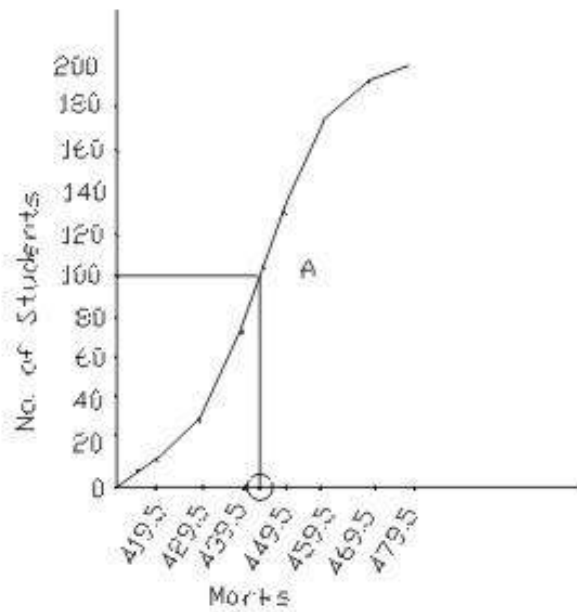
1. Presenting the data graphically in the form of 'less than' ogive or 'more than' ogive.
2. Presenting the data graphically and simultaneously in the form of 'less than' and 'more than' ogives. The two ogives are drawn together.

### 1. Less than Ogive approach

| Marks           | Cumulative Frequency (C.M) |
|-----------------|----------------------------|
| Less than 419.5 | 14                         |
| Less than 429.5 | 34                         |
| Less than 439.5 | 76                         |
| Less than 449.5 | 130                        |
| Less than 459.5 | 175                        |
| Less than 469.5 | 193                        |
| Less than 479.5 | 200                        |

Steps involved in calculating median using less than Ogive approach -

1. Convert the series into a 'less than ' cumulative frequency distribution as shown above.
2. Let N be the total number of students who's data is given. N will also be the cumulative frequency of the last interval. Find the  $(N/2)^{\text{th}}$  item(student) and mark it on the y-axis. In this case the  $(N/2)^{\text{th}}$  item (student) is  $200/2 = 100^{\text{th}}$  student.
3. Draw a perpendicular from 100 to the right to cut the Ogive curve at point A.
4. From point A where the Ogive curve is cut, draw a perpendicular on the x-axis. The point at which it touches the x-axis will be the median value of the series as shown in the graph.

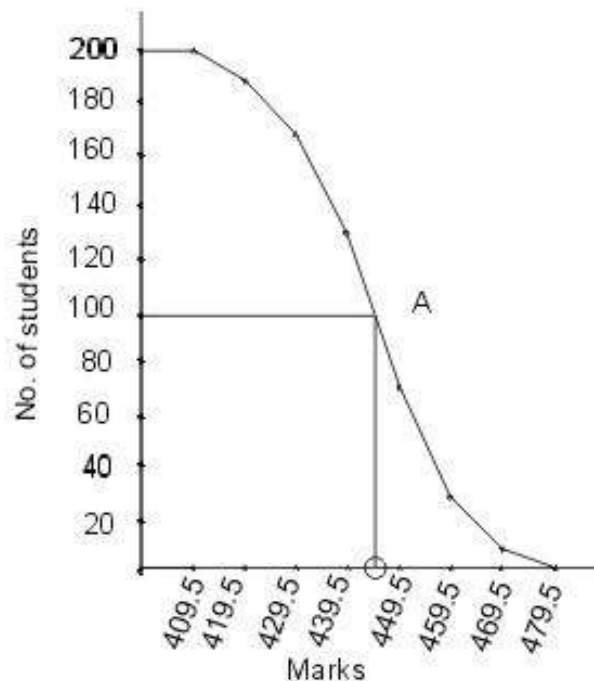


**More than Ogive approach:**

| Marks           | Cumulative Frequency (C.M) |
|-----------------|----------------------------|
| More than 409.5 | 200                        |
| More than 419.5 | 186                        |
| More than 429.5 | 166                        |
| More than 439.5 | 124                        |
| More than 449.5 | 70                         |
| More than 459.5 | 25                         |
| More than 469.5 | 7                          |
| More than 479.5 | 0                          |

Steps involved in calculating median using more than Ogive approach -

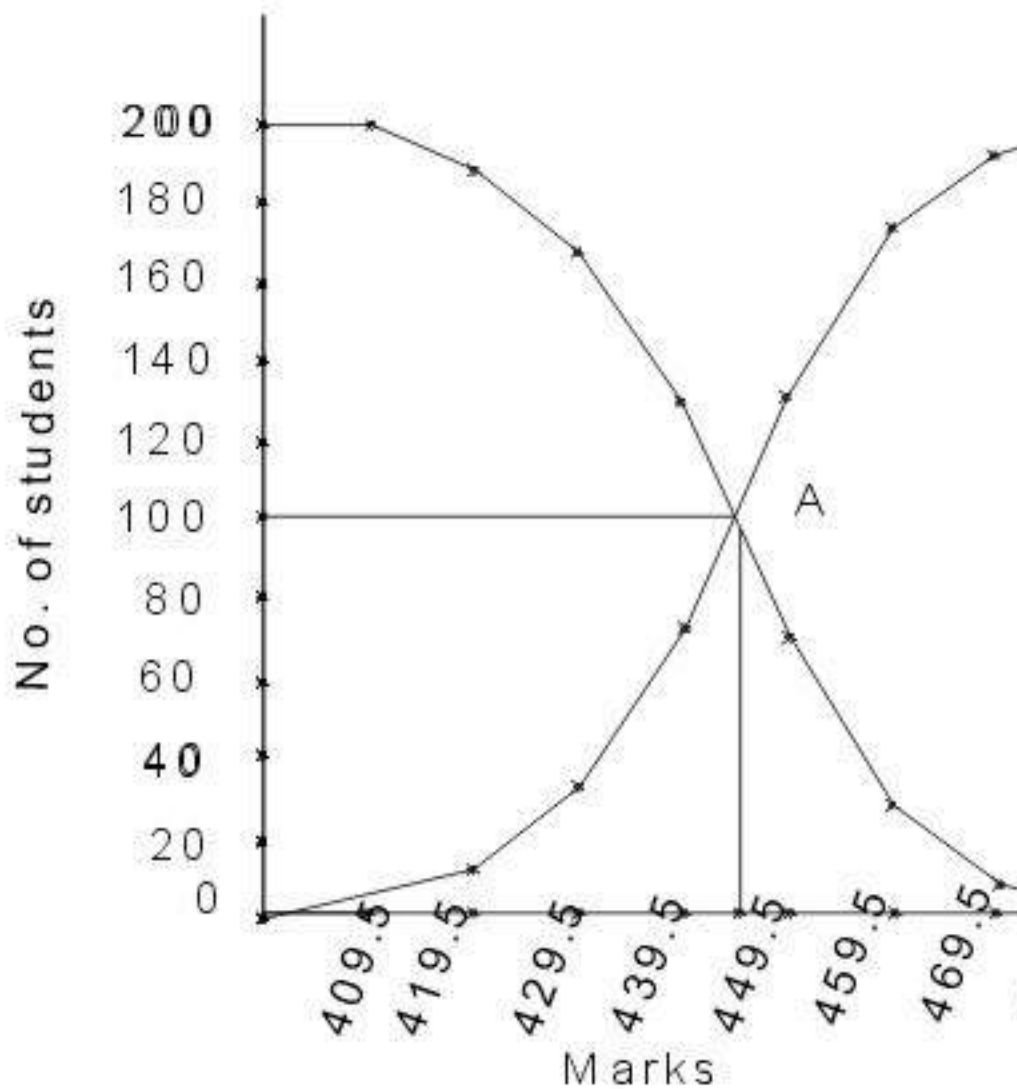
1. Convert the series into a 'more than ' cumulative frequency distribution as shown above.
2. Let  $N$  be the total number of students who's data is given will also be the cumulative frequency of the last interval. Find the  $(N/2)^{\text{th}}$  item(student) and mark it on the y-axis. In this case the  $(N/2)^{\text{th}}$  item (student) is  $200/2 = 100^{\text{th}}$  student.
3. Draw a perpendicular from 100 to the right to cut the Ogive curve at point A.
4. From point A where the Ogive curve is cut, draw a perpendicular on the x-axis. The point at which it touches the x-axis will be the median value of the series as shown in the graph.



## 2. Less than and more than Ogive approach:

Another way of graphical determination of median is through simultaneous graphic presentation of both the less than and more than Ogives.

1. Mark the point A where the Ogive curves cut each other.
2. Draw a perpendicular from A on the x-axis. The corresponding value on the x-axis would be the median value.



**Thank You**



**# Find the mode of the following data:**

|                    |      |       |       |       |        |
|--------------------|------|-------|-------|-------|--------|
| Marks              | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 |
| Number of Students | 4    | 10    | 10    | 4     | 2      |

**Solution:** Since highest frequency is 10 which have two classes such as 20-40 and 40-60. So we can use empirical formula to find mode. Empirical formula is Mode = 3 Median - 2 Mean

Calculation of mean, median and mode

| Marks  | Number of Students<br>f | Mid Values<br>X | fX               | Cumulative Frequency<br>(C.f) |
|--------|-------------------------|-----------------|------------------|-------------------------------|
| 0-20   | 4                       | 10              | 40               | 4                             |
| 20-40  | 10                      | 30              | 300              | 14                            |
| 40-60  | 10                      | 50              | 500              | 24                            |
| 60-80  | 4                       | 70              | 280              | 28                            |
| 80-100 | 2                       | 90              | 180              | 30                            |
| Total  | N=30                    |                 | $\sum fX = 1300$ |                               |

$$\text{Mean, } \bar{X} = \frac{\sum fX}{N}$$

**Mean:** We know that,  $\bar{X} = \frac{1300}{30}$   
 $\bar{X} = 43.33$

**Median:**

$$\begin{aligned} \text{Median} &= \text{The size of the } \frac{N}{2} \text{th item} \\ &= \text{The size of the } \frac{30}{2} = 15\text{th item} \end{aligned}$$

So Median lies in the class = 40-60

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} \times i$$

$$\text{Here, } L = 40$$

$$f = 10$$

$$p.c.f = 14$$

$$i = 20$$

$$\text{Median} = 40 + \frac{\frac{30}{2} - 14}{10} \times 20$$

$$\text{Median} = 40 + 2$$

$$\therefore \text{Median} = 42$$

**Mode:** By using empirical formula,

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

$$\text{Mode} = 3(42) - 2(43.33)$$

$$\text{Mode} = 126 - 86.66$$

$$\text{Mode} = 39.34 \text{ Ans.}$$

**Problem-2(Exercise 16-Page-128)**

Calculate mean, median and mode from the following data pertaining to the profits (in crore Tk) of 125 companies

| Profits (TK in Crore) | No. of companies |
|-----------------------|------------------|
| less than 10          | 4                |
| less than 20          | 16               |
| less than 30          | 40               |
| less than 40          | 76               |
| less than 50          | 96               |
| less than 60          | 112              |
| less than 70          | 120              |
| less than 80          | 125              |

Calculation of mean, median and mode

| Profits(TK.in Crore) | No.of companies<br>f | Mid -Point<br>x | fX               | Cumulative<br>Frequency(c.f) |
|----------------------|----------------------|-----------------|------------------|------------------------------|
| 0-10                 | 4                    | 5               | 20               | 4                            |
| 10-20                | 12                   | 15              | 180              | 16                           |
| 20-30                | 24                   | 25              | 600              | 40                           |
| 30-40                | 36                   | 35              | 1260             | 76                           |
| 40-50                | 20                   | 45              | 900              | 96                           |
| 50-60                | 16                   | 55              | 880              | 112                          |
| 60-70                | 8                    | 65              | 520              | 120                          |
| 70-80                | 5                    | 75              | 375              | 125                          |
| Total                | $N = \sum f = 125$   |                 | $\sum fX = 4735$ |                              |

Mean: We know that, Mean,  $\bar{X} = \frac{\sum fX}{N}$

$$\bar{X} = \frac{4735}{125}$$

$$\therefore \text{Mean}, \bar{X} = 37.88$$

Median: we know that, Median = The size of the  $\frac{N}{2}$ th item

$$= \text{The size of the } \frac{125}{2} = 62.5 \text{th item}$$

$\therefore$  Median lies in the class = 30 – 40

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} i$$

$$\text{Here, } L = 30, f = 36, p.c.f = 40, i = 10$$

$$\text{Median} = 30 + \frac{\frac{125}{2} - 40}{36} \times 10$$

$$\text{Median} = 30 + 6.25$$

$$\therefore \text{Median} = 36.25$$

Mode: Since highest frequency is 36.

$\therefore$  Modal class = 30 - 40

$$\text{We know that, Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} i$$

$$\text{Here, } L = 30, \Delta_1 = 36 - 24 = 12, \Delta_2 = 36 - 20 = 16, i = 10$$

$$\text{Mode} = 30 + \frac{12}{12 + 16} \times 10$$

$$\text{Mode} = 30 + 4.29$$

$$\text{Mode} = 34.29$$

**Mean=37.88, Median=36.25 & Mode=34.29 [Ans.]**

### **Problem-3( Exercise 27-Page-130)**

Calculate mean, median and mode for the following data pertaining to marks in statistics out of 140 marks for 80 students in a class:

|                 |    |    |    |    |    |     |     |
|-----------------|----|----|----|----|----|-----|-----|
| Marks more than | 0  | 20 | 40 | 60 | 80 | 100 | 120 |
| No. of students | 80 | 76 | 50 | 28 | 18 | 9   | 3   |

Calculation of mean, median and mode

| Marks   | No. of students<br>(f) | Mid-Point<br>X | fX               | Cumulative<br>Frequency(c.f) |
|---------|------------------------|----------------|------------------|------------------------------|
| 0-20    | 4                      | 10             | 40               | 4                            |
| 20-40   | 26                     | 30             | 780              | 30                           |
| 40-60   | 22                     | 50             | 1100             | 52                           |
| 60-80   | 10                     | 70             | 700              | 62                           |
| 80-100  | 9                      | 90             | 810              | 71                           |
| 100-120 | 6                      | 110            | 660              | 77                           |
| 120-140 | 3                      | 130            | 390              | 80                           |
| Total   | $N = \sum f = 80$      |                | $\sum fX = 4480$ |                              |

Mean : Mean,  $\bar{X} = \frac{\sum fX}{N}$

$$\bar{X} = \frac{4480}{80}$$

Mean,  $\bar{X} = 56$

Median : we know that, Median = The size of the  $\frac{N}{2}$ th item  
 = The size of the  $\frac{80}{2} = 40$ th item

$\therefore$  Median lies in the class = 40 – 60

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} i$$

Here,  $L = 40$ ,  $f = 22$ ,  $p.c.f = 30$ ,  $i = 20$

$$\text{Median} = 40 + \frac{\frac{80}{2} - 30}{22} \times 20$$

$$\text{Median} = 40 + 9.09$$

$$\therefore \text{Median} = 49.09$$

### **Mode:**

Since highest frequency is 26. So modal class = 20-40

$$\text{We know that, Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i$$

Here,  $L = 20$ ,  $\Delta_1 = 26 - 4 = 22$ ,  $\Delta_2 = 26 - 22 = 4$ ,  $i = 20$

$$\Rightarrow \text{Mode} = 20 + \frac{22}{22 + 4} \times 20$$

$$\Rightarrow \text{Mode} = 20 + \frac{440}{26}$$

$$\Rightarrow \text{Mode} = 20 + 16.92$$

$$\therefore \text{Mode} = 36.92 [\text{Ans.}]$$

### **Problem 8**

BBA (Honours) 2006; BBA (Honours) 2010

The expenditure of 1000 families is given as under:

| Expenditure (Taka in thousand) | 40 – 59 | 60 – 79 | 80 – 99 | 100 – 119 | 120 – 139 |
|--------------------------------|---------|---------|---------|-----------|-----------|
| No. of families                | 50      | ?       | 500     | ?         | 50        |

Median and mean of the distribution are Tk 87.50 each. Calculate the missing frequencies  $f_1$  and  $f_2$ .

**Solution:**

**Calculation of Missing Frequencies**

| <b>Output (in units)</b> | <b>f</b>        | <b>cf</b>         |
|--------------------------|-----------------|-------------------|
| 40 – 59                  | 50              | 50                |
| 60 – 79                  | $f_1$           | $50 + f_1$        |
| 80 – 99                  | 500             | $550 + f_1$       |
| 100 – 119                | $f_2$           | $550 + f_1 + f_2$ |
| 120 – 139                | 50              | $600 + f_1 + f_2$ |
|                          | <b>N = 1000</b> |                   |

Median =87.50 which lies in the class 80-99. But the real class limit is 79.5-99.5.

$$\therefore \text{Median} = L + \frac{\frac{N}{2} - p.c.f.}{f} \times c$$

$$\Rightarrow 87.50 = 79.5 + \frac{500 - (50 + f_1)}{500} \times 20$$

$$\Rightarrow 87.50 - 79.5 =$$

$$\Rightarrow 8 = \frac{450 - f_1}{25}$$

$$\Rightarrow 8 \times 25 = 450 - f_1$$

$$\Rightarrow 200 = 450 - f_1$$

$$\Rightarrow f_1 = 450 - 200$$

$$\Rightarrow f_1 = 250$$

Again,  $600 + f_1 + f_2 = N$

$$\text{Or, } 600 + 250 + f_2 = 1000$$

$$\text{Or, } 850 + f_2 = 1000$$

$$\therefore f_2 = 150$$

$$\therefore f_1 = 250, f_2 = 150 \quad [\text{Ans.}]$$

**THANK YOU**

## Number System

**Natural numbers:** The numbers 1, 2, 3, 4, ----- which are used for counting, are natural numbers.

**Integer:** The integers are whole numbers positive, negative or zero.

Example: -1, -2, 0, 2, 3, 4, etc.

**Prime numbers:** The numbers greater than 1 and which have two common factor 1 and the number itself is called Prime number.

Example: 2, 3, 5 etc.

**Rational number:** The number which can be expressed in the form  $\frac{p}{q}$  where p is any integer and q is an integer not equal to zero.

Example :  $-5, 3, -\frac{2}{3}, \frac{3}{5}$  etc.

**Irrational number:** The number which cannot be expressed in the form  $\frac{p}{q}$  where p is any integer and q is an integer not equal to zero.

Example :  $\sqrt{2}, \sqrt{3}, \sqrt{5}$  etc.

**Real numbers:** A number that is rational or irrational number is called real number.

Example :  $-2, -\frac{2}{3}, -3, 2, 3, 4, \frac{3}{5}, \sqrt{2}, \sqrt{3}$  etc.

**Imaginary number:** Square root of negative numbers are called imaginary number.

Example :  $\sqrt{-2}, \sqrt{-3}, 2i, 3i$  etc.

**Complex number:** If A and B are real numbers then  $A+iB$  is known as a complex number which has 'A' the real part and 'B' the imaginary part.

Example:  $2+3i, -5+\frac{2}{3}i$  etc.



**Composite numbers:** Composite numbers are whole numbers that can be divided by numbers other than itself and 1.

Example: 4, 6, 8, 10, 12 etc.

**Even number:** The number which is divisible by 2 is called even number.

Example: 2, 4, 6, 8 etc.

**Odd number:** The number which cannot be divisible by 2 is called odd number.

Example: 1, 3, 5, etc.

Number System:

# Prove that  $\sqrt{2}/\sqrt{3}/\sqrt{5}/\sqrt{7}$  is an irrational number (Page - 24)

Let  $\sqrt{3}$  be a rational number so that  $\sqrt{3} = \frac{p}{q}$ , where  $q \neq 0$  and  $p, q$  are two integers with no common factor except 1.

$$\therefore \sqrt{3} = \frac{p}{q}$$

$$3 = \frac{p^2}{q^2}$$

$$\frac{2}{3}$$

$$\frac{4 \times 2}{4 \times 3}$$

$$\Rightarrow 3q^2 = p^2 \text{ [Taking square on both sides]}$$

Since  $3q^2$  is divisible by 3, so  $p^2$  and  $p$  both are divisible by 3.

Again, let  $p = 3m$

$$\Rightarrow p^2 = 9m^2 \text{ [Taking square on both sides]}$$

$$\Rightarrow 3q^2 = 9m^2 \text{ [Since } p^2 = 3q^2]$$

$$\Rightarrow q^2 = \underline{3m^2}$$

$$\Rightarrow 3m^2 = \underline{q^2}$$

Since  $3m^2$  is divisible by 3, so  $q^2$  and  $q$  both are divisible by 3. Here  $p$  and  $q$  are divisible by 3.

Thus  $p$  and  $q$  have at least one common factor 3. This is a contradiction on our assumption that

$\sqrt{3}$  is a rational number.

Hence  $\sqrt{3}$  is an irrational number (Proved)

Demo2 (Page-26) Simplify and show if  $\frac{3+2i}{2-5i} + \frac{3-2i}{2+5i}$  is a rational number

*Solution:* Given  $= \frac{3+2i}{2-5i} + \frac{3-2i}{2+5i}$

$$= \frac{(2+5i)(3+2i) + (2-5i)(3-2i)}{(2-5i)(2+5i)}$$

$$= \frac{6 + \cancel{4i} + \cancel{15i} + 10i^2 + 6 - \cancel{4i} - \cancel{15i} + 10i^2}{(2)^2 - (5i)^2}$$

$$= \frac{12 + 20i^2}{4 - 25i^2}$$

$$= \frac{12 + 20(-1)}{4 - 25(-1)} \text{ [Since } i^2 = -1 \text{]}$$

$$= \frac{12 - 20}{4 + 25}$$

$$= -\frac{8}{29}$$

So,  $\frac{3+2i}{2-5i} + \frac{3-2i}{2+5i}$  is a rational number (Shown)

$$\frac{2}{3} + \frac{3}{4} = \frac{12}{12}$$

$$-\frac{4}{5} - \frac{4}{5} = -\frac{8}{5}$$

2x3

Problem 24 (Page-65):

Express the following in the

form of  $A+iB$ :  $\frac{3+2i}{2+3i} + \frac{1+5i}{1-2i}$

$$\begin{aligned}
 \text{Solution : Given} &= \frac{3+2i}{2+3i} + \frac{1+5i}{1-2i} \\
 &= \frac{(3+2i)(1-2i) + (1+5i)(2+3i)}{(2+3i)(1-2i)} \quad \checkmark \\
 &= \frac{3-6i+2i-4i^2 + 2+3i+10i+15i^2}{2-4i+3i-6i^2} \\
 &= \frac{5+9i+11i^2}{2-i-6i^2} \\
 &= \frac{5+9i+11(-1)}{2-i-6(-1)} \quad [\text{Since } i^2 = -1] \\
 &= \frac{5+9i-11}{2-i+6} \\
 &= \frac{9i-6}{8-i} \quad | \\
 &= \frac{(9i-6)(8+i)}{(8-i)(8+i)} \\
 &= \frac{72i+9i^2-48-6i}{8^2-i^2} \\
 &= \frac{72i+9(-1)-48-6i}{64-(-1)} \quad [\text{Since } i^2 = -1] \\
 &= \frac{72i-9-48-6i}{64+1} \\
 &= \frac{-57+66i}{65} \quad \checkmark \\
 &= -\frac{57}{65} + i\frac{66}{65}, \text{ Which is the form of } A+iB \text{ [Ans.]}
 \end{aligned}$$

$$\begin{array}{r}
 2 \times 2 \\
 \hline
 3 \times 2
 \end{array}$$

**THANK YOU**



1. Prove that for two non-zero positive numbers,

$$A.M \geq G.M \geq H.M$$

Where A.M = Arithmetic Mean, G.M = Geometric Mean and H.M = Harmonic Mean

When does equality hold?

*Proof* : Suppose  $x_1$  &  $x_2$  are two positive & non-zero quantities. Then by definition,

$$A.M = \frac{x_1 + x_2}{2} \text{ ----- (i)}$$

$$G.M = \sqrt{x_1 x_2} \text{ ----- (ii)}$$

$$H.M = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}$$

$$H.M = \frac{2}{\frac{x_1 + x_2}{x_1 x_2}}$$

$$\therefore H.M = \frac{2x_1 x_2}{x_1 + x_2} \text{ ----- (iii)}$$

Let us take the expression,

$(\sqrt{x_1} - \sqrt{x_2})^2$ , which is always positive.

$$\text{i.e. } (\sqrt{x_1} - \sqrt{x_2})^2 \geq 0$$

$$\text{or, } x_1 + x_2 - 2\sqrt{x_1 x_2} \geq 0$$

$$\text{or, } x_1 + x_2 \geq 2\sqrt{x_1 x_2} \text{ ----- (iv)}$$

$$\text{or, } \frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2} \text{ [Dividing both sides by 2]}$$

$$A.M \geq G.M \text{ [From (i) \& (ii)] ----- (v)}$$

Again, from (iv),

$$x_1 + x_2 \geq 2\sqrt{x_1 x_2}$$

$$\text{or, } (x_1 + x_2) \frac{\sqrt{x_1 x_2}}{(x_1 + x_2)} \geq 2\sqrt{x_1 x_2} \frac{\sqrt{x_1 x_2}}{x_1 + x_2} \text{ [Multiplying both sides by } \frac{\sqrt{x_1 x_2}}{x_1 + x_2}]$$

$$\text{or, } \sqrt{x_1 x_2} \geq \frac{2x_1 x_2}{x_1 + x_2}$$

$$\therefore G.M \geq H.M \text{ [From (ii) \& (iii)] ----- (vi)}$$

From (v) & (vi), we have

$$A.M \geq G.M \geq H.M$$

Equality hold, if  $x_1 = x_2$

Hence the proof.

2. Prove that for two non - zero & positive quantities

$$G.M = \sqrt{A.M \times H.M}$$

Proof : Suppose  $x_1$  &  $x_2$  are two positive and non - zero quantities. Then by definition,

$$A.M = \frac{x_1 + x_2}{2} \text{----- (i)}$$

$$G.M = \sqrt{x_1 x_2} \text{----- (ii)}$$

$$H.M = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}$$

$$\text{Or, } H.M = \frac{2}{\frac{x_1 + x_2}{x_1 x_2}}$$

$$\text{Or, } H.M = \frac{2x_1 x_2}{x_1 + x_2} \text{----- (iii)}$$

Multiplying equation (i) & (iii), we get,

$$A.M \times H.M = \frac{(x_1 + x_2)}{2} \frac{2x_1 x_2}{(x_1 + x_2)}$$

$$\text{Or, } A.M \times H.M = x_1 x_2$$

$$\text{Or, } A.M \times H.M = (\sqrt{x_1 x_2})^2$$

$$\text{Or, } (\sqrt{x_1 x_2})^2 = A.M \times H.M$$

$$\text{Or, } G.M^2 = A.M \times H.M$$

$$\therefore G.M = \sqrt{A.M \times H.M} \text{ [ Taking square root on both sides]}$$

Hence the proof.

3. Prove that Arithmetic Mean depends on change of origin and scale.

*Proof :* Let the variable  $x$  having  $n$  values where  $x$  takes  $x_1, x_2, x_3, \dots, x_n$ . Then by definition,

$$\text{Arithmetic mean, } \bar{x} = \frac{\sum_{i=1}^n x_i}{n} \text{----- (i)}$$

Let us take new variable  $d_i$  where  $d_i = \frac{x_i - A}{C}$  [Here,  $A$  = origin and  $C$  = Scale ]

$$\Rightarrow x_i - A = C d_i$$

$$\Rightarrow x_i = A + C d_i$$

$$\Rightarrow \sum_{i=1}^n x_i = \sum_{i=1}^n (A + C d_i) \text{ [Taking Summation on both sides]}$$

$$\Rightarrow \sum_{i=1}^n x_i = nA + C \sum_{i=1}^n d_i$$

$$\Rightarrow \frac{\sum_{i=1}^n x_i}{n} = \frac{nA}{n} + C \frac{\sum_{i=1}^n d_i}{n} \text{ [Dividing both sides by } n]$$

$$\Rightarrow \bar{x} = A + C \bar{d} \text{ [Since } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \text{ so } \bar{d} = \frac{\sum_{i=1}^n d_i}{n} ]$$

So, Arithmetic Mean depends on change of origin and scale. (Proved)

4. Prove that If  $\bar{X}_1$  and  $\bar{X}_2$  are the means having sizes  $N_1$  and  $N_2$  respectively. Then

$$\text{Combined mean, } \bar{X}_{12} = \frac{N_1 \bar{X}_1 + N_2 \bar{X}_2}{N_1 + N_2}$$

Proof : Let,  $X_{11}, X_{12}, \dots, X_{1N_1}$  be the set of  $N_1$  observations of 1st group and  $X_{21}, X_{22}, \dots, X_{2N_2}$  be the set of  $N_2$  observations of 2nd group. Then by definition,

$$\begin{aligned} \text{Arithmetic Mean, } \bar{X}_1 &= \frac{\sum_{i=1}^{N_1} X_{1i}}{N_1} \\ \Rightarrow \sum_{i=1}^{N_1} X_{1i} &= N_1 \bar{X}_1 \text{----- (i)} \end{aligned}$$

$$\begin{aligned} \text{And } \bar{X}_2 &= \frac{\sum_{i=1}^{N_2} X_{2i}}{N_2} \\ \Rightarrow \sum_{i=1}^{N_2} X_{2i} &= N_2 \bar{X}_2 \text{----- (ii)} \end{aligned}$$

$$\text{Then Combined mean, } \bar{X}_{12} = \frac{(X_{11} + X_{12} + \dots + X_{1N_1}) + (X_{21} + X_{22} + \dots + X_{2N_2})}{N_1 + N_2}$$

$$\Rightarrow \bar{X}_{12} = \frac{\sum_{i=1}^{N_1} X_{1i} + \sum_{i=1}^{N_2} X_{2i}}{N_1 + N_2}$$

$$\Rightarrow \bar{X}_{12} = \frac{N_1 \bar{X}_1 + N_2 \bar{X}_2}{N_1 + N_2} \text{ [From (i) \& (ii)]}$$

Hence the Proof.



5. Prove that mean of first  $n$  natural numbers is  $\frac{n+1}{2}$

Proof : We know that first  $n$  natural numbers are  $1, 2, 3, \dots, n$

Let  $X_1, X_2, \dots, X_n$  be the first  $n$  natural numbers

where  $X_1 = 1, X_2 = 2, \dots, X_n = n$ . Then by definition,

$$\begin{aligned} \text{Mean, } \bar{X} &= \frac{\sum_{i=1}^n X_i}{n} \\ \Rightarrow \bar{X} &= \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \\ \Rightarrow \bar{X} &= \frac{1 + 2 + 3 + \dots + n}{n} \\ \Rightarrow \bar{X} &= \frac{\frac{n(n+1)}{2}}{n} \\ \therefore \bar{X} &= \frac{n+1}{2} \text{ [Proved]} \end{aligned}$$

6. If  $U = X + Y$  then prove that  $\bar{U} = \bar{X} + \bar{Y}$

Proof : We have,  $U = X + Y$

So,  $U_i = X_i + Y_i$

$$\sum U_i = \sum (X_i + Y_i) \text{ [Taking summation on both sides]}$$

$$\Rightarrow \frac{\sum U_i}{N} = \frac{\sum X_i}{N} + \frac{\sum Y_i}{N} \text{ [Dividing both sides by N]}$$

$$\therefore \bar{U} = \bar{X} + \bar{Y} \text{ [Proved]}$$

7. If  $G_1$  &  $G_2$  are the geometric means of two sets of observations, then we have to prove that combined geometric mean is

$$\text{Log } G = \frac{n_1 \text{Log } G_1 + n_2 \text{Log } G_2}{n_1 + n_2}$$

Where,  $n_1$  = Total number of observations of group 1

$n_2$  = Total number of observations of group 2

Proof: Let the product of the  $n_1$  observations in the first group be denoted by  $X_1$  & the product of the  $n_2$  observations in the 2nd group be denoted by  $X_2$ .

Then by definition,  $G_1 = (X_1)^{\frac{1}{n_1}} \text{-----}(i)$

$$G_2 = (X_2)^{\frac{1}{n_2}} \text{-----}(ii)$$

The combined set now consists of the observations  $X_1$  &  $X_2$ , the number of observations,  
 $n_1 + n_2 = N$  (Say)

So, geometric mean  $G$  of the combined set is

$$G = (X_1 \cdot X_2)^{\frac{1}{N}}$$

$$\Rightarrow \text{Log } G = \text{Log } (X_1 \cdot X_2)^{\frac{1}{N}}$$

$$\Rightarrow \text{Log } G = \frac{1}{N} \text{Log } (X_1 \cdot X_2)$$

$$\therefore \text{Log } G = \frac{\text{Log } X_1 + \text{Log } X_2}{n_1 + n_2} [\text{Since } \text{Log}(a \cdot b) = \text{Log } a + \text{Log } b \text{ \& } N = n_1 + n_2] (\text{Proved})$$

**Thank You**

# Measures of Dispersion/Variation

## Dispersion:

The word dispersion has a technical meaning in statistics. The average measures the center of the data. It is one aspect observations. Another feature of the observations is as to how the observations are spread about the center. The observation may be close to the center or they may be spread away from the center. If the observation are close to the center (usually the arithmetic mean or median), we say that dispersion, scatter or variation is small. If the observations are spread away from the center, we say dispersion is large.

### Definition of dispersion:

Some important definitions of dispersion are given below:

- (i) "Dispersion or spread is the degree of the scatter or variation of the variable about a central value".-Brooks and Dick
- (ii) "Dispersion is the measure of the variations of the item."- A.L Bowley
- (iii) " The degree to which numerical data tend to spread about an average value is called the variation or dispersion of the data."- Spiegel.
- (iv) "The measurement of the scatterness of the mass of figures in a series about an
- (v) Average is called measure of variation or dispersion." -Simpson & Kafka.

Suppose we have two groups of students who have obtained the following marks in a test. The arithmetic means of the two groups are also given below:

**Group A:** 46, 48, 50, 52, 54

$$\overline{X}_A = 50$$

**Group B:** 30, 40, 50, 60, 70

$$\overline{X}_B = 50$$

In a group A and B arithmetic means are equal i.e.  $\overline{X}_A = \overline{X}_B = 50$  But in group A the observations are concentrated on the center. All students of group A have almost the same level of performance. We say that there is consistence in the observations in group A. In group B the mean is 50 but the observations are not closed to the center. One observation is as small as 30 and one observation is as large as 70. Thus there is greater dispersion in group B.

### **Objectives of Measuring Dispersion:**

Measures of variations are calculated to serve the following purposes:

1. To judge the reliability of measures of central tendency.
2. To compare two or more series with regard to their variability.
3. To control the variability itself.
4. To facilitate the use of other statistical measure.

### **Characteristics of a good measure of dispersion:**

An ideal measure of dispersion is expected to possess the following properties

1. It should be rigidly defined
2. It should be based on all the items.
3. It should not be unduly affected by extreme items
4. It should lend itself for algebraic manipulation.
5. It should be simple to understand and easy to calculate

### **Measures of Dispersion**

For the study of dispersion, we need some measures which show whether the dispersion is small or large. There are two types of measure of dispersion which are:

**(a)** Absolute Measure of Dispersion

**(b)** Relative Measure of Dispersion

#### **Absolute Measures of Dispersion:**

These measures give us an idea about the amount of dispersion in a set of observations. They give the answers in the same units as the units of the original observations. When the observations are in kilograms, the absolute measure is also in kilograms. If we have two sets of observations, we cannot always use the absolute measures to compare their dispersion. We shall explain later as to when the absolute measures can be used for comparison of dispersion in two or more than two sets of data. The absolute measures which are commonly used are:

1. The Range
2. The Quartile Deviation
3. The Mean Deviation
4. The Standard deviation and Variance

#### **Relative Measure of Dispersion:**

These measures are calculated for the comparison of dispersion in two or more than two sets of observations. These measures are free of the units in which the original data is measured. If the original data is in dollar or kilometers, we do not use these units with relative measure of dispersion. These measures are a sort of ratio and are called coefficients. Each absolute measure of dispersion can be converted into its relative measure. Thus the relative measures of dispersion are:

1. Coefficient of Range or Coefficient of Dispersion.
2. Coefficient of Quartile Deviation or Quartile Coefficient of Dispersion.
3. Coefficient of Mean Deviation or Mean Deviation of Dispersion.
4. Coefficient of Variation

## **Distinguish between absolute measure and relative measure**

The Absolute Measure of dispersion is basically the measure of variation from the mean such as standard deviation. On the other hand the relative measure of dispersion is basically the position of a certain variable with reference to or as compared with the other variables such as Coefficient of Variation. Absolute measures of Dispersion are expressed in same units in which original data is presented but these measures cannot be used to compare the variations between the two series. Relative measures are not expressed in units but it is a pure number. It is the ratios of absolute dispersion to an appropriate average such as co-efficient of Standard Deviation or Co-efficient of Mean Deviation. Absolute measure does not expressed in percentage relative measure is usually expressed in percentage.

### **1. Range and Coefficient of Range:**

Range is defined as the difference between the maximum and the minimum observation of the given data. If  $X_m$  denotes the maximum observation  $X_0$  denotes the minimum observation then the range is defined as

$$\text{Range} = X_m - X_0$$

#### **Merits and Demerits of Range:**

##### **Merits:**

1. It is simple to understand.
2. It is easy to calculate.
3. In certain types of problems like quality control, weather forecasts, share price analysis, etc., range is most widely used.

##### **Demerits:**

1. It is very much affected by the extreme items.
2. It is based on only two extreme observations.
3. It cannot be calculated from open-end class intervals.
4. It is not suitable for mathematical treatment.
5. It is a very rarely used measure.

#### **Coefficient of Range:**

It is relative measure of dispersion and is based on the value of range. It is also called range coefficient of dispersion. It is defined as:

$$\text{Coefficient of range} = \frac{X_m - X_0}{X_m + X_0} \times 100$$

The range  $X_m - X_0$  is standardized by the total  $X_m + X_0$

### **2. Quartile Deviation and its Coefficient Quartile Deviation:**

It is based on the lower quartile  $Q_1$  and the upper quartile  $Q_3$ . The difference  $Q_3 - Q_1$  is called the inter quartile range. The difference  $Q_3 - Q_1$  divided by 2 is called semi-inter-quartile range or the quartile deviation.

$$\text{Quartile Deviation (Q.D)} = \frac{Q_3 - Q_1}{2}$$

The quartile deviation is a slightly better measure of absolute dispersion than the range. But it ignores the observation on the tails. If we take difference samples from a population and calculate their quartile deviations, their values are quite likely to be sufficiently different. This is called sampling fluctuation. It is not a popular measure of dispersion. The quartile deviation calculated from the sample data does not help us to draw any conclusion (inference) about the quartile deviation in the population.

### **Merits and Demerits of Quartile Deviation**

#### **Merits:**

1. It is Simple to understand and easy to calculate
2. It is not affected by extreme values.
3. It can be calculated for data with open end classes also.

#### **Demerits:**

1. It is not based on all the items. It is based on two Positional values  $Q_1$  and  $Q_3$  and ignores the extreme 50% of the items
2. It is not amenable to further mathematical treatment.
3. It is affected by sampling fluctuations.

### **Coefficient of Quartile Deviation:**

A relative measure of dispersion based on the quartile deviation is called the coefficient of quartile deviation. It is defined as

$$\text{Coefficient of Quartile Deviation} = \frac{\frac{Q_3 - Q_1}{2}}{\frac{Q_3 + Q_1}{2}} \times 100 = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

It is pure number free of any units of measurement. It can be used for comparing the dispersion in two or more than two sets of data.

### **3. Mean Deviation and its Coefficient:**

- The mean deviation or the average deviation is defined as the mean of the absolute deviations of observations from some suitable average which may be the arithmetic

mean, the median or the mode. The difference (x-average) is called deviation and when we ignore the negative sign, this deviation is written as  $|X - average|$  and is read as mod deviations. The mean of these mod or absolute deviations is called the mean deviation or the mean absolute deviation. Thus for sample data in which the suitable average is the  $\bar{X}$ , the mean deviation(M.D) is given by the relation:

$$M.D = \frac{\sum |X - \bar{X}|}{n} \text{ [Without frequency]}$$

For frequency distribution, the mean deviation is given by

$$M.D = \frac{\sum f |X - \bar{X}|}{\sum f}$$

$$M.D = \frac{\sum f |X - \bar{X}|}{N} [\because \sum f = N]$$

When the mean deviation is calculated about the median, the formula becomes

$$M.D(\text{about median}) = \frac{\sum f |X - \text{Median}|}{N}$$

The mean deviation about the mode is

$$M.D(\text{about mode}) = \frac{\sum f |X - \text{Mode}|}{N}$$

The mean deviation is a better measure of absolute dispersion than the range and the quartile deviation. A drawback in the mean deviation is that we use the absolute deviations which do not seem logical. The reason for this is that it is always equal to zero. Even if we use median or mode in place of  $\bar{X}$ , even then the summation will be zero or approximately zero with the result that the mean deviation would always be either zero or close to zero. Thus the very definition of the mean deviation is possible only on the absolute deviations.

## **Merits and Demerits of M.D:**

### **Merits:**

1. It is simple to understand and easy to compute.
2. It is rigidly defined.
3. It is based on all items of the series.
4. It is not much affected by the fluctuations of sampling.
5. It is less affected by the extreme items.
6. It is flexible, because it can be calculated from any average.

7. It is better measure of comparison.

**Demerits:**

1. It is not a very accurate measure of dispersion.
2. It is not suitable for further mathematical calculation.
3. It is rarely used. It is not as popular as standard deviation.
4. Algebraic positive and negative signs are ignored. It is mathematically unsound and illogical.

**Coefficient of the Mean Deviation:**

A relative measure of dispersion based on the mean deviation is called the coefficient of the mean deviation or the coefficient of dispersion. It is defined as the ratio of the mean deviation to the average used in the calculation of the mean deviation. Thus

$$\text{Coefficient of } M.D(\text{about mean}) = \frac{\text{Mean Deviation from Mean}}{\text{Mean}} \times 100$$

$$\text{Coefficient of } M.D(\text{about median})$$

$$= \frac{\text{Mean Deviation from Median}}{\text{Median}} \times 100$$

$$\text{Coefficient of } M.D(\text{about Mode}) = \frac{\text{Mean Deviation from Mode}}{\text{Mode}} \times 100$$

**4. Standard Deviation**

The standard deviation is a measure of the spread of scores within a set of data. Usually, we are interested in the standard deviation of a population. However, as we are often presented with data from a sample only, we can estimate the population standard deviation from a sample standard deviation. These two standard deviations - sample and population standard deviations - are calculated differently. In statistics, we are usually presented with having to calculate sample standard deviations although the formula for a population standard deviation will also be shown.

**When to use the sample or population standard deviation**

We are normally interested in knowing the population standard deviation because our population contains all the values we are interested in. Therefore, you would normally calculate the population standard deviation if: (1) you have the entire population or (2) you have a sample of a larger population, but you are only interested in this sample and do not wish to generalize your findings to the population. However, in statistics, we are usually presented with a sample from which we wish to estimate (generalize to) a population, and the standard deviation is no exception to this. Therefore, if all you have is a sample, but you wish to make a statement about the population standard deviation from which the sample is drawn, you need to use the sample standard deviation. Confusion can often arise as to which standard deviation to use due to the name "sample" standard deviation incorrectly being interpreted as meaning the standard deviation of the sample itself and not the estimate of the population standard deviation based on the sample.



### **What type of data should you use when you calculate a standard deviation?**

The standard deviation is used in conjunction with the mean to summarize continuous data, not categorical data. In addition, the standard deviation, like the mean, is normally only appropriate when the continuous data is not significantly skewed or has outliers.

#### **Definition of Standard Deviation:**

The standard deviation is defined as **the positive square root of the mean of the square deviations taken from arithmetic mean of the data.**

For the sample data the standard deviation is denoted by  $s$  and is defined as:

$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n-1}} \text{ [Without frequency]}$$

For frequency distribution the formula becomes

$$s = \sqrt{\frac{\Sigma f(X - \bar{X})^2}{n-1}}$$

For a population data the standard deviation is denoted by  $\sigma$  (sigma) and is defined as:

$$\sigma = \sqrt{\frac{\Sigma(X - \mu)^2}{N}}$$

$$\sigma = \sqrt{\frac{\Sigma f(X - \mu)^2}{N}} \text{ [With frequency]}$$

The standard deviation is in the same units as the units of the original observations. If the original observations are in grams, the value of the standard deviation will also be in grams.

#### **Merits and Demerits of Standard Deviation:**

##### **Merits:**

1. It is rigidly defined and its value is always definite and based on all the observations and the actual signs of deviations are used.
2. As it is based on arithmetic mean, it has all the merits of arithmetic mean.
3. It is the most important and widely used measure of dispersion.
4. It is possible for further algebraic treatment.
5. It is less affected by the fluctuations of sampling and hence stable.
6. It is the basis for measuring the coefficient of correlation and sampling.

**Demerits:**

1. It is not easy to understand and it is difficult to calculate.
2. It gives more weight to extreme values because the values are squared up.
3. As it is an absolute measure of variability, it cannot be used for the purpose of comparison.

**Coefficient of Variation:**

The most important of all the relative measure of dispersion is the coefficient of variation. This word is variation not variance. There is no such thing as coefficient of variance. The coefficient of variation c.v is defined as:

$$\text{Coefficient of Variation, } C.V = \frac{\sigma}{\bar{X}} \times 100$$

Thus C.V is the value of  $\sigma$  when  $\bar{X}$  is assumed equal to 100. It is a pure number and the unit of observations is not mentioned with its value. It is written in percentage form like 20% or 25%. When its value is 20%, it means that when the mean of the observations is assumed equal to 100, their standard deviation will be 20. The C.V is used to compare the dispersion in different sets of data particularly the data which differ in their means or differ in the units of measurement. The wages of workers may be in dollars and the consumption of meat in their families may be in kilograms. The standard deviation of wages in dollars cannot be compared with the standard deviation of amounts of meat in kilograms. Both the standard deviations need to be converted into coefficient of variation for comparison. Suppose the value of C.V for wages is 10% and the values of C.V for kilograms of meat is 25%. This means that the wages of workers are consistent.

**Properties of Standard deviation:****1. Combined Variance**

Like combined mean, the combined variance or standard deviation can be calculated for different sets of data. Suppose we have two sets of data containing  $N_1$  and  $N_2$  observations with means  $\bar{X}_1$  and  $\bar{X}_2$ , and variances  $\sigma_1^2$  and  $\sigma_2^2$ . If  $\bar{X}_{12}$  is the combined mean and  $\sigma_{12}^2$  is the combined variance of  $N_1 + N_2$  observations, then combined variance is given by,

$$\sigma_{12}^2 = \frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1d_1^2 + N_2d_2^2}{N_1 + N_2}$$

$$\text{Where } d_1^2 = (\bar{X}_1 - \bar{X}_{12})^2 \text{ \& } d_2^2 = (\bar{X}_2 - \bar{X}_{12})^2$$

$$\text{And } \bar{X}_{12} = \frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1 + N_2}$$

The combine standard deviation  $\sigma_{12}$  can be calculated by taking the square root of  $\sigma_{12}^2$

2. Standard deviation is independent of change of origin but not of scale.
3. Standard deviation is the minimum root-mean-square deviation,

$$\text{i.e., } \sqrt{\frac{1}{N} \sum (X - \bar{X})^2} \leq \sqrt{\frac{1}{N} \sum (X - A)^2}$$

Where, A= Arbitrary value

4. For a given set of observations, standard deviation (S.D) is never less than mean deviation (M.D),  
i.e.,  $S.D \geq M.D$

### Various Methods of Measuring Dispersion:

The various measures of dispersion are as follows:

#### (1) Absolute Measures:

- a) The Range.
- b) The Mean Deviation.
- c) The Standard Deviation.
- d) The Quartile Deviation.
- e) The Lorenz Curve.

#### (2) Relative Measures:

- a) The Coefficient of Variation.
- b) The Coefficient of Mean Deviation
- c) The Coefficient of Quartile Deviation.

They are briefly discussed below:

#### (i) Absolute Measures :

- a) **The Range:** The range is the absolute difference between the largest and smallest values in a set of data. It is very rough and elementary measure of variation.
- b) **The Mean Deviation:** The mean deviation is the mean difference of the items in a set of data from their average. It is simply an average of absolute deviations of individual items from the central value of a series.
- c) **The Standard Deviation:** The arithmetic mean of the squares of the deviations of the given observations from their arithmetic mean is known as variance. The positive square root of variance is the standard deviation. It measures the absolute variation of a distribution.
- d) **The Quartile Deviation:** The quartile deviation is the half of the difference between the first quartile and third quartile. This is also called semi- interquartile range. It is notable that quartiles divide the observations into 4 equal parts when the observations are arranged in order of magnitude.
- e) **The Lorenz Curve:** The Lorenz curve is a cumulative percentage curve in which the percentage of items is combined with the percentage of other things as wealth, profits, turnover etc.

#### (ii) Relative Measures of Dispersion:

- a) **Coefficient of variation:** The Coefficient of variation is defined by

$$CV = \frac{\sigma}{x} \times 100$$

Where  $\sigma$  is the standard deviation and  $\bar{X}$  is the mean.

- b) **Coefficient of Mean Deviation:** The coefficient of mean deviation is the ratio of the mean deviation measured from certain measure of central location to the corresponding measure of central location and, expressed as percentage Symbolically,

$$(i) \quad C.M.D (\bar{X}) = \frac{M.D.(\bar{X})}{\bar{X}} \times 100$$

$$(ii) \quad C.M.D. (Med) = \frac{M.D.(Med)}{Med} \times 100$$

$$(iii) \quad C.M.D. (M_0) = \frac{M.D.}{M_0} \times 100$$

- c) **Coefficient of Quartile Deviation:** The coefficient of quartile deviation is defined by

$$\text{Coefficient of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

Where  $Q_3$  is the third quartile and  $Q_1$  is the first quartile:

### **Which of these I like best?**

If observations are few in numbers, or certain extreme values, then the standard deviation should be avoided. If they have gaps around the quartiles, then the quartile deviation should be avoided. Despite, I like the standard deviation. Because, practically all advanced statistical methods deal with variability, and center around the standard deviation.

### **Comparison between absolute and relative measures of dispersion:**

The measures of absolute dispersion are quite suitable for measuring the scatter of the items around the average for a particular distribution but are not suitable for comparison between two or more distributions in terms of their variability. Absolute measures of dispersion give us the variability of the data in terms of absolute units of measurement. If two given distributions have got two different units of measurement their variability cannot be compared in terms of absolute units. The relative measures of dispersion are the adaptation of the measures of absolute dispersion permitting comparison among different distributions. It is notable that the most important and commonly used relative measure of dispersion has been given by Karl Pearson which is known as the coefficient of variation.

### **The Requisites of a Good Measure of Dispersion:**

A good measure of dispersion should possess, as far as possible, the following properties:

- i. It should be simple to understand.
- ii. It should be easy to compute.
- iii. It should be rigidly defined.
- iv. It should be based on each and every observation of the distribution.
- v. It should be amenable to further algebraic treatment.
- vi. It should have sampling stability.
- vii. It should be not be unduly affected by extreme observations.

### **Preference for standard deviation:**

Standard deviation is used as the measure of variation. Coefficient of variation is more useful when the two distributions are entirely different, and the units of measurement are also different. When the relative variation is started in terms of the arithmetic mean, and the standard deviation, the resulting percentage is known as the coefficient of variation or coefficient of variability. Hence coefficient of variation is preferred to standard deviation.

### **Needs/importance to study of Dispersion.**

The study of measures of dispersion is needed for the following four basic purposes:

- i. To determine the reliability of an average.
- ii. To serve as a basis for the control of variability.
- iii. To compare two or more series with regard to their variability and
- iv. To facilitate the use of other statistical measures.

### **Distinction between Variance and Coefficient of Variance:**

The arithmetic mean of the squares of the deviations of the given observation from their arithmetic mean is known as variance. Any measure of variance can be used in conjunction with any average in computing relation variance. Coefficient of variance is more useful when the two distributions are entirely different and the units of measurement are also different. When the relative variance is started in terms of the arithmetic mean and the standard deviation, the resulting percentage is known as the coefficient of variance. A measure of dispersion helps us in studying the important characteristic of a distribution, i.e., the extent to which the observations vary from one another and from some average value.

### **Distinction between Variance and Coefficient of Variance:**

| <b>Standard Deviation</b>                  | <b>Variance</b>                              |
|--------------------------------------------|----------------------------------------------|
| (1) It is a absolute measure.              | (1) It is relative measure.                  |
| (2) It is formatted.                       | (2) It is not formatted.                     |
| (3) It is relatively difficult to make.    | (3) It is relatively easier to make.         |
| (4) It has limited option.                 | (4) It has many options.                     |
| (5) It is widely used.                     | (5) Its uses are limited.                    |
| (6) It is suitable for discrete variables. | (6) It is suitable for continuous variables. |
| (8) It is difficult to understand.         | (8) It is easy to understand.                |
| (10) It is not independent.                | (10) It is independent.                      |

### **Some Useful Definitions:**

**Mean:** When the sum of observations is divided by the number of observations then it is called simple mean.

**Uses:** Mean has the following uses:

1. It is commonly used to locate central value of a data set.
2. It is treated as the basis of mean.

**Median:** Median is a point in the middle of a data distribution that splits the distribution in two halves.

**Uses:** Median has the following uses:

1. It is used to overcome extreme values problem of data set.
2. It is used for easy graphing.

**Mode:** Mode is the most frequent value of a data set.

**Uses:** Mode has the following uses:

1. It is applicable to qualitative distribution.
2. It is applicable to open-end class intervals.

**Quartile Deviation:** The quartile deviation is the half of the difference between the first quartile and third quartile. This is also called semi-interquartile range. It is notable that quartiles divide the observations into 4 equal parts when the observations are arranged in order of magnitude.

**Uses:** Quartile deviation has the following uses:

1. It is used to find the appropriate quarterly deviations of a data set.
2. It is used to measure length of a data set.

**Standard Deviation:** The arithmetic mean of the squares of the deviations of the given observations from their arithmetic mean is known as variance. The positive square root of variance is the standard deviation. It measures the absolute variation of a distribution.

**Uses:** Median has the following uses:

1. It is used to find the appropriate deviation of the observations.
2. It is used to measure the stability of a data set.

**Variance:** Variance is the squared value of standard deviation.

**Uses:** Variance has the following uses:

1. It is applicable to compare between data sets.
2. It is used to measure appropriate deviation of data set.

## Symbols

| Symbols        | Meaning                                      |
|----------------|----------------------------------------------|
| $Q_1$          | first quartile                               |
| $Q_3$          | third quartile                               |
| $D_1$          | first docile                                 |
| $D_7$          | seventh docile                               |
| $P_1$          | first percentile                             |
| $P_{99}$       | ninety ninth percentile                      |
| $F_n$          | nth fractile                                 |
| TV             | target value of a range                      |
| $  $           | absolute value                               |
| UCL            | upper class limit                            |
| LCL            | lower class limit                            |
| CR             | coefficient of range                         |
| MD             | median deviation                             |
| CMD            | coefficient of median deviation              |
| QD             | quartile deviation                           |
| CQD            | coefficient of quartile deviation            |
| IQR            | interquartile range                          |
| $\sigma$       | sigma (standard deviation)                   |
| S.D.           | standard deviation                           |
| C.S.D.         | combined standard deviation                  |
| CV             | coefficient of variation                     |
| CCV            | combined coefficient of variation            |
| $d_1$          | 1 <sup>st</sup> deviation from combined mean |
| $d_2$          | 2 <sup>nd</sup> deviation from combined mean |
| $\sigma_{12}$  | SD of two data sets                          |
| $\sigma_{123}$ | SD of three data sets                        |
| Md. D.         | median deviation                             |
| Mo. D.         | mode deviation                               |
| AD             | average deviation                            |
| C50%           | central 50%                                  |
| FNS            | five number summary                          |
| $X_{\min}$     | minimum value of data set                    |
| $X_{\max}$     | maximum value of data set                    |

## Formulas

(1) Range = Highest Value (H) – Lowest Value (L)

$$(2) \text{ Coefficient of Range} = \frac{H - L}{H + L} \times 100$$

(3) For mean/average deviation with ungrouped data,

$$\text{Mean Deviation (MD)} = \frac{\sum |X - \bar{X}|}{N}$$

(4) For mean deviation with grouped data,

$$\text{Mean Deviation (MD)} = \frac{\sum f |X - \bar{X}|}{N}$$

(5) For median deviation with ungrouped data,

$$\text{Median Deviation (MD)} = \frac{\sum |X - \text{Median}|}{N}$$

(7) For mode deviation with ungrouped data,

$$\text{Mode Deviation (MD)} = \frac{\sum |X - \text{Mode}|}{N}$$

(8) For coefficient of mean deviation,

$$\text{Coefficient of Mean Deviation (CMD)} = \frac{\text{MD}}{\text{Mean}} \times 100$$

(9) For coefficient of median deviation,

$$\text{Coefficient of Median Deviation (CMD)} = \frac{\text{MD}}{\text{Median}} \times 100$$

(10) For coefficient of mean deviation,

$$\text{Coefficient of Mode Deviation (CMD)} = \frac{\text{MD}}{\text{Mode}} \times 100$$

(11) For inter quartile range,

$$\text{Inter Quartile Range (IQR)} = Q_3 - Q_1 \text{ or } Q_1 \text{ to } Q_3$$

(12) For quartile deviation (grouped or ungrouped data),

$$\text{Quartile Deviation (QD)} = \frac{Q_3 - Q_1}{2}$$

(13) For coefficient of quartile deviation,

$$\text{Coefficient of Quartile Deviation (CQD)} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

(14) For mean deviation by standard deviation,

$$\text{Mean Deviation (MD)} = \frac{4}{5} \times \text{SD}$$

(15) For quartile deviation by standard deviation,



$$\text{Quartile Deviation (QD)} = \frac{2}{3} \times \text{SD}$$

(16) For standard deviation with ungrouped data,

$$\text{Standard Deviation (SD)} = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

(17) For SD with ungrouped data and frequency,

$$\text{Standard Deviation (SD)} = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$$

(18) For SD with grouped data and frequency,

$$\text{SD } (\sigma) = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times c$$

(19) For variance on respective data,

$$\text{Variance } (\sigma^2) = (\text{SD})^2 \text{ or } \sigma^2$$

(20) For coefficient of variation,

$$\text{Coefficient of Variation (CV)} = \frac{\text{SD}}{\text{Mean}} \times 100$$

(21) For combined SD of two data sets,

$$\text{Combined SD } (\sigma_{12}) = \sqrt{\frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1d_1^2 + N_2d_2^2}{N_1 + N_2}}$$

$$\text{Where } d_1 = |\bar{X}_{12} - \bar{X}_1|, d_2 = |\bar{X}_{12} - \bar{X}_2|$$

(22) For combined SD of three data sets,

$$\sigma_{123} = \sqrt{\frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_3\sigma_3^2 + N_1d_1^2 + N_2d_2^2 + N_3d_3^2}{N_1 + N_2 + N_3}}$$

$$\text{Where } d_1 = |\bar{X}_{123} - \bar{X}_1|, d_2 = |\bar{X}_{123} - \bar{X}_2|, d_3 = |\bar{X}_{123} - \bar{X}_3|$$

# Thank You

**Prove that the standard deviation of the first n natural numbers is  $\sigma = \sqrt{\frac{n^2-1}{12}}$**

Proof: The first n natural numbers are 1,2,3,.....,n.

Let,  $x_1, x_2, \dots, x_n$  stand for the 1st n natural numbers with mean  $\bar{x}$ .

$$\bar{x} = \frac{\sum x}{n} = \frac{1+2+\dots+n}{n} = \frac{\frac{n(n+1)}{2}}{n}$$

$$\therefore \bar{x} = \frac{n+1}{2}$$

$$\begin{aligned} \text{Sum of the square of 1st n natural number is } \sum x^2 &= 1^2 + 2^2 + \dots + n^2 \\ &= \frac{n(n+1)(2n+1)}{6} \end{aligned}$$

$$\begin{aligned} \text{Thus the standard deviation } \sigma &= \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2} \\ &\Rightarrow \sigma = \sqrt{\frac{\frac{n(n+1)(2n+1)}{6}}{n} - \left(\frac{n+1}{2}\right)^2} \\ &\Rightarrow \sigma = \sqrt{\frac{n(n+1)(2n+1)}{6n} - \left(\frac{n+1}{2}\right)^2} \\ &\Rightarrow \sigma = \sqrt{\frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2} \\ &\Rightarrow \sigma = \sqrt{\left(\frac{n+1}{2}\right) \left[\frac{(2n+1)}{3} - \frac{(n+1)}{2}\right]} \\ &\Rightarrow \sigma = \sqrt{\frac{(n+1)}{2} \left[\frac{2(2n+1) - 3(n+1)}{6}\right]} \\ &\Rightarrow \sigma = \sqrt{\frac{(n+1)}{2} \left[\frac{4n+2-3n-3}{6}\right]} \\ &\Rightarrow \sigma = \sqrt{\frac{(n+1)}{2} \left[\frac{(n-1)}{6}\right]} \\ &\therefore \sigma = \sqrt{\frac{n^2-1}{12}} \quad [\text{Proved}] \end{aligned}$$

# If  $\bar{x}$  and  $s$  denote respectively the mean and standard deviation of a set of  $n$  non-negative quantities, then prove that  $\bar{x} \sqrt{(n-1)} \geq s$  and hence prove that  $100\sqrt{(n-1)} \geq CV$

Proof: By definition,

$$\sum_{i=1}^n x_i = n\bar{x} \text{-----}(i)$$

$$ns^2 = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} \text{-----}(ii)$$

We know,

$$\left(\sum_{i=1}^n x_i\right)^2 = \sum_{i=1}^n x_i^2 + \sum_{i \neq j} x_i x_j \text{-----}(iii)$$

Since,  $x_i$  's are non – negative,

$$\sum_{i \neq j} x_i x_j \geq 0$$

Hence from (iii)

$$\begin{aligned} \left(\sum_{i=1}^n x_i\right)^2 &\geq \sum_{i=1}^n x_i^2 \\ \Rightarrow \left(\sum_{i=1}^n x_i\right)^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} &\geq \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} \text{[Subtracting both sides by } \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}] \end{aligned}$$

$$\Rightarrow \left(\sum_{i=1}^n x_i\right)^2 \left(1 - \frac{1}{n}\right) \geq ns^2 \text{ [From (ii)]}$$

$$\Rightarrow \frac{n-1}{n} (\bar{x})^2 \geq ns^2$$

$$\Rightarrow (\bar{x})^2 (n-1) \geq s^2$$

$$\Rightarrow \bar{x} \sqrt{n-1} \geq s \text{ [ Taking square root of both sides] (Proved)}$$

$$\Rightarrow \sqrt{n-1} \geq \frac{s}{\bar{x}}$$

$$\Rightarrow 100\sqrt{n-1} \geq \frac{s}{\bar{x}} \times 100 \text{ [Multiplying both sides by 100]}$$

$$\Rightarrow 100\sqrt{n-1} \geq CV \text{ [Since, } CV = \frac{s}{\bar{x}} \times 100 \text{ ] (Proved)}$$

**# Prove that for two unequal positive observations,  $M.D = S.D = R/2$**

**Where M.D= Mean Deviation**

**S.D= Standard Deviation**

**R = Range**

Proof : Let the two observations be  $x_1$  and  $x_2$  ( $x_1 > x_2$ ).

By definition,

$$\text{Mean, } \bar{x} = \frac{x_1 + x_2}{2} \text{ -----(i)}$$

Range,  $R = \text{Highest Value} - \text{Lowest Value}$

$$\therefore R = x_1 - x_2 \text{ -----(ii)}$$

$$\text{Standard Deviation, SD} = \sqrt{\frac{\sum_{i=1}^2 (x_i - \bar{x})^2}{2}} \text{ -----(iii)}$$

$$\begin{aligned} \text{And Mean Deviation, MD} &= \frac{\sum_{i=1}^2 |x_i - \bar{x}|}{2} \\ &= \frac{|x_1 - \bar{x}| + |x_2 - \bar{x}|}{2} \\ &= \frac{|x_1 - \frac{x_1 + x_2}{2}| + |x_2 - \frac{x_1 + x_2}{2}|}{2} \text{ [From (i)]} \\ &= \frac{|\frac{2x_1 - x_1 - x_2}{2}| + |\frac{2x_2 - x_1 - x_2}{2}|}{2} \\ &= \frac{\frac{x_1 - x_2}{2} + \frac{x_1 - x_2}{2}}{2} \\ &= \frac{2(\frac{x_1 - x_2}{2})}{2} \\ \Rightarrow \text{MD} &= \frac{x_1 - x_2}{2} = \frac{R}{2} \text{ [From (ii)] -----(iv)} \end{aligned}$$

$$\begin{aligned} \text{From (iii), SD} &= \sqrt{\frac{\sum_{i=1}^2 (x_i - \bar{x})^2}{2}} \\ \Rightarrow \text{SD}^2 &= \frac{\sum_{i=1}^2 (x_i - \bar{x})^2}{2} \text{ [Taking square on both sides]} \\ &= \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2}{2} \\ &= \frac{(x_1 - \frac{x_1 + x_2}{2})^2 + (x_2 - \frac{x_1 + x_2}{2})^2}{2} \\ &= \frac{(\frac{x_1 - x_2}{2})^2 + (\frac{x_1 - x_2}{2})^2}{2} \end{aligned}$$

$$\Rightarrow SD^2 = \frac{2(\frac{x_1 - x_2}{2})^2}{2}$$

$$\Rightarrow SD^2 = (\frac{x_1 - x_2}{2})^2$$

$$\Rightarrow SD = \frac{x_1 - x_2}{2} \text{ [ Taking square root on both sides]}$$

$$\therefore SD = \frac{R}{2} \text{ [From (ii)] -----(v)}$$

From (iv) and (v), we can write

$$MD = SD = \frac{R}{2} \text{ (Proved)}$$

## Important Problems (For Mid-Term Exam) in the Dispersion Chapter:

Calculate (i) Mean Deviation (ii) Standard deviation of the following data:

| Income<br>(TK. In Thousand) | Number of Employees |
|-----------------------------|---------------------|
| 10-20                       | 10                  |
| 20-30                       | 12                  |
| 30-30                       | 15                  |
| 40-50                       | 20                  |
| 50-60                       | 22                  |
| 60-70                       | 14                  |
| 70-80                       | 7                   |

Solution (i):

### Calculation of Mean Deviation

| Income<br>(TK.in<br>thousand) | Number of<br>Employees<br>f | Mid<br>Point<br>X | fx               | $ x - \bar{x} $<br>$=  x - 45.2 $ | $f x - \bar{x} $               |
|-------------------------------|-----------------------------|-------------------|------------------|-----------------------------------|--------------------------------|
| 10-20                         | 10                          | 15                | 150              | 30.2                              | 302                            |
| 20-30                         | 12                          | 25                | 300              | 20.2                              | 242.4                          |
| 30-40                         | 15                          | 35                | 525              | 10.2                              | 153                            |
| 40-50                         | 20                          | 45                | 900              | 0.2                               | 4                              |
| 50-60                         | 22                          | 55                | 1210             | 9.8                               | 215.6                          |
| 60-70                         | 14                          | 65                | 910              | 19.8                              | 277.2                          |
| 70-80                         | 7                           | 75                | 525              | 29.8                              | 208.6                          |
| Total                         | N=100                       |                   | $\sum fx = 4520$ |                                   | $\sum f x - \bar{x}  = 1402.8$ |

$$\begin{aligned}\text{Mean, } \bar{x} &= \frac{\sum fx}{N} \\ \Rightarrow \bar{x} &= \frac{4520}{100} \\ \Rightarrow \bar{x} &= 45.2\end{aligned}$$

We know that, Mean Deviation, M.D =  $\frac{\sum f|x - \bar{x}|}{N}$

$$= \frac{1402.8}{100}$$

$$= 14.028$$

So, Mean Deviation, M.D = TK. 14.028 thousand

Solution (ii):

### Calculation of Standard deviation

| Income<br>(TK. in thousand) | Number of<br>Employees<br>f | Mid Point<br>X | fx               | fx <sup>2</sup>      |
|-----------------------------|-----------------------------|----------------|------------------|----------------------|
| 10-20                       | 10                          | 15             | 150              | 2250                 |
| 20-30                       | 12                          | 25             | 300              | 7500                 |
| 30-40                       | 15                          | 35             | 525              | 18375                |
| 40-50                       | 20                          | 45             | 900              | 40500                |
| 50-60                       | 22                          | 55             | 1210             | 66550                |
| 60-70                       | 14                          | 65             | 910              | 59150                |
| 70-80                       | 7                           | 75             | 525              | 39375                |
| Total                       | N=100                       |                | $\sum fx = 4520$ | $\sum fx^2 = 233700$ |

$$\text{Mean, } \bar{x} = \frac{\sum fx}{N}$$

$$\Rightarrow \bar{x} = \frac{4520}{100}$$

$$\Rightarrow \bar{x} = 45.2$$

We know that , Standard deviation,  $\sigma = \sqrt{\frac{\sum fx^2}{N} - (\bar{x})^2}$

$$= \sqrt{\frac{233700}{100} - (45.2)^2}$$

$$= \sqrt{2337 - 2043.04}$$

$$= \sqrt{293.96}$$

$$= 17.15$$

So, Standard deviation,  $\sigma$  = TK. 17.15 thousand [Ans.]



# Measures of Skewness and Kurtosis

## Moments:

### Definition of moments:

Moments can be defined as the arithmetic mean of various powers of deviations taken from the mean of a distribution. These moments are known as central moments. The first four moments about arithmetic mean or central moments are defined below.

|                                        | Individual series                           | Discrete series                              |
|----------------------------------------|---------------------------------------------|----------------------------------------------|
| First moments about the mean; $\mu_1$  | $\frac{\sum (X - \bar{X})}{n} = 0$          | $\frac{\sum f(X - \bar{X})}{N} = 0$          |
| Second moments about the mean; $\mu_2$ | $\frac{\sum (X - \bar{X})^2}{n} = \sigma^2$ | $\frac{\sum f(X - \bar{X})^2}{N} = \sigma^2$ |
| Third moments about the mean ; $\mu_3$ | $\frac{\sum (X - \bar{X})^3}{n}$            | $\frac{\sum f(X - \bar{X})^3}{N}$            |
| Fourth moment about the Mean ; $\mu_4$ | $\frac{\sum (X - \bar{X})^4}{n}$            | $\frac{\sum f(X - \bar{X})^4}{N}$            |

$\mu$  is a Greek letter, pronounced as ‘mu’. If the mean is a fractional value, then it becomes a difficult task to work out the moments. In such cases, we can calculate moments about a working origin and then change it into moments about the actual mean. The moments about an origin are known as raw moments.

The first four raw moments – individual series.

$$\mu_1' = \frac{\sum (X - A)}{n}, \mu_2' = \frac{\sum (X - A)^2}{n}, \mu_3' = \frac{\sum (X - A)^3}{n}, \mu_4' = \frac{\sum (X - A)^4}{n}$$

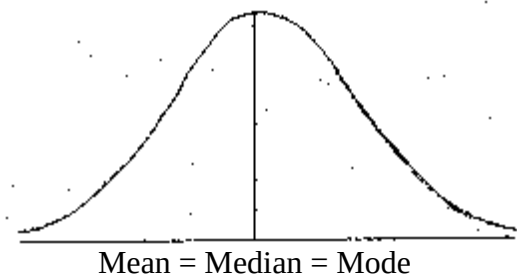
The first four raw moments – discrete series:

$$\mu_1' = \frac{\sum f(X - A)}{N}, \mu_2' = \frac{\sum f(X - A)^2}{N}, \mu_3' = \frac{\sum f(X - A)^3}{N}, \mu_4' = \frac{\sum f(X - A)^4}{N}$$

## Skewness:

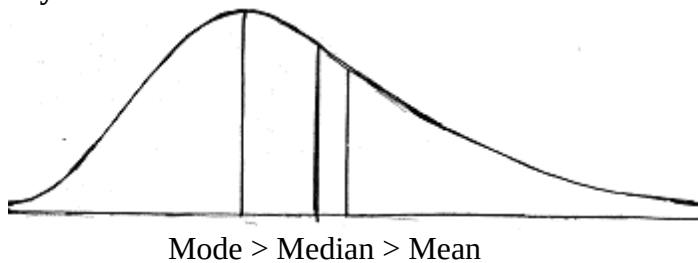
Skewness means ‘lack of symmetry’. We study skewness to have an idea about the shape of the curve which we can draw with the help of the given data. If in a distribution mean = median = mode, then that distribution is known as symmetrical distribution. If in a distribution mean  $\neq$  median  $\neq$  mode, then it is not a symmetrical distribution and it is called a skewed distribution and such a distribution could either be positively skewed or negatively skewed.

### a) Symmetrical distribution:



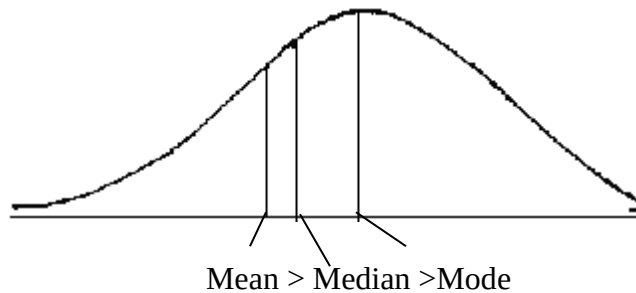
It is clear from the above diagram that in a symmetrical distribution the values of mean, median and mode coincide. The spread of the frequencies is the same on both sides of the center point of the curve

**b) Positively skewed distribution:**



It is clear from the above diagram, in a positively skewed distribution, the value of the mean is maximum and that of the mode is least, the median lies in between the two. In the positively skewed distribution the frequencies are spread out over a greater range of values on the right hand side than they are on the left hand side.

**c) Negatively skewed distribution:**



It is clear from the above diagram, in a negatively skewed distribution, the value of the mode is maximum and that of the mean is least. The median lies in between the two. In the negatively skewed distribution the frequencies are spread out over a greater range of values on the left hand side than they are on the right hand side.

**Measures of skewness:**

The important measures of skewness are

- (i) Karl – Pearson' s coefficient of skewness
- (ii) Bowley' s coefficient of skewness
- (iii) Measure of skewness based on moments

**Karl – Pearson' s Coefficient of skewness:**

According to Karl – Pearson, the absolute measure of skewness = mean – mode. This measure is not suitable for making valid comparison of the skewness in two or more distributions because the unit of measurement may be different in different series. To avoid this difficulty use relative measure of skewness called Karl – Pearson's coefficient of skewness given by:

$$\text{Karl – Pearson's Coefficient of Skewness, } SK_p = \frac{\text{Mean} - \text{Mode}}{S.D}$$

In case of mode is ill – defined, the coefficient can be determined by the formula:

$$\text{Coefficient of skewness, } SK_p = \frac{3(\text{Mean} - \text{Median})}{S.D}$$

### **Bowley's Coefficient of skewness:**

In Karl – Pearson's method of measuring skewness the whole of the series is needed. Prof. Bowley has suggested a formula based on relative position of quartiles. In a symmetrical distribution, the quartiles are equidistant from the value of the median; ie., Median – Q<sub>1</sub> = Q<sub>3</sub> – Median. But in a skewed distribution, the quartiles will not be equidistant from the median. Hence Bowley has suggested the following formula:

$$\text{Bowley's Coefficient of Skewness, } SK = \frac{Q_3 + Q_1 - 2\text{Median}}{Q_3 - Q_1}$$

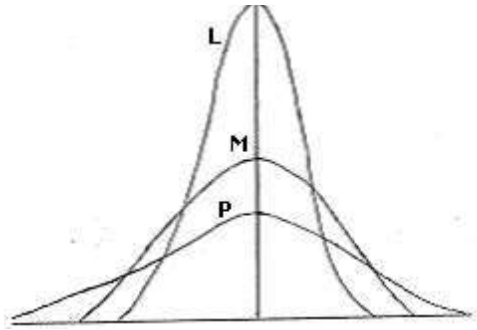
### **Measure of skewness based on moments:**

The measure of skewness based on moments is denoted by  $\beta_1$  and is given by:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

### **Kurtosis:**

The expression 'Kurtosis' is used to describe the peakedness of a curve. The three measures – central tendency, dispersion and skewness describe the characteristics of frequency distributions. But these studies will not give us a clear picture of the characteristics of a distribution. As far as the measurement of shape is concerned, we have two characteristics – skewness which refers to asymmetry of a series and kurtosis which measures the peakedness of a normal curve. All the frequency curves expose different degrees of flatness or peakedness. This characteristic of frequency curve is termed as kurtosis. Measure of kurtosis denote the shape of top of a frequency curve. Measure of kurtosis tell us the extent to which a distribution is more peaked or more flat topped than the normal curve, which is symmetrical and bell-shaped, is designated as Mesokurtic. If a curve is relatively more narrow and peaked at the top, it is designated as Leptokurtic. If the frequency curve is more flat than normal curve, it is designated as platykurtic. If  $\mu_3$  is negative, then  $\beta_1$  is negative



L= Leptokurtic, M=Mesokurtic P= platykurtic

### Measure of Kurtosis:

The measure of kurtosis of a frequency distribution based moments is denoted by  $\beta_2$  and is given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

If  $\beta_2 = 3$ , the distribution is said to be normal and the curve is mesokurtic.

If  $\beta_2 > 3$ , the distribution is said to be more peaked and the curve is leptokurtic.

If  $\beta_2 < 3$ , the distribution is said to be flat topped and the curve is platykurtic.

### Skewness

Skewness means 'lack of symmetry' i.e., departure from symmetry of a distribution. The measures of skewness indicate the difference between the manner in which the observation are distributed in a particular distribution compared with a symmetrical (or normal) distribution. The three types of skewness are as under:



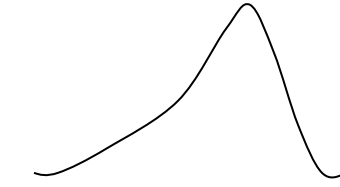
$$\bar{X} = \text{Med} = M_0$$

(i) Symmetrical skewed distribution.



$$\bar{X} > \text{Med} < M_0$$

(ii) Positively skewed distribution.



$$M_0 > \text{Med} > \bar{X}$$

(iii) Negatively skewed distribution.

### Moments

Moments are popularly used to describe the characteristics of a distribution. They present la many of the most commonly used descriptive statistical measures such as central tendency, Variation, skewness and Kurtosis. The moments are generally decoted by the Greek letter  $\mu$  (read as mu).

The rth central/ corrected moment is defined by

$$\mu_r = 1/N \sum (X - \bar{X})^r \quad [\text{for ungrouped data}]$$

$$\mu_r = 1/N \sum (X - \bar{X})^r \quad [\text{for grouped data}]$$

The rth moment is defined by

$$\mu'_r = 1/N \sum (X - \bar{X})^r \quad [\text{for ungrouped data}]$$

$$\mu'_r = 1/N \sum f(X - \bar{X})^r \quad [\text{for grouped data}]$$

Where,

r = No. of moment

$\bar{X}$  = Actual mean.

A = Assumed mean.

## Kurtosis

The degree of peakness or flatness of a distribution relative to a normal distribution is called kurtosis.

The measure of kurtosis is defined by

$$\beta_2 = \mu_4 / \mu_2^2, \quad \gamma_2 = \beta_2 - 3$$

When  $\beta_2 = 3$  (or  $\gamma_2 = 0$ ), then the curve is called mesokurtic.

When  $\beta_2 > 3$  then the curve is called leptokurtic.

When  $\beta_2 < 3$  then the curve is called platykurtic.

## Significance of Skewness, and Kurtosis

Skewness and kurtosis are very important to statistical analysis. Skewness indicate the nature of distributions, i.e., where the mean, median, and mode lie in a numerical information series. Decision madders apply skewness to find their required measures of central value i.e., mean, median or mode. The measure of kurtosis, the fourth device in describing a frequency distribution, can be used to show the degree of concentration, either the values concentrated in the area around the mode ( a peaked curve) or decentralized form the mode of both tails of the frequency curve (a flat topped curve). Hence the concepts of kurtosis and skewness are used in analyzing business data.

## Relationship between Raw Moments and Central Moments

We have  $\mu_r = 1/N \sum (X - \bar{X})^r$

$$= 1/N \sum \{(X - A) - (\bar{X} - A)\}^r$$

$$= 1/N$$

$$[\sum \{(X - A)^r - (r_1)d \sum (X - A)^{r-1} + (r_2)d^2 \sum (X - A)^{r-2} \dots + (-1)^r d^r]$$

$$\text{Where, } d = (\bar{X} - A).$$

Now according to definition of moments, we have

$$\mu'_r = \mu'_r - (r)d\mu'_{r-1} + (r_2)d^2\mu'_{r-2} + (r_1)d^3\mu'_{r-3} + \dots + (-1)^r d^r \dots\dots(1)$$

We know,  $\mu'_1 = 1/N \sum f(X - A) = (\bar{X} - A) = d$

Putting r = 1, 2, 3, 4, .....etc., and d =  $\mu'_1$  in (1), we have

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

It is notable that the moments are independent on change of origin but not of scale. If we calculate the third moment and the fourth moment about the origin, then the third moment tells us about skewness and the fourth moment tells us about the kurtosis.

### Distinction between Skewness and Kurtosis

| Skewness                                           | Kurtosis                                              |
|----------------------------------------------------|-------------------------------------------------------|
| (1) It measures the form of the data distribution. | (1) It measures the spread of the data distribution.  |
| (2) It is zero for symmetric data distribution.    | (2) It is normal for symmetric data distribution.     |
| (3) It may be positive, negative or zero           | (3) It may be platykurtic, mesokurtic or leptokurtic. |
| (4) It has three formulas for findings.            | (4) It has two formulas for findings.                 |
| (5) It is widely used.                             | (5) Its uses are limited.                             |

### Distinction between Skewness and dispersion

| Skewness                                                   | Dispersion                                           |
|------------------------------------------------------------|------------------------------------------------------|
| (1) It is relative measure.                                | (1) It is a absolute measure.                        |
| (2) It is not formatted.                                   | (2) It is formatted.                                 |
| (3) It is relatively difficult to make.                    | (3) It is relatively easier to make.                 |
| (4) It has limited option.                                 | (4) It has many options.                             |
| (5) Its uses are limited.                                  | (5) It is widely used.                               |
| (6) It is used to find the shape of the data distribution. | (6) It is used to compare among different data sets. |

## Symbols

| Symbols   | Meaning                       |
|-----------|-------------------------------|
| $\mu$     | mu                            |
| $\mu'_1$  | first raw moment              |
| $\mu'_2$  | second raw moment             |
| $\mu'_3$  | third raw moment              |
| $\mu'_4$  | fourth raw moment             |
| $\mu_1$   | first central moment          |
| $\mu_2$   | second central moment         |
| $\mu_3$   | third central moment          |
| $\mu_4$   | fourth central moment         |
| $\beta_1$ | beta one (coeff. of skewness) |

|            |                                   |
|------------|-----------------------------------|
| $\gamma_1$ | gamma one (coeff. of skewness)    |
| $\beta_2$  | beta two (coeff. of kurtosis)     |
| $Sk_p$     | Karl Pearson's coeff. of skewness |
| $Sk_B$     | Bowley's coeff. of skewness       |
| $Sk_K$     | Kelly's coeff. of skewness        |

## Problems & Solutions

### Problem 1

BBA (Professional) 2005

You are given the following data pertaining to kilowatt hours of electricity consumed by 100 persons in Dhaka:

|                               |      |       |       |       |       |
|-------------------------------|------|-------|-------|-------|-------|
| Consumption (In k-watt hours) | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| No. of users                  | 6    | 25    | 36    | 20    | 13    |

Calculate: (i) Arithmetic mean; (ii) Standard deviation; (iii) Coefficient of skewness.

**Solution:**

#### Calculation of A.M., S.D. and $Sk_p$

| Marks   | f       | M.P.(X) | fX                 | fX <sup>2</sup>       |
|---------|---------|---------|--------------------|-----------------------|
| 0 – 10  | 6       | 5       | 30                 | 150                   |
| 10 – 20 | 25      | 15      | 375                | 5625                  |
| 20 – 30 | 36      | 25      | 900                | 22500                 |
| 30 – 40 | 20      | 35      | 700                | 24500                 |
| 40 – 50 | 13      | 45      | 585                | 26325                 |
|         | N = 100 |         | $\Sigma fX = 2590$ | $\Sigma fX^2 = 79100$ |

$$(i) \text{ Mean} = \frac{\Sigma fX}{N} = \frac{2590}{100} = 25.9 \quad [\text{Ans.}]$$

$$(ii) \text{ S.D.} = \sqrt{\frac{\Sigma fX^2}{N} - \left(\frac{\Sigma fX}{N}\right)^2} = \sqrt{\frac{79100}{100} - \left(\frac{2590}{100}\right)^2} = 10.963 \quad [\text{Ans.}]$$

(iii) For highest frequency 36, mode lies in the class 20 – 30.

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 20 + \frac{11}{11+16} \times 10 = 24.07$$

$$\therefore Sk_p = \frac{(\text{Mean} - \text{mode})}{\sigma} = \frac{(25.9 - 24.07)}{10.963} = 0.167 \quad [\text{Ans.}]$$

### Problem 2

BBA (Professional) 2006

Find the first two central moments from the following data and hence find coefficient variation:  
(10, 5, 0, -4, -1, 6)

**Solution:**

**Calculation of  $\mu_1$  and  $\mu_2$  and C.V.**

| X               | $(X - \bar{X})$           | $(X - \bar{X})^2$                |
|-----------------|---------------------------|----------------------------------|
| 10              | 7.33                      | 53.73                            |
| 5               | 2.33                      | 5.43                             |
| 0               | -2.67                     | 7.13                             |
| -4              | -6.67                     | 44.49                            |
| -1              | -3.67                     | 13.47                            |
| 6               | 3.35*                     | 11.22                            |
| $\Sigma X = 16$ | $\Sigma(X - \bar{X}) = 0$ | $\Sigma(X - \bar{X})^2 = 135.47$ |

\*Adjusted

$$\therefore \text{Mean } (\bar{X}) = \frac{\Sigma X}{N} = \frac{16}{6} = 2.67$$

$$(i) \text{ The first central moment } (\mu_1) = \frac{\Sigma(X - \bar{X})}{N} = \frac{0}{6} = 0 \quad [\text{Ans.}]$$

$$\text{The second central moment } (\mu_2) = \frac{\Sigma(X - \bar{X})^2}{N} = \frac{135.47}{6} = 22.58 \quad [\text{Ans.}]$$

$$(ii) \text{ Here SD } (\sigma) = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}} = \sqrt{\frac{135.47}{6}} = 4.75$$

$$\therefore \text{C.V.} = \frac{\sigma}{\bar{X}} \times 100 = \frac{4.75}{2.67} \times 100 = 177.9\% \quad [\text{Ans.}]$$

**Problem 3**

BBA (Professional) 2007

Calculate coefficient of skewness of the above data (Given that standard deviation is 4.12).  
Based on the above result, what is your conclusion about the skewness?

**Solution:** We are given,  
Standard deviation ( $\sigma$ ) = 4.12  
Mean = 30.09  
Mode = 34

$$\therefore \text{Coefficient of skewness } (Sk_p) = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{30.09 - 34}{4.12} = -0.949 \quad [\text{Ans.}]$$

**Problem 4**

BBA (Professional) 2008

Find coefficient of skewness from the following data:

25 -5 18 30 16 29

**Solution:** Given

$$\text{Mean } (\bar{X}) = \frac{\Sigma X}{n} = \frac{-5 + 16 + 18 + 25 + 29 + 30}{6} = 18.83$$

$$\text{Median (Md)} = \text{Mid-value} = (18 + 25)/2 = 21.5$$

$$\text{Mode (Mo)} = 3(\text{median}) - 2(\text{mean}) = 3(21.5) - 2(18.83) = 26.84$$



$$\begin{aligned}
 \text{SD } (\sigma) &= \sqrt{\frac{\Sigma(X - \bar{X})^2}{n}} \\
 &= \sqrt{\frac{(-5-18.83)^2 + (16-18.83)^2 + (18-18.83)^2 + (25-18.83)^2 + (29-18.83)^2 + (30-18.83)^2}{n}} \\
 &= 11.85
 \end{aligned}$$

$$\therefore \text{Coefficient of skewness (Sk}_p\text{)} = \frac{\text{Mean} - \text{Mode}}{\text{SD}} = \frac{18.83 - 26.84}{11.85} = -0.68 \text{ [Ans]}$$

For any set of observations  $x_1, x_2, \dots, x_n$ , prove that  $\beta_2 \geq \beta_1 + 1$

Proof : We know that, for ungrouped data,

$$\mu_2 = \frac{1}{n} \sum (x_i - \bar{x})^2 \quad \text{--- (i)}$$

$$\mu_3 = \frac{1}{n} \sum (x_i - \bar{x})^3 \quad \text{--- (ii)}$$

$$\& \mu_4 = \frac{1}{n} \sum (x_i - \bar{x})^4 \quad \text{--- (iii)}$$

Consider the following expression,

$$\{a(x_i - \bar{x})^2 + b(x_i - \bar{x}) + c\}^2$$

where a, b, c are real arbitrary constants.  $\therefore$  the square term of the above expression is always positive. Thus performing the squares, summing and dividing by n all through.

$$\begin{aligned} & \frac{a^2 \sum (x_i - \bar{x})^4}{n} + \frac{b^2 \sum (x_i - \bar{x})^2}{n} + c^2 + \frac{2ab \sum (x_i - \bar{x})^3}{n} + \frac{2bc \sum (x_i - \bar{x})}{n} + \frac{2ac \sum (x_i - \bar{x})^2}{n} \geq 0 \\ \Rightarrow & a^2 \mu_4 + b^2 \mu_2 + c^2 + 2ab \mu_3 + \frac{2bc \times 0}{n} + 2ac \mu_2 \geq 0 \quad [\text{Since } \sum (x_i - \bar{x}) = 0] \end{aligned}$$

Choosing  $a = 1, b = -\frac{\mu_3}{\mu_2}$  and  $c = -\mu_2$ , the above expression becomes

$$\mu_4 - \frac{\mu_3^2}{\mu_2} - \mu_2^2 \geq 0$$

Dividing both sides by  $\mu_2^2$

$$\Rightarrow \frac{\mu_4}{\mu_2^2} - \frac{\mu_3^2}{\mu_2^3} - 1 \geq 0$$

$$\Rightarrow \beta_2 - \beta_1 - 1 \geq 0$$

$$\therefore \beta_2 \geq \beta_1 + 1 \text{ (Proved)}$$

# Correlation Analysis

## Introduction:

The term correlation is used by a common man without knowing that he is making use of the term correlation. For example when parents advice their children to work hard so that they may get good marks, they are correlating good marks with hard work.

The study related to the characteristics of only variable such as height, weight, ages, marks, wages, etc., is known as univariate analysis. The statistical Analysis related to the study of the relationship between two variables is known as Bi-Variate Analysis. Sometimes the variables may be inter-related. In health sciences we study the relationship between blood pressure and age, consumption level of some nutrient and weight gain, total income and medical expenditure, etc., the nature and strength of relationship may be examined by correlation and Regression analysis.

Thus **the degree of relationship between two or more variables is called Correlation. As for example: relation between height of father and son, yield and rainfall, wage and price index, share and debentures etc.**

## Uses of correlation:

1. It is used in physical and social sciences.
2. It is useful for economists to study the relationship between variables like price, quantity etc. Businessmen estimates costs, sales, price etc. using correlation.
3. It is helpful in measuring the degree of relationship between the variables like income and expenditure, price and supply, supply and demand etc.
4. Sampling error can be calculated.
5. It is the basis for the concept of regression.

## Types of Correlation:

Correlation is classified into various types. The most important ones are

- i) Positive and negative.
- ii) Simple , Multiple and partial
- iii) Linear and non-linear.

**(i) Positive and Negative Correlation:** Whether correlation is positive (direct) or negative (inverse) would depend upon the direction of change of the variable. If both the variables are varying in the same direction, i.e., if one variable is increasing the other on an average is also increasing or, if one variable is decreasing the other on an average is also decreasing, correlation is said to be positive. Price and supply, height and weight, yield and rainfall, are some examples of positive correlation.

On the other hand, the variables are varying in opposite direction, i.e., as one variable is increasing the other on an average is decreasing or vice versa, correlation is said to be negative. Price and demand, yield of crop and price, are examples of negative correlation.

(ii) Simple, Multiple and partial: The degree of relationship between two variables is called simple correlation. Yield of crop per acre and the amount of rainfall, it is a problem of simple correlation.

**Multiple correlations:** The degree of relationship among more than two variables is called multiple correlations. Yield of crop per acre and both the amount of rainfall and the amount of fertilizers used, it is a problem of multiple correlation.

Partial correlation: The degree of relationship between two variables where the other(s) variable(s) is/are held constant. The relationship between Yield of crop per acre and the amount of rainfall where the other variable such as the amount of fertilizers are held constant.

(iii) Linear and non linear correlation: If the amount of change in one variable tends to bear a constant ratio to the amount of change in other variable then the correlation is said to be linear. The following example shows the linear correlation.

|    |   |    |    |    |    |
|----|---|----|----|----|----|
| X: | 1 | 2  | 3  | 4  | 5  |
| Y: | 5 | 10 | 15 | 20 | 25 |

Here the ratio of change between the two variables is the same. If we plot these points on a graph we get a straight line.

If the amount of change in one variable does not bear a constant ratio to the amount of change in other variable then the correlation is said to be non linear.

## Methods of studying Correlation

The following are the Methods of studying Correlation:

- (1) Scatter Diagram Method
- (2) Karl Pearson's Coefficient of Correlation
- (3) Spearman's Rank Correlation Coefficient
- (4) Method of Least Square

1. Scatter diagram method:

### Scatter Diagram:

It is the simplest method of studying the relationship between two variables diagrammatically. One variable is represented along the horizontal axis and the second variable along the vertical axis. For each pair of observations of two variables, we put a dot in the plane. There are as many dots in the plane as the number of paired observations of two variables. The direction of dots shows the scatter or concentration of various points. This will show the type of correlation.

1. If all the plotted points form a straight line from lower left hand corner to the upper right hand corner then there is Perfect positive correlation. We denote this as  $r = +1$

2. If all the plotted dots lie on a straight line falling from upper left hand corner to lower right hand corner, there is a perfect negative correlation between the two variables. In this case the coefficient of correlation takes the value  $r = -1$ .
3. If the plotted points in the plane form a band and they show a rising trend from the lower left hand corner to the upper right hand corner the two variables are highly positively correlated.
4. If the points fall in a narrow band from the upper left hand corner to the lower right hand corner, there will be a high degree of negative correlation.
5. If the plotted points in the plane are spread all over the diagram there is no correlation between the two variables.

## 2. Karl Pearson's Coefficient of Correlation:

Coefficient of correlation measures the strength and direction of a linear association between two variables.

Karl Pearson, a great biometrician and statistician, suggested a mathematical method for measuring the magnitude of linear relationship between the two variables. It is most widely used method in practice and it is known as Pearsonian coefficient of correlation. It is denoted by  $r$  and

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2} \sqrt{\sum (Y - \bar{Y})^2}}$$

The coefficient of correlation can also be calculated from the original set of observations (i.e., without taking deviations from mean) by applying the following formula:

$$\begin{aligned} r &= \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{N}} \sqrt{\sum Y^2 - \frac{(\sum Y)^2}{N}}} \\ &= \frac{N \sum XY - \sum X \sum Y}{\sqrt{N \sum X^2 - (\sum X)^2} \sqrt{N \sum Y^2 - (\sum Y)^2}} \end{aligned}$$

## Properties of Correlation:

1. Correlation coefficient lies between  $-1$  and  $+1$

Proof: We Know that,

$$\text{Correlation Coefficient, } r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2} \sqrt{\sum (Y - \bar{Y})^2}}$$

$$\text{Let, } a = \frac{(X - \bar{X})}{\sqrt{\sum (X - \bar{X})^2}} \text{ and } b = \frac{(Y - \bar{Y})}{\sqrt{\sum (Y - \bar{Y})^2}}$$

$$\sum a^2 = 1, \sum b^2 = 1 \& \sum ab = r$$

$$\text{Then, } \sum (a+b)^2 = \sum a^2 + 2\sum ab + \sum b^2 \geq 0$$

$$\begin{aligned} & \text{[Since square term is always positive]} \\ & = 1 + 2r + 1 = 2 + 2r = 2(1+r) \geq 0 \end{aligned}$$

$$0r, 1+r \geq 0 \text{------(i)}$$

$$\text{Similarly, } \sum (a-b)^2 = \sum a^2 - 2\sum ab + \sum b^2 \geq 0$$

$$= 1 - 2r + 1 = 2 - 2r = 2(1-r) \geq 0$$

$$\text{or, } 1-r \geq 0 \text{------(ii)}$$

From (i) and (ii), we can write,  $-1 \leq r \leq 1$  (Proved)

2. Correlation Coefficient is independent of change of origin and scale.
3. Correlation coefficient is the geometric mean of two regression coefficients.
4. Two independent variables are uncorrelated but the converse is not true.

### Interpreting the coefficient of correlation

1.  $r=1$ , it means there is perfect positive correlation between the variables.
2.  $r=-1$ , it means there is perfect negative correlation between the variables.
3.  $r=0$ , it means there is no correlation between the variables, i.e. the variables are uncorrelated.
4.  $r=0.3$ , it means there is low degree positive correlation between the variables.
5.  $r=-0.7$ , it means there is high degree negative correlation between the variables.
6.  $r=0.02$ , it means there is slightly positive correlation between the variables.
7.  $r=0.5$  it means there is moderate positive correlation between the variables.,

### Probable Error in Coefficient Of Correlation & Its Interpretation:

The probable error of the coefficient of correlation helps in interpreting its value. With the help of probable error it is possible to determine the liability of the value of the coefficient so far as it depends on the condition of random sampling.

**The formula for finding out the probable error of the Karl Pearson's coefficient of correlation is:**

Probable error of coefficient of correlation

$$P.E.r = 0.6745 \frac{1-r^2}{\sqrt{N}}$$

- (i) If the value of r is less than the probable error, there is no evidence of correlation.
- (ii) If the value of r is more than six times of the probable error it is significant correlation.
- (iii) If the probable error is not much and if the coefficient of correlation is .5 or more it is generally considered to be significant.
- (iv) Probable error as a measure for interpreting coefficient of correlation should be used only when the number of pairs of observations is large. If n is small probable error may give misleading conclusions.
- (v) Probable error as a measure for interpreting coefficient of correlation should be used only when a sample study is being made and the sample is unbiased and representative. With the help of sample studies, the correlation in the universe is estimated.

### **Rank Correlation:**

It is studied when no assumption about the parameters of the population is made. This method is based on ranks. It is useful to study the qualitative measure of attributes like honesty, color, beauty, intelligence, character, morality etc. Spearman Rank Correlation Coefficient is a non-parametric measure of correlation, using ranks to calculate the correlation. The individuals in the group can be arranged in order and there on, obtaining for each individual a number showing his/her rank in the group. This method was developed by Edward Spearman in 1904. It is defined by

$$R = 1 - \frac{6 \sum D^2}{N(N^2 - 1)} = 1 - \frac{6 \sum D^2}{N^3 - N}$$

Where, R denotes rank correlation coefficient and D refers to the difference of ranks between paired items in two series.

### **Proof:**

Let us suppose that a group of n individuals is arranged in order of merit or proficiency in possession of two characteristics A and B. these ranks in two characteristics will, in general, be different. For example, if we consider the relation between intelligence and beauty, it is not necessary that a beautiful individual is intelligent also. Let  $(x_i, y_i)$ ;  $i=1, 2, \dots, n$  be the ranks of the  $i$ th individual in two characteristics A and B respectively. Pearsonian coefficient of

correlation between the ranks  $x_i$ 's and  $y_i$  's is called the rank correlation coefficient between A and B for that group of individuals.

Assuming that no two individuals are bracketed equal in either classification, each of the variables X and Y takes the values 1,2,.....,n

$$\begin{aligned}\bar{x} = \bar{y} &= \frac{1}{n}(1+2+3+\dots+n) = \frac{n+1}{2} \\ \sigma_{x^2} &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 = \frac{1}{n}(1^2+2^2+\dots+n^2) - \left(\frac{n+1}{2}\right)^2 \\ &= \frac{n(n+1)(2n+1)}{6n} - \left(\frac{n+1}{2}\right)^2 = \frac{n^2-1}{12} \\ \sigma_{x^2} &= \frac{n^2-1}{12} = \sigma_{y^2}\end{aligned}$$

In general  $x_i \neq y_i$ . Let  $d_i = x_i - y_i$

$$d_i = (x_i - \bar{x}) - (y_i - \bar{y}) \quad (\because \bar{x} = \bar{y})$$

Squaring and summing over  $i$  from 1 to  $n$ , we get

$$\begin{aligned}\sum d_i^2 &= \sum [(x_i - \bar{x}) - (y_i - \bar{y})]^2 \\ &= \sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2 - 2 \sum (x_i - \bar{x})(y_i - \bar{y})\end{aligned}$$

Dividing both side by  $n$ , we get

$$\frac{1}{n} \sum d_i^2 = \sigma_x^2 + \sigma_y^2 - 2 \text{Cov}(X, Y) = \sigma_x^2 + \sigma_y^2 - 2\rho\sigma_x\sigma_y$$

Where  $\rho$  is the rank correlation coefficient between A and B.

$$\therefore \frac{1}{n} \sum d_i^2 = 2\sigma_x^2 - 2\rho\sigma_x^2$$

$$\Rightarrow 1 - \rho = \frac{\sum d_i^2}{2n\sigma_x^2}$$

$$\Rightarrow \rho = 1 - \frac{\sum d_i^2}{2n\sigma_x^2} = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$$

Which is the spearman's formula for the rank correlation coefficient.



## What is covariance?

Covariance is a quantitative measure of the degree to which the deviation of one variable (X) from its mean is related to the deviation of another variable (Y) from its mean. To simplify, **covariance measures the joint variability of two random variables**. There are two types of covariance. (1) **Positive covariance** (2) **Negative covariance**

**(1)Positive covariance:** Indicates that two variables tend to move in the same direction.

**(2)Negative covariance:** Reveals that two variables tend to move in inverse directions.

### Formula of Covariance

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{N}$$

### Coefficient of Determination

The coefficient of determination ( $r^2$  or r-squared) is a statistical measure in a regression model that determines the proportion of variance in the dependent variable that can be explained by the independent variable.

$r^2 = 0.81$ . What does it mean?

$r^2 = 0.81$ . It means 81% of the variation in the dependent variable has been explained by the independent variable.

# Thank You

**Problem 1.**

BBA (Professional) 2002

Determine the value of 'r' from the following data:

|   |    |    |    |   |    |    |    |   |    |    |
|---|----|----|----|---|----|----|----|---|----|----|
| X | 2  | 4  | 3  | 8 | 6  | 5  | 6  | 7 | 5  | 4  |
| Y | 15 | 20 | 16 | 6 | 14 | 12 | 12 | 9 | 11 | 15 |

**Solution:****Calculation of the value of 'r'**

| x       | y        | x <sup>2</sup>        | y <sup>2</sup>         | xy        |
|---------|----------|-----------------------|------------------------|-----------|
| 2       | 15       | 4                     | 225                    | 30        |
| 4       | 20       | 16                    | 400                    | 80        |
| 3       | 16       | 9                     | 256                    | 48        |
| 8       | 6        | 64                    | 36                     | 48        |
| 6       | 14       | 36                    | 196                    | 84        |
| 5       | 12       | 25                    | 144                    | 60        |
| 6       | 12       | 36                    | 144                    | 72        |
| 7       | 9        | 49                    | 81                     | 63        |
| 5       | 11       | 25                    | 121                    | 55        |
| 4       | 15       | 16                    | 225                    | 60        |
| Σx = 50 | Σy = 130 | Σx <sup>2</sup> = 280 | Σy <sup>2</sup> = 1828 | Σxy = 600 |

$$\begin{aligned}
 \therefore r &= \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}} \\
 &= \frac{10(600) - (50)(130)}{\sqrt{10(280) - (50)^2} \cdot \sqrt{10(1828) - (130)^2}} \\
 &= -0.78 \text{ [Ans.]}
 \end{aligned}$$

**Problem 2**

BBA (Honours) 2004; BBA (Honours) 2008

Find the correlation coefficient between age and playing habits of the following students and comment:

|                 |     |     |     |     |     |    |
|-----------------|-----|-----|-----|-----|-----|----|
| Age             | 15  | 16  | 17  | 18  | 18  | 20 |
| No. of students | 250 | 200 | 150 | 120 | 100 | 80 |
| Regular player  | 200 | 150 | 90  | 48  | 30  | 12 |

**Solution:****Calculation of Correlation Coefficient**

| x        | y (%)        | x <sup>2</sup>         | y <sup>2</sup>          | xy         |
|----------|--------------|------------------------|-------------------------|------------|
| 15       | 200/250 = 80 | 225                    | 6400                    | 1200       |
| 16       | 150/200 = 75 | 256                    | 5625                    | 1200       |
| 17       | 90/150 = 60  | 289                    | 3600                    | 1020       |
| 18       | 48/120 = 40  | 324                    | 1600                    | 720        |
| 19       | 30/100 = 30  | 361                    | 900                     | 570        |
| 20       | 12/80 = 15   | 400                    | 225                     | 300        |
| Σx = 105 | Σy = 300     | Σx <sup>2</sup> = 1855 | Σy <sup>2</sup> = 18350 | Σxy = 5010 |

$$\therefore r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}}$$

$$= \frac{6(5010) - (105)(300)}{\sqrt{6(1855) - (105)^2} \cdot \sqrt{6(18350) - (300)^2}}$$

$$= -0.991 \quad [\text{Ans.}]$$

### Problem 3

BBA (Honours) 2007

Following figures give the rainfall in inches for the year and the production in months for the Rabi crop and Kharif crop. Calculate the Karl Pearson's coefficient of correlation between rainfall and production:

|                   |    |    |    |    |    |    |    |
|-------------------|----|----|----|----|----|----|----|
| Rainfall          | 20 | 22 | 24 | 26 | 28 | 30 | 32 |
| Rabi Production   | 16 | 18 | 20 | 32 | 40 | 38 | 35 |
| Kharif Production | 16 | 17 | 20 | 16 | 20 | 22 | 20 |

**Solution:**

#### Calculation of Correlation Coefficient

| Rainfall<br>x   | Total Production<br>(Rabi + Kharif)<br>y | xy                | x <sup>2</sup>               | y <sup>2</sup>                |
|-----------------|------------------------------------------|-------------------|------------------------------|-------------------------------|
| 20              | 16 + 16 = 32                             | 640               | 400                          | 1024                          |
| 22              | 18 + 17 = 35                             | 770               | 484                          | 1225                          |
| 24              | 20 + 20 = 40                             | 960               | 576                          | 1600                          |
| 26              | 32 + 16 = 48                             | 1248              | 676                          | 2304                          |
| 28              | 40 + 20 = 60                             | 1680              | 784                          | 3600                          |
| 30              | 38 + 22 = 60                             | 1800              | 900                          | 3600                          |
| 32              | 25 + 20 = 45                             | 1440              | 1024                         | 2025                          |
| <b>Σx = 182</b> | <b>Σy = 320</b>                          | <b>Σxy = 8538</b> | <b>Σx<sup>2</sup> = 4844</b> | <b>Σy<sup>2</sup> = 15378</b> |

$$\therefore r = \frac{N\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{N\Sigma x^2 - (\Sigma x)^2} \sqrt{N\Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{7(8538) - (182)(320)}{\sqrt{7(4844) - (182)^2} \sqrt{7(15378) - (320)^2}}$$

$$= +0.75 \quad [\text{Ans.}]$$

**Comment:**  $r = +0.75$  shows highly positive correlation between rainfall and production of crop.

### Problem 4

BBA (Professional) 2003

The coefficient of rank correlation of the marks obtained by 10 students in statistics and accountancy was found to be 80. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly take as 7 instead of 9. Find the correct coefficient of rank correlation.

**Solution:** Given  $N = 10$

$$R = 0.80$$

$$\therefore R = 1 - \frac{6\Sigma D^2}{N^3 - N}$$

$$\Rightarrow 0.80 = 1 - \frac{6\Sigma D^2}{10^3 - 10}$$

$$\Rightarrow \frac{6\Sigma D^2}{990} = 1 - 0.80$$

$$\Rightarrow \frac{\Sigma D^2}{165} = 0.20$$

$$\Rightarrow \Sigma D^2 = 33, \text{ but this is not correct.}$$

$$\therefore \text{Correct } \Sigma D^2 = 33 - 7^2 + 9^2 = 65$$

$$\therefore R = 1 - \frac{6\Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 65}{10^3 - 10} = 1 - 0.394 = 0.606 \quad [\text{Ans.}]$$

### Problem 5

BBA (Professional) 2004

A computer while calculating the correlation coefficient between the variables x and y obtained following results:

$$N = 30, \Sigma x = 120, \Sigma x^2 = 600, \Sigma y = 90, \Sigma y^2 = 250, \Sigma xy = 335.$$

It was, however, later discovered at the time of checking that it had copied down two pairs of observations as:

| X  | Y  |
|----|----|
| 8  | 10 |
| 12 | 7  |

While the correct values were

| X  | Y  |
|----|----|
| 8  | 12 |
| 10 | 8  |

Obtain the correct value of the correlation coefficient between x and y.

**Solution:** Here, Correct  $\Sigma x = 120 - 8 - 12 + 8 + 10 = 118$

$$\text{Correct } \Sigma y = 90 - 10 - 7 + 12 + 8 = 93$$

$$\text{Correct } \Sigma x^2 = 600 - (8)^2 - (12)^2 + (8)^2 + (10)^2 = 556$$

$$\text{Correct } \Sigma y^2 = 250 - (10)^2 - (7)^2 + (12)^2 + (8)^2 = 309$$

$$\text{Correct } \Sigma xy = 335 - (8 \times 10) - (12 \times 7) + (8 \times 12) + (10 \times 8) = 347$$

$\therefore$  The correct value of the correlation coefficient between x and y is

$$\begin{aligned} r &= \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{n(\Sigma x^2) - (\Sigma x)^2} \cdot \sqrt{n(\Sigma y^2) - (\Sigma y)^2}} \\ &= \frac{30(347) - (118)(93)}{\sqrt{30(556) - (118)^2} \cdot \sqrt{30(309) - (93)^2}} \\ &= -0.43 \quad [\text{Ans.}] \end{aligned}$$

**Problem 6: Calculate rank correlation coefficient between income and expenditure of 10 employees:**

| Income (Taka in Thousand) | Expenditure(Taka in Thousand) |
|---------------------------|-------------------------------|
| 70                        | 65                            |
| 40                        | 40                            |
| 60                        | 58                            |
| 50                        | 49                            |
| 20                        | 20                            |
| 80                        | 79                            |
| 90                        | 88                            |
| 85                        | 82                            |
| 45                        | 40                            |
| 55                        | 50                            |

**Solution:**

|       | Serial Number | Income (Taka in Thousand)<br>X | Expenditure (Taka in Thousand)<br>Y | Rank of X<br>R <sub>1</sub> | Rank of Y<br>R <sub>2</sub> | D <sup>2</sup><br>=(R <sub>1</sub> -R <sub>2</sub> ) <sup>2</sup> |
|-------|---------------|--------------------------------|-------------------------------------|-----------------------------|-----------------------------|-------------------------------------------------------------------|
|       | 1             | 70                             | 66                                  | 7                           | 7                           | 0                                                                 |
|       | 2             | 40                             | 46                                  | 2                           | 2                           | 0                                                                 |
|       | 3             | 60                             | 58                                  | 6                           | 5                           | 1                                                                 |
|       | 4             | 50                             | 49                                  | 4                           | 3                           | 1                                                                 |
|       | 5             | 20                             | 20                                  | 1                           | 1                           | 0                                                                 |
|       | 6             | 80                             | 86                                  | 8                           | 10                          | 4                                                                 |
|       | 7             | 90                             | 81                                  | 10                          | 8                           | 4                                                                 |
|       | 8             | 85                             | 82                                  | 9                           | 9                           | 0                                                                 |
|       | 9             | 45                             | 50                                  | 3                           | 4                           | 1                                                                 |
|       | 10            | 55                             | 60                                  | 5                           | 6                           | 1                                                                 |
| Total | N=10          |                                |                                     |                             |                             | Σ D <sup>2</sup> = 11                                             |

We know that Rank Correlation Coefficient,

$$\begin{aligned}
 R &= 1 - \frac{6 \sum D^2}{N^3 - N} \\
 &= 1 - \frac{6 \times 11}{10^3 - 10} \\
 &= 1 - \frac{66}{990} = 0.93
 \end{aligned}$$

**Problem 7**

BBA (Honours) 2009

Ten competitors in a music contest are ranked by three Judges in the following order:

|              |   |   |   |   |   |    |    |   |   |   |
|--------------|---|---|---|---|---|----|----|---|---|---|
| First Judge  | 4 | 6 | 9 | 5 | 1 | 3  | 10 | 8 | 7 | 2 |
| Second Judge | 4 | 8 | 7 | 5 | 6 | 9  | 10 | 3 | 2 | 1 |
| Third Judge  | 7 | 8 | 6 | 1 | 5 | 10 | 9  | 2 | 3 | 4 |

Use the rank correlation coefficient to discuss which pair of Judges has the nearest approach to common tastes in music?

**Solution:****Calculation of Rank correlation coefficient**

| Judge 1<br>$R_1$ | Judge 2<br>$R_2$ | Judge 3<br>$R_3$ | $(R_1 - R_2)^2$<br>$= D^2$ | $(R_2 - R_3)^2$<br>$= D^2$ | $(R_1 - R_3)^2$<br>$= D^2$ |
|------------------|------------------|------------------|----------------------------|----------------------------|----------------------------|
| 4                | 4                | 7                | 0                          | 9                          | 9                          |
| 6                | 8                | 8                | 4                          | 0                          | 4                          |
| 9                | 7                | 6                | 4                          | 1                          | 9                          |
| 5                | 5                | 1                | 0                          | 16                         | 16                         |
| 1                | 6                | 5                | 25                         | 1                          | 16                         |
| 3                | 9                | 10               | 36                         | 1                          | 49                         |
| 10               | 10               | 9                | 0                          | 1                          | 1                          |
| 8                | 3                | 2                | 25                         | 1                          | 36                         |
| 7                | 2                | 3                | 25                         | 1                          | 16                         |
| 2                | 1                | 4                | 1                          | 9                          | 4                          |
| N = 10           |                  |                  | $\Sigma D^2 = 120$         | $\Sigma D^2 = 40$          | $\Sigma D^2 = 160$         |

$$R(1 \& 2) = 1 - \frac{6\Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 120}{10^3 - 10} = 1 - 0.73 = 0.27$$

$$R(2 \& 3) = 1 - \frac{6\Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 40}{10^3 - 10} = 1 - 0.24 = 0.76$$

$$R(1 \& 3) = 1 - \frac{6\Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 160}{10^3 - 10} = 1 - 0.97 = 0.03$$

**Comment:** Since Judge 2 and Judge 3 have the highest positive the rank correlation coefficient, so they have the nearest approach to common tastes in music.

**Problem 8**

BBA (Professional) 2008

A psychologist wanted to compare two methods A and B of teaching. He selected a random sample of 22 students. He grouped them into 11 pairs so that the students in a pair have approximately equal scores on an intelligence test. In each pair one student was taught by method A and the other by method B and examined after the course. The marks obtained by them are tabulated below:

|      |    |    |    |    |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|----|----|----|----|
| Pair | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |
| A    | 24 | 29 | 19 | 14 | 30 | 19 | 27 | 30 | 20 | 28 | 11 |
| B    | 37 | 35 | 16 | 26 | 23 | 27 | 19 | 20 | 16 | 11 | 21 |

Find the rank correlation coefficient between the two sets of scores.

**Solution: (a)**

**Calculation of Rank correlation coefficient**

| Ranking No. | A  | Method A (R <sub>1</sub> ) | B  | Method B (R <sub>2</sub> ) | (R <sub>1</sub> -R <sub>2</sub> ) <sup>2</sup> =D <sup>2</sup> |
|-------------|----|----------------------------|----|----------------------------|----------------------------------------------------------------|
| 1           | 24 | 6                          | 37 | 11                         | 25                                                             |
| 2           | 29 | 9                          | 35 | 10                         | 1                                                              |
| 3           | 19 | 3.5                        | 16 | 2.5                        | 1                                                              |
| 4           | 14 | 2                          | 26 | 8                          | 36                                                             |
| 5           | 30 | 10.5                       | 23 | 7                          | 12.25                                                          |
| 6           | 19 | 3.5                        | 27 | 9                          | 30.25                                                          |
| 7           | 27 | 7                          | 19 | 4                          | 9                                                              |
| 8           | 30 | 10.5                       | 20 | 5                          | 30.25                                                          |
| 9           | 20 | 5                          | 16 | 2.5                        | 6.25                                                           |
| 10          | 28 | 8                          | 11 | 1                          | 49                                                             |
| 11          | 11 | 1                          | 21 | 6                          | 25                                                             |
| N=11        |    |                            |    |                            | ΣD <sup>2</sup> =225                                           |

**In the method A**

19 is repeated 2 times.

So m<sub>1</sub>=2

30 is repeated 2 times.

So m<sub>2</sub>=2

**In the method B**

16 is repeated 2 times.

So m<sub>3</sub>=2

$$R = 1 - \frac{6 \left\{ \Sigma D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \frac{1}{12} (m_3^3 - m_3) \right\}}{N^3 - N} \left[ \begin{array}{l} m_1, m_2 \text{ and } m_3 \text{ stands for number} \\ \text{of repeated items} \end{array} \right]$$

$$= 1 - \frac{6 \left\{ 225 + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) \right\}}{11^3 - 11}$$

$$= -0.0295 \text{ [Ans.]}$$

**Problem 9**

BBA (Professional) 2003

You are provided with the following set of data:

Advertising Expenditure  
(X) in Tk lakh

Sales  
(Y) in Tk lakh

|   |    |
|---|----|
| 5 | 10 |
| 4 | 8  |
| 3 | 3  |
| 4 | 8  |
| 5 | 7  |

Find out coefficient of determination and hence comment.

**Solution:**

**Calculation of Coefficient of Determination ( $r^2$ )**

| x               | y               | $x^2$             | $y^2$              | xy                |
|-----------------|-----------------|-------------------|--------------------|-------------------|
| 5               | 10              | 25                | 100                | 50                |
| 4               | 8               | 16                | 64                 | 32                |
| 3               | 3               | 9                 | 9                  | 9                 |
| 4               | 8               | 16                | 64                 | 32                |
| 5               | 7               | 25                | 49                 | 35                |
| $\Sigma x = 21$ | $\Sigma y = 36$ | $\Sigma x^2 = 91$ | $\Sigma y^2 = 286$ | $\Sigma xy = 158$ |

$$\begin{aligned} \text{(i) Here } r &= \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{n(\Sigma x^2) - (\Sigma x)^2} \cdot \sqrt{n(\Sigma y^2) - (\Sigma y)^2}} \\ &= \frac{5(158) - (21)(36)}{\sqrt{5(91) - (21)^2} \cdot \sqrt{5(286) - (36)^2}} \\ &= 0.78 \\ \therefore \text{ The coefficient of determination } (r^2) &= (r)^2 \\ &= (0.78)^2 \\ &= 0.61 \quad [\text{Ans.}] \end{aligned}$$

Comment:  $r^2 = 0.61$ .

It means 61% of the variation in the dependent variable has been explained by the independent variable.

**Thank You**



## Regression

According to M.M.Blair “Regression is the measure of the average relationship between two or more variables in terms of the original units of the data.”

### Utility of Regression Analysis

Some of the Utility of Regression Analysis are as follows:

1. Regression analysis helps in establishing a functional relationship between two or more variables.
2. Since most of the problems of economic analysis are based on cause and effect relationship, the regression analysis is a highly valuable tool in economics & business research.
3. Regression analysis is widely used in statistical estimation of demand curves, supply curves, production functions, cost functions, consumption functions etc.

### Standard Error:

### Distinguish between Correlation & regression

| Correlation                                                                                                      | Regression                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. The degree of relationship between two or variables is called correlation.                                    | 1. Regression means stepping back or returning to the average value, i.e. it expresses average relationship between two or more variables.                                                                                 |
| 2. Correlation need not imply cause & effect relationship.                                                       | 2. Regression analysis clearly indicates the cause & effect relationship. The variable(s) constituting cause(s) is taken as independent variable(s) & the variable constituting the effect is taken as dependent variable. |
| 3. It may be a case of nonsense correlation.                                                                     | 3. There is nothing like nonsense regression.                                                                                                                                                                              |
| 4. Correlation coefficient between X & Y & Correlation coefficient between Y & X are same i.e. $r_{XY} = r_{YX}$ | 4. But Regression coefficient of X on Y and Regression coefficient of Y on X are not same i.e. $b_{XY} \neq b_{YX}$ .                                                                                                      |
| 5. Correlation Coefficient lies between -1 & +1.                                                                 | 5. Regression coefficient lies between $-\infty$ to $+\infty$ .                                                                                                                                                            |
| 6. Correlation analysis studies only linear relationship.                                                        | 6. Regression analysis studies linear as well as non linear relationship.                                                                                                                                                  |

### Regression coefficients

In the regression line of Y on X, viz.,

$$Y = a + bX$$

The coefficient ‘b’ which is the slope of the line of regression of Y on X is called the coefficient of regression of Y on X. It represents the increment in the value of the dependent variable Y for a unit change in the value of the independent variable X. For notational convenience, Coefficient of regression of Y on X is denoted as  $b_{YX}$ .

The regression line of Y on X, viz.,

$$X = a + bY$$

The coefficient ‘b’ which is the slope of the line of regression of X on Y is called the coefficient of regression of X on Y. It represents the increment in the value of the dependent variable X for a unit change in the value of the independent variable Y. For notational convenience; Coefficient of regression of X on Y is denoted as  $b_{XY}$ .

## Properties of Regression coefficients

1. The coefficient of correlation is the geometric mean of the two regression coefficients.

Symbolically:  $r = \pm\sqrt{b_{YX}b_{XY}}$

Proof: Regression coefficient of Y on X ,  $b_{YX} = r \frac{\sigma_Y}{\sigma_X}$  .....(i)

Regression coefficient of X on Y ,  $b_{XY} = r \frac{\sigma_X}{\sigma_Y}$  .....(ii)

Multiplied equation (i) & (ii), we get,

$$b_{YX} \times b_{XY} = r \frac{\sigma_Y}{\sigma_X} \times r \frac{\sigma_X}{\sigma_Y}$$

$$r^2 = b_{YX} \times b_{XY}$$

$$r = \pm\sqrt{b_{YX} * b_{XY}} \text{ (Proved)}$$

2. If one of the regression coefficients is greater than unity the other must be less than unity.

3. Both the regression coefficients will have the same sign, i.e. they will be either positive or negative.

4. The coefficient of correlation will have the same sign as that of regression coefficients, i.e., if regression coefficients have positive sign, then r is positive. If regressions coefficient have negative sign then r is negative.

For Example: If  $b_{YX} = -1$  &  $b_{XY} = -10$  Then  $r = \pm\sqrt{-0.1 \times -10} = -1$

5. The average value of the two regression coefficients would be greater than the value of the correlation coefficient.

6. Regression coefficients are independent of change of origin but not of scale.

## What Is Standard Error? | How to Calculate (Guide with Examples)

The standard error of the mean, or simply **standard error**, indicates how different the population mean is likely to be from a sample mean. It tells you how much the sample mean would vary if you were to repeat a study using new samples from within a single population.

The standard error of the mean (SE or SEM) is the most commonly reported type of standard error. But you can also find the standard error for other statistics, like medians or proportions. The standard error is a common measure of sampling error—the difference between a population parameter and a sample statistic.

# Standard error vs standard deviation

Standard error and standard deviation are both measures of variability:

- The **standard deviation** describes variability **within a single sample**.
- The **standard error** estimates the variability **across multiple samples** of a population.

## • Why standard error matters

- In statistics, data from samples is used to understand larger populations. Standard error matters because it helps you estimate how well your sample data represents the whole population.
- With probability sampling, where elements of a sample are randomly selected, you can collect data that is likely to be representative of the population. However, even with probability samples, some sampling error will remain. That's because a sample will never perfectly match the population it comes from in terms of measures like means and standard deviations.
- By calculating standard error, you can estimate how representative your sample is of your population and make valid conclusions.
- A high standard error shows that sample means are widely spread around the population mean—your sample may not closely represent your population. A low standard error shows that sample means are closely distributed around the population mean—your sample is representative of your population.
- You can decrease standard error by increasing sample size. Using a large, random sam

# THANK YOU

## **PROBLEMS OF REGRESSION**

Problem 31(By Abdul Aziz –Page-7-58): Husband's age and wife's age are given below:

|               |    |    |    |    |    |    |    |    |    |    |
|---------------|----|----|----|----|----|----|----|----|----|----|
| Husband's Age | 22 | 23 | 23 | 24 | 26 | 27 | 27 | 28 | 30 | 30 |
| Wife's Age    | 18 | 20 | 21 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |

- (i) Determine the regression equation of husband's age on wife's age
- (ii) Determine the regression equation of wife's age on husband's age
- (iii) Compute the probable age of husband's when wife's age is 30.
- (iv) Compute the Coefficient of regression of husband's age on wife's age.
- (v) Calculate the regression coefficient of wife's age on husband's age
- (vi) Calculate the coefficient of correlation
- (vii) Determine the coefficient of determination and interpret.
- (viii) Two regression coefficients are 3.4 and 0.7 respectively. Is the statement correct? Give reasons.

Solution: Let, Husband's Age= $x$

& Wife's Age= $y$

### **Calculation Table**

|       | SL.Number | $x$            | $y$            | $x^2$             | $y^2$             | $xy$             |
|-------|-----------|----------------|----------------|-------------------|-------------------|------------------|
|       | 1         | 22             | 18             | 484               | 324               | 396              |
|       | 2         | 23             | 20             | 529               | 400               | 460              |
|       | 3         | 23             | 21             | 529               | 441               | 483              |
|       | 4         | 24             | 20             | 576               | 400               | 480              |
|       | 5         | 26             | 21             | 676               | 441               | 546              |
|       | 6         | 27             | 22             | 729               | 484               | 594              |
|       | 7         | 27             | 23             | 729               | 529               | 621              |
|       | 8         | 28             | 24             | 784               | 576               | 672              |
|       | 9         | 30             | 25             | 900               | 625               | 750              |
|       | 10        | 30             | 26             | 900               | 676               | 780              |
| Total | $n=10$    | $\sum x = 260$ | $\sum y = 220$ | $\sum x^2 = 6836$ | $\sum y^2 = 4896$ | $\sum xy = 5782$ |

Solution (i) : We know that, Mean  $\bar{x} = \frac{\sum x}{n} = \frac{260}{10} = 26$

$$\& \bar{y} = \frac{\sum y}{n} = \frac{220}{10} = 22$$

Regression Coefficient of x on y be

$$b_{xy} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$$
$$\Rightarrow b_{xy} = \frac{5782 - \frac{260 \times 220}{10}}{4896 - \frac{(220)^2}{10}}$$

$$\Rightarrow b_{xy} = 1.107$$

Let the Regression equation of x on y be

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\Rightarrow x - 26 = 1.107(y - 22)$$

$$\Rightarrow x = 26 + 1.107y - 24.354$$

$$\Rightarrow x_c = 1.646 + 1.107y$$

So the required Regression equation of husband's age (x) on wife's age(y) is

$$x_c = 1.646 + 1.107y$$

Solution (ii) : Regression Coefficient of y on x be

$$b_{yx} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$
$$\Rightarrow b_{yx} = \frac{5782 - \frac{260 \times 220}{10}}{6836 - \frac{(260)^2}{10}}$$

$$\Rightarrow b_{yx} = 0.816$$

Let the Regression Equation of y on x be

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - 22 = 0.816(x - 26)$$

$$\Rightarrow y = 22 + 0.816x - 21.216$$

$$\Rightarrow y_c = 0.784 + 0.816x$$

So, the required Regression Equation of Wife's age (y) on husband's (x)age

$$y_c = 0.784 + 0.816x [\text{Ans.}]$$

Solution (iii) : When Wife's age,  $y = 30$  then the probable age of husband age( $x$ ) is

$$x_c = 1.64 + 1.107y$$

$$\Rightarrow x_c = 1.64 + 1.107(30) = 34.856 [\text{Ans.}]$$

Solution(iv) : We have already find out Regression coefficient t of husband's age( $x$ ) on Wife's age( $y$ ) which is as follows :

$$b_{xy} = 1.107 [\text{Ans.}]$$

Solution (v) : We have also find out Regression coefficient t of wife's age ( $y$ ) on husband's age( $x$ ) which is as follows :

$$b_{yx} = 0.816$$

Solution (vi) : Correlation Coefficient t,  $r = \pm \sqrt{b_{xy} \cdot b_{yx}}$

$$= \sqrt{1.107 \times 0.816} \quad [\text{Since both Regression Coefficients are positive, so } r \text{ is positive}]$$

$$= 0.95$$

Comment :  $r = 0.95$ . It means there is high degree positive correlation between the variables.

Solution (vii) : Coefficient of determination,  $r^2 = (0.95)^2 = 0.9025$

Interpretation :  $r^2 = 0.9025$  It means 90.25% of the variation in the dependent variable has been explained by the independent variable.

Solution (viii) : Let us consider two regression coefficients are  $b_{xy} = 3.4$  and  $b_{yx} = 0.7$  respectively

$$\therefore \text{The Coefficient of Correlation, } r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{3.4 \times 0.7} = 1.543 > 1 \text{ which}$$

is impossible. Because the value of correlation coefficient lies between -1 and +1.

So the given statement is not correct.

## Problem 2

You are given below the following information about advertisement and sales.

Adv. Exp. (X)

Sales (Y)

(Lakh Tk)

(Lakh Tk)

|                         |      |     |
|-------------------------|------|-----|
| Mean                    | 20   | 120 |
| S.D.                    | 5    | 25  |
| Correlation coefficient | +0.8 |     |

- (i) Calculate the two regression equations.  
(ii) If the advertisement expenses is Rs. 25 lakh, what will be the estimated sales?

**Solution:** (i)(a) Regression equation of X on Y is

$$\begin{aligned}
 X - \bar{X} &= b_{xy} (Y - \bar{Y}) \\
 X - \bar{X} &= r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y}) \\
 \Rightarrow X - 20 &= .16(Y - 120) \\
 \Rightarrow X - 20 &= .16Y - 19.2 \\
 \Rightarrow X &= .16Y - 0.8 \text{ [Ans]}
 \end{aligned}
 \quad \left| \quad \begin{aligned}
 &\text{where,} \\
 b_{xy} &= r \frac{\sigma_x}{\sigma_y} = .8 \frac{5}{25} = 0.16
 \end{aligned}
 \right.$$

$$\begin{aligned}
 Y - \bar{Y} &= b_{yx} (X - \bar{X}) \\
 \Rightarrow Y - \bar{Y} &= r \frac{\sigma_y}{\sigma_x} (X - \bar{X}) \\
 \Rightarrow Y - 120 &= 4(X - 20) \\
 \Rightarrow Y - 120 &= 4X - 80 \\
 \Rightarrow Y &= 4X + 40 \text{ [Ans]}
 \end{aligned}
 \quad \left| \quad \begin{aligned}
 &\text{Where} \\
 b_{yx} &= r \frac{\sigma_y}{\sigma_x} = 0.8 \frac{25}{5} = 4
 \end{aligned}
 \right.$$

- (ii) When X = 25, then Y will be

$$\begin{aligned}
 Y &= 4(25) + 40 \\
 &= 140 \text{ [Ans.]}
 \end{aligned}$$

.991 [Ans.]

### Problem 3

The following table gives the aptitude test scores and productivity indices of 10 workers selected at random:

|                    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|
| Aptitude scores    | 60 | 62 | 65 | 70 | 50 | 53 | 73 | 65 | 82 | 72 |
| Productivity index | 68 | 60 | 62 | 80 | 85 | 40 | 52 | 62 | 60 | 81 |



Estimate: (i) The productivity index of a worker whose test score is 92.  
(ii) The test score of a worker whose productivity index is 75.

**Solution:**

### Calculation of Regression Equations

| x                | y                | $x^2$                | $y^2$                | xy                  |
|------------------|------------------|----------------------|----------------------|---------------------|
| 60               | 68               | 3600                 | 4624                 | 4080                |
| 62               | 60               | 3844                 | 3600                 | 3720                |
| 65               | 62               | 4225                 | 3844                 | 4030                |
| 70               | 80               | 4900                 | 6400                 | 5600                |
| 50               | 85               | 2500                 | 7225                 | 4250                |
| 53               | 40               | 2809                 | 1600                 | 2120                |
| 73               | 52               | 5329                 | 2704                 | 3796                |
| 65               | 62               | 4225                 | 3844                 | 4030                |
| 82               | 60               | 6724                 | 3600                 | 4920                |
| 72               | 81               | 5184                 | 6561                 | 5832                |
| $\Sigma x = 652$ | $\Sigma y = 650$ | $\Sigma x^2 = 43340$ | $\Sigma y^2 = 44002$ | $\Sigma xy = 42378$ |

Here  $\bar{x} = \frac{\Sigma x}{n} = \frac{652}{10} = 65.2$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{650}{10} = 65$$

$$b_{yx} = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}} = \frac{42378 - \frac{652 \times 650}{10}}{43340 - \frac{(652)^2}{10}} = -0.0024$$

$$b_{xy} = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\Sigma y^2 - \frac{(\Sigma y)^2}{n}} = \frac{42378 - \frac{652 \times 650}{10}}{44002 - \frac{(650)^2}{10}} = -0.0011$$

(i) The regression equation of y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - 65 = -0.0024(x - 65.2)$$

$$\Rightarrow y - 65 = -0.0024x + 0.16$$

$$\Rightarrow y = -0.0024x + 65.16$$

When  $x = 92$ , then  $y = -0.0024x + 65.16 = -0.0024(92) + 65.16 = 64.94$  [Ans.]

(ii) The regression equation of  $x$  on  $y$  is

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\Rightarrow x - 65.2 = -0.0011(y - 65)$$

$$\Rightarrow x - 65.2 = -0.0011y + 0.07$$

$$\Rightarrow x = -0.0011y + 65.27$$

When  $y = 75$ , then  $x = -0.0011y + 65.27 = -0.0011(75) + 65.27 = 65.19$  [Ans.]

#### Problem 4

You are given variance of  $X = 9$ . The regression equation are,  $8X - 10Y + 66 = 0$  and  $40X - 18Y = 214$ , find on the basis of the above information:

- (i) The average (mean) values of  $X$  and  $Y$ ;
- (ii) Coefficient of correlation between  $X$  and  $Y$ ;
- (iii) Standard deviation of  $Y$ .

**Solution:** Given Variance of  $X$  ( $\sigma^2$ ) = 9

$$\therefore \text{S.D. } (\sigma) = \sqrt{9} = 3$$

$$8X - 10Y + 66 = 0 \dots\dots\dots (1)$$

$$40X - 18Y = 214 \dots\dots\dots (2)$$

(i) Solving (1) and (2), we get

$$(1) \times 5 - (2) \Rightarrow 40X - 50Y + 330 = 0$$

$$40X - 18Y - 214 = 0$$

$$\underline{\begin{array}{r} (-) \quad (+) \quad (+) \\ \hline \end{array}}.$$

$$-32Y + 544 = 0$$

$$\Rightarrow 32Y = 544$$

$$\Rightarrow Y = 17$$

$$\therefore 8X - 10Y + 66 = 0$$

$$\Rightarrow 8X - 10(17) + 66 = 0$$

$$\Rightarrow 8X - 170 + 66 = 0$$

$$\Rightarrow 8X = 104$$

$$\Rightarrow X = 13$$

$$\therefore \bar{X} = 13, \text{ and } \bar{Y} = 17 \text{ [Ans.]}$$

(ii) We know,

$$\text{The correlation coefficient (r)} = \sqrt{b_{xy} \times b_{yx}}$$

To find  $b_{xy}$ , we have

$$40X - 18Y = 214$$

$$\Rightarrow 40X = 18Y + 214$$

$$\Rightarrow X = \frac{18}{40}Y + \frac{24}{40}$$

$$\therefore b_{xy} = \frac{18}{40}$$

To find  $b_{yx}$ , we have

$$8X - 10Y + 66 = 0$$

$$\Rightarrow 10Y = 8X + 66$$

$$\Rightarrow Y = \frac{8}{10}X + \frac{66}{10}$$

$$\therefore b_{yx} = \frac{8}{10}$$

$$\therefore \text{The correlation coefficient (r)} = \sqrt{b_{xy} \times b_{yx}}$$

$$= \sqrt{\frac{18}{40} \times \frac{8}{10}}$$

$$= \sqrt{0.36}$$

$$= 0.60 \text{ [Ans.]}$$

(iii) We know,

$$r \frac{\sigma_y}{\sigma_x} = b_{yx}$$

$$\Rightarrow 0.60 \frac{\sigma_y}{3} = \frac{8}{10}$$

$$\Rightarrow \sigma_y = \frac{24}{6} = 4$$

$$\Rightarrow \sigma_y = 4 \text{ [Ans.]}$$

**Thank You**

1. The coefficient of correlation lies between -1 and +1. Symbolical  $-1 \leq r \leq 1$

Proof : We know that,

$$\text{Coefficient of Correlation, } r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} \text{----- (i)}$$

$$\begin{aligned} \text{Let, } a &= \frac{X - \bar{X}}{\sqrt{\sum (X - \bar{X})^2}} & \& \ b = \frac{Y - \bar{Y}}{\sqrt{\sum (Y - \bar{Y})^2}} \\ a^2 &= \frac{(X - \bar{X})^2}{\sum (X - \bar{X})^2} & \& \ b^2 = \frac{Y - \bar{Y}}{\sum (Y - \bar{Y})} \text{ [Taking square on both sides]} \\ \sum a^2 &= \frac{\sum (X - \bar{X})^2}{\sum (X - \bar{X})^2} & \& \ \sum b^2 = \frac{\sum (Y - \bar{Y})^2}{\sum (Y - \bar{Y})^2} \\ \sum a^2 &= 1 & \& \ \sum b^2 = 1 \end{aligned}$$

$$ab = \frac{(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}}$$

$$\sum ab = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}}$$

$$\sum ab = r \text{ [From (i)]}$$

Then  $\sum (a \pm b)^2 \geq 0$  [Square term can not be negative]

$$\Rightarrow \sum a^2 + \sum b^2 \pm 2 \sum ab \geq 0$$

$$\Rightarrow 1 + 1 \pm 2r \geq 0$$

$$\Rightarrow 2 \pm 2r \geq 0$$

$$\Rightarrow 2(1 \pm r) \geq 0$$

$$\Rightarrow 1 \pm r \geq 0$$

Taking positive sign,  $1 + r \geq 0$  or,  $r \geq -1$  or,  $-1 \leq r$  ----- (ii)

Taking negative sign,  $1 - r \geq 0$  or,  $1 \geq r$  or,  $r \leq 1$  ----- (iii)

From the equation (ii) & (iii), we can write,  $-1 \leq r \leq 1$  (Proved)

2. The coefficient of correlation is independent of change of origin and scale.

Proof : By change of origin we mean subtracting some constant from the given value of X and Y and by change of scale we mean dividing or multiplying every value of X and Y by some constant.

$$\text{We know that, correlation coefficient, } r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} \text{----- (i)}$$

$$\text{Let the new variable U & V where } U = \frac{X - a}{i} \text{ \& } V = \frac{Y - b}{c}$$

Here, a and b are the origin & i & c are the scale.

$$\Rightarrow X - a = iU \text{ \& } Y - b = cV$$

$$\Rightarrow X = a + iU \text{ \& } Y = b + cV$$

$$\Rightarrow \bar{X} = a + i\bar{U} \text{ \& } \bar{Y} = b + c\bar{V}$$

$$\Rightarrow X - \bar{X} = a + iU - a - i\bar{U} \text{ \& } Y - \bar{Y} = b + cV - b - c\bar{V}$$

$$\Rightarrow X - \bar{X} = iU - i\bar{U} \text{ \& } Y - \bar{Y} = cV - c\bar{V}$$

$$U - \bar{U} = (U - \bar{U}) \text{ \& } V - \bar{V} = (V - \bar{V})$$

$$X = a + iU \text{ \& } Y = b + cV$$

$$\bar{X} = a + i\bar{U} \text{ \& } \bar{Y} = b + c\bar{V}$$

$$X - \bar{X} = iU - i\bar{U} \text{ \& } Y - \bar{Y} = cV - c\bar{V}$$

$$X - \bar{X} = i(U - \bar{U}) \text{ \& } Y - \bar{Y} = c(V - \bar{V})$$

Substituting these values in (i), we get,

$$r = \frac{\sum ic(U - \bar{U})(V - \bar{V})}{\sqrt{\sum [i(U - \bar{U})]^2 \sum [c(V - \bar{V})]^2}}$$

$$r = \frac{ic \sum (U - \bar{U})(V - \bar{V})}{ic \sqrt{\sum (U - \bar{U})^2 \sum (V - \bar{V})^2}}$$

$$r = \frac{\sum (U - \bar{U})(V - \bar{V})}{\sqrt{\sum (U - \bar{U})^2 \sum (V - \bar{V})^2}}$$

Thus the formula for r remains unchanged. Hence the value of r is independent of change of origin and scale. (Proved)

3. Two independent variables are uncorrelated but the converse is not true.

Proof : If X and Y are independent variables, then

$$\text{Cov}(X, Y) = 0$$

$$r(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} = 0$$

Hence two independent variables are uncorrelated but the converse of the theorem is not true, i.e., two uncorrelated variables may not be independent as the following example illustrates:

|    |     |    |    |   |   |    | Total         |
|----|-----|----|----|---|---|----|---------------|
| X  | -3  | -2 | -1 | 1 | 2 | 3  | $\sum X = 0$  |
| Y  | 9   | 4  | 1  | 1 | 4 | 9  | $\sum Y = 28$ |
| XY | -27 | -8 | -1 | 1 | 8 | 27 | $\sum XY = 0$ |

$$\bar{X} = \frac{\sum X}{n} = 0, \text{Cov}(X, Y) = \frac{\sum XY}{n} - \bar{X} \bar{Y} = 0$$

$$r(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 0$$

But, in the above example, the variables X and Y are uncorrelated. But we find that X and Y are not independent but they are connected by the relation  $Y=X^2$ . Hence two uncorrelated variables need not necessarily be independent. (Proved)

Prove that correlation coefficient is the geometric mean of two regression coefficients.

Proof : We know that, Regression Coefficient of x on y,  $b_{xy} = r \frac{\sigma_x}{\sigma_y}$  ----- (i)

And Regression Coefficient of y on x,  $b_{yx} = r \frac{\sigma_y}{\sigma_x}$  ----- (ii)

(i) × (ii) ⇒

$$b_{xy} \times b_{yx} = r \frac{\sigma_x}{\sigma_y} \times r \frac{\sigma_y}{\sigma_x}$$

$$\Rightarrow r^2 = b_{xy} \times b_{yx}$$

$$\Rightarrow r = \pm \sqrt{b_{xy} \cdot b_{yx}} \text{ [If Regression coefficients are Positive then } r \text{ is Positive and}$$

Regression Coefficients are negative then r is negative].

So, Correlation Coefficient is the geometric mean of two Regression Coefficients. (Proved)

Estimation of Parameters of simple linear regression model.

Or, Estimation of regression coefficient by using the method of least square.

By the method of least square we mean to minimize or making least the squared error.

Let us consider the simple linear equation

$$y_i = a + bx_i + e_i \text{ [} i = 1, 2, 3, \dots, n \text{]}$$

$$\Rightarrow e_i = y_i - a - bx_i \text{ [Where, } a = \text{Intercept (Parameter)}$$

$b = \text{Regression Coefficient (Parameter)}$

$y = \text{Dependent Variable}$

$x = \text{Independent Variable}$

$e = \text{Error term}$ ]

$$\Rightarrow e_i^2 = (y_i - a - bx_i)^2 \text{ [Taking square on both sides]}$$

$$\Rightarrow \sum e_i^2 = \sum (y_i - a - bx_i)^2 \text{ [Taking Summation on both sides]}$$

$$\Rightarrow \text{Let } \sum e_i^2 = L = \sum (y_i - a - bx_i)^2$$

Let us differentiate L with respect to a and b and equating to zero.

$$\frac{\partial L}{\partial a} = 0$$

$$\Rightarrow \frac{\partial}{\partial a} \sum (y_i - a - bx_i)^2 = 0$$

$$\Rightarrow -2\sum (y_i - a - bx_i) = 0$$

$$\Rightarrow \sum (y_i - a - bx_i) = 0$$

$$\Rightarrow \sum y_i - \sum a - b \sum x_i = 0$$

$$\Rightarrow \sum y_i - na - b \sum x_i = 0$$

$$\Rightarrow \sum y_i = na + b \sum x_i$$

$$\Rightarrow \frac{\sum y_i}{n} = \frac{na}{n} + b \frac{\sum x_i}{n} \text{ [Dividing both sides by } n]$$

$$\Rightarrow \bar{y} = a + b\bar{x} \text{ [Since } \bar{x} = \frac{\sum x_i}{n} \text{ and } \bar{y} = \frac{\sum y_i}{n}]$$

$$\Rightarrow a = \bar{y} - b\bar{x} \text{ ----- (i)}$$



$$\begin{aligned}
& \text{And } \frac{\partial L}{\partial b} = 0 \\
& \Rightarrow \frac{\partial}{\partial b} \sum (y_i - a - bx_i)^2 = 0 \\
& \Rightarrow 2 \left\{ \sum (y_i - a - bx_i) \right\} \times (-x_i) = 0 \\
& \Rightarrow -2 \left\{ \sum (y_i - a - bx_i) \times x_i \right\} = 0 \\
& \Rightarrow \left\{ \sum (y_i - a - bx_i) \right\} \times x_i = 0 \\
& \Rightarrow \sum x_i y_i - a \sum x_i - b \sum x_i^2 = 0 \\
& \Rightarrow \sum x_i y_i - a \sum x_i = b \sum x_i^2 \\
& \Rightarrow \sum x_i y_i - (\bar{y} - b\bar{x}) \sum x_i = b \sum x_i^2 \\
& \Rightarrow \sum x_i y_i - (\bar{y} - b\bar{x}) n\bar{x} = b \sum x_i^2 \quad \left[ \text{Since } \bar{x} = \frac{\sum x_i}{n} \right. \\
& \qquad \qquad \qquad \left. \Rightarrow n\bar{x} = \sum x_i \right] \\
& \Rightarrow \sum x_i y_i - n\bar{x}\bar{y} + b n \bar{x}^2 = b \sum x_i^2 \\
& \Rightarrow \sum x_i y_i - n\bar{x}\bar{y} = b \sum x_i^2 - b n \bar{x}^2 \\
& \Rightarrow \sum x_i y_i - n\bar{x}\bar{y} = b (\sum x_i^2 - n\bar{x}^2) \\
& \Rightarrow b = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} \text{-----(ii)}
\end{aligned}$$

Where  $a$  is the intercept of the model and  $b$  is the regression coefficient of  $y$  on  $x$  which is also denoted by  $b_{yx}$ .

Prove that Regression coefficients are independent of origin but not of scale.

*Proof :* We know that, RegressionCoefficient of y on x is

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Let,  $u = \frac{x - a}{h}$  and  $v = \frac{y - b}{k}$  where a, b are the origin and h, k are the scale.

$$\Rightarrow x - a = hu \quad \text{and} \quad y - b = kv$$

$$\Rightarrow x = a + hu \quad \text{and} \quad y = b + kv$$

$$\Rightarrow \bar{x} = a + h\bar{u} \quad \text{and} \quad \bar{y} = b + k\bar{v}$$

$$b_{yx} = \frac{\sum (a + hu - a - h\bar{u})(b + kv - b - k\bar{v})}{\sum (a + hu - a - h\bar{u})^2}$$

$$\Rightarrow b_{yx} = \frac{\sum (hu - h\bar{u})(kv - k\bar{v})}{\sum (hu - h\bar{u})^2}$$

$$\Rightarrow b_{yx} = \frac{hk \sum (u - \bar{u})(v - \bar{v})}{h^2 \sum (u - \bar{u})^2}$$

$$\Rightarrow b_{yx} = \frac{k \sum (u - \bar{u})(v - \bar{v})}{h \sum (u - \bar{u})^2}$$

$$\Rightarrow b_{yx} = \frac{k}{h} b_{uv}$$

Similarly, we have, RegressionCoefficient of x on y,  $b_{xy} = \frac{h}{k} b_{vu}$

Hence the Proof.